7b	Does the substance demonstrate **activation of the Nrf2-Keap1-ARE toxicity pathway** in an EU/OECD adopted *in vitro* test (e.g. B.60/OECD TG 442d)? *(Key event 2 of the AOP*) *In vitro* test methods that have been validated and are considered scientifically valid but are not yet adopted by the EU and/or OECD may also be used if the provisions defined in Annex XI to the REACH Regulation are met.	**YES/NO:** Consider classifying as Skin sensitiser (Cat. 1A or not[e])[68]. If not conclusive, use this information for *Weight-of-Evidence* analysis under point 8.
7c	Does the substance demonstrate **induction of the cell surface markers** (CD54 and/or CD86) of monocytic cells in a validated *in vitro* test (e.g. h-CLAT)? (*Key event 3 of the AOP*) *In vitro* test methods that have been validated and are considered scientifically valid but are not yet adopted by the EU and/or OECD may also be used if the provisions defined in Annex XI to the REACH Regulation are met.	**YES/NO:** Consider classifying as Skin sensitiser (Cat. 1A or not[e])[68]. If not conclusive, use this information for *Weight-of-Evidence* analysis under point 8.
7d	Is any additional testing/generation of data considered necessary in order to conclude on classification, or e.g. to explain the inconsistent data obtained in previous elements or to address the *Key event 4 of the AOP* (T-cell proliferation) with an *in vitro* test? [d]	**YES/NO**: Consider performing the test and use this information for *Weight-of-Evidence* analysis.
Weight-of-Evidence analysis[e]		
8	The "elements" described above may be arranged as appropriate. Taking all existing and relevant data (elements 1-7) into account, is there sufficient information to meet the respective information requirement of Section 8.3 of Annex VII and to make a decision on whether classification and labelling are warranted? For specific guidance on *Weight of Evidence see below*.	**YES**: Classify as Skin Sensitiser Cat. 1A or not[e] or consider no classification[69]. **NO:** Consider the next element of the strategy.
Generation of new in vivo data for sensitisation as a last resort (Annex VII to the REACH Regulation)		
9	Does the substance demonstrate sensitising properties in an EU/OECD adopted *in vivo* test, the LLNA (EU B.42/OECD TG 429[f])? →	**YES**: Classify according to CLP criteria (Skin Sensitiser Cat. 1A or not[e]). **NO:** No classification needed.

Notes to the testing and assessment figure on skin sensitisation:

a) Data from case reports, occupational experience, poison information centres, Human Patch Tests or from clinical studies.

b) It is worthwhile to apply the OECD QSAR Toolbox (see Section R.7.3.4.1) to check whether there are existing data available for the substance of interest or existing and good quality data available on skin sensitisation for potential analogue substances. It should be noted that in case read-across or a category approach is to be used, adequate justification must be provided (for further information on ECHA's read-across assessment framework (RAAF) see http://echa.europa.eu/support/grouping-of-substances-and-read-across). The use of available and suitable (Q)SAR models for skin sensitisation is also recommended. In case substance metabolism and/or abiotic transformation (e.g. auto-oxidation or hydrolysis) leads to the generation of new chemical species the use of the QSAR Toolbox may be helpful in finding relevant data that can be used.

c) When (a) non-animal testing approach(es) is (are) used, information needs to be generated to address a sufficient number of the key events specified in elements 7a to 7c in order to conclude on the skin sensitisation endpoint, including whether the substance can be presumed to produce significant sentisitisation in humans (Cat. 1A or not) (Basketter *et al.,* 2015b). Additional information obtained from e.g. (Q)SARs can be used to support the conclusions reached. The information obtained from the assessment of one of the key events may be used to select the next most appropriate *in chemico/in vitro* test.

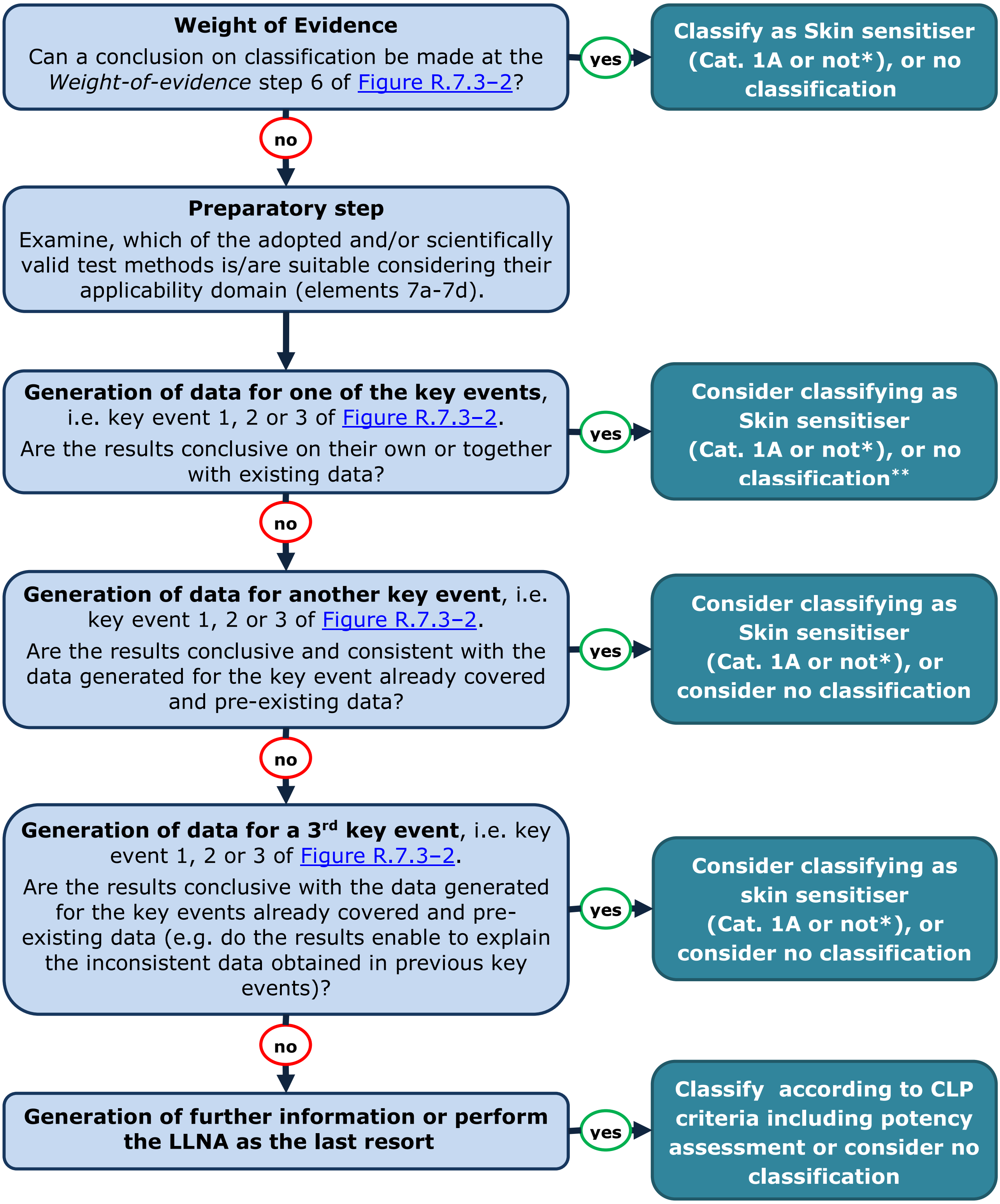
d) At this point in time (at the time of publication of this Guidance), there are no validated or adopted *in vitro* test methods available to address T-cell proliferation. Developments may occur in the future in the field of *in chemico/in vitro* test methods that may be able to address the limitations of the currenlty adopted and/or validated test methods and could provide more confidence in the results already obtained.

e) To reach a conclusion on (non-)classification, the following questions should be addressed:

> i) Does the evidence enable to conclude that the substance is not a skin sensitiser? If so, conclude on no classification.
>
> ii) Does the evidence enable to conclude that the substance is presumed to produce significant sensitisation in humans i.e. Cat. 1A? If so, classify accordingly.
>
> iii) Does the evidence enable to conclude that the substance is a skin sensitiser and significant sensitisation in humans i.e. Cat. 1A **can** be excluded? If so, it is presumed that the substance would be a moderate skin sensitiser i.e. Cat. 1B and it is recommended to classify accordingly.
>
> In case none of these conditions are met, e.g. when Cat. 1A **cannot** be excluded, further testing needs to be performed, *in vivo* testing being the last resort.

f) Note: for the LLNA variants, i.e. EU B.50/OECD TG 442A and B.51/442B, there are currently no CLP criteria available for predicting skin sensitisation potency. However, dose-response relationship information may provide some information on skin sensitisation potency that can be used within a *Weight-of-Evidence* approach. It is recommended that, when new *in vivo* data are generated, the "standard" LLNA according to EU B.42 / OECD TG 429 be used, if possible.

An overview of how to use the information on the different Key events is given in Figure R.7.3–3 below.



* For a sensitising substance, in case a sufficiently reliable conclusion can be made to exclude significant sensitisation in humans (Cat. 1A), it is presumed that the substance would be a moderate skin sensitiser (Cat. 1B).

** Concluding with "no classification" based on the data generated for one of the key events is only possible when additional information is available to support the conclusion.

Figure R.7.3–3 Snapshot of the Testing and Assessment Strategy - How to use the data on key events. Note: the order of key events is not specified and can be arranged as appropriate

Predictive capacity of the existing *in vivo* and non-animal tests when compared to human data

Different approaches have been presented in the scientific literature on the predictivity of non-animal testing approaches (see Paragraph below on "How to perform and report *Weight-of-Evidence* analysis based on non-animal approaches"). E.g. Urbisch *et al*. (2015) compared the predictive capacity of the LLNA and that of non-animal *(in chemico/in vitro)* testing strategies towards skin sensitisers in humans. The authors showed that for LLNA *vs.* historical predictive testing in humans, the accuracy of prediction was 82%, with a sensitivity (i.e. true positive rate) of 91% and a specificity (i.e. true negative rate) of 64%. For non-animal test methods used in combination the accuracy was 90% with a sensitivity and a specificity of 90% (n~100 substances). While the nature of the data is only partly disclosed and these data have not been assessed independently, there is some indication that, when *in chemico* and *in vitro* methods are used in combination, their ability to predict human data seems to be comparable to that of the LLNA in the identification of human sensitisers and non-sensitisers (i.e. Cat. 1 *vs.* non-classified substances). However, the individual tests on their own were not as sensitive as the LLNA.

How to deal with the lack of or limited metabolic capacity of non-animal test methods

The *in chemico* DPRA does not have any metabolic capacity and the *in vitro* Keratinosens^{TM} assay and h-CLAT assay have only a limited metabolic capacity in the test systems. Due to the lack of or the limited metabolic capacity, these test methods may not correctly identify sensitisers that would require metabolic activation to exert their sensitisation activity and therefore they may provide false negative results. The above non-animal test methods may also provide false negative results for substances requiring abiotic transformation (e.g. auto-oxidation or hydrolysis), especially in case of slow oxidation rates. The issue of not identifying pre-haptens correctly is not solely related to *in chemico/in vitro* methods, but can also occur in *in vivo* tests. Currently, a modification of the *in chemico* DPRA is under development (Peroxidase Peptide Reactivity Assay (PPRA)) for a better identification of pro-haptens (Merckel *et al.*, 2013), however work is still ongoing to assess the added value of this assay.

An analysis concerning the ability of non-animal testing methods to detect pro- and pre-haptens was performed on the occasion of an EURL ECVAM expert meeting held in 2015 and discussed with a group of experts (Casati *et al.*, 2016).
The experts noted that many of the substances believed to be pro-haptens were actually also pre-haptens (e.g. geraniol). It was also noted that the majority of the non-direct acting haptens were pre-haptens which were generally identified with the DPRA and cell-based assays. A problem was noted with slow oxidizers, which were not correctly identified, however their identification would also fail by using *in vivo* test methods. Substances that are exclusively pro-haptens, which were not identified by the DPRA, were generally correctly identified by one of the cell-based assays (the h-CLAT detecting the majority of those). The outcome of the analysis was that, by using non-animal test methods, a comparable prediction of skin sensitisation hazard to the one of the LLNA was obtained. The expert group concluded that, in light of this analysis, unless there is a compelling scientific argument that a substance could be an exclusively metabolically activated pro-hapten, the negative results obtained from non-animal test methods could be seen as acceptable. *In silico* tools such as TIMES-SS or OECD QSAR Toolbox could be used to support such argumentation.

Due to these uncertainties it is strongly recommended to evaluate all available toxicokinetic information (see Section R.7.12 of Chapter R.7c the *Guidance on IR&CSA*) and to run computational tools such as the OECD QSAR Toolbox or TIMES-SS that can partially cover for the lack of metabolic or abiotic transformation (e.g. auto-oxidation or hydrolysis) information. However the user should not rely solely on the results from these tools to exclude the possibility that metabolic activation or abiotic transformation (e.g. auto-oxidation or hydrolysis) may take place. These softwares have modules for simulating (skin) metabolism and abiotic transformation (e.g. auto-oxidation or hydrolysis) of substances. TIMES-SS currently does not have a skin sensitisation model that combines for example hydrolysis with the skin sensitisation endpoint. In case the substance is predicted to be a non-sensitiser but the simulated metabolites or products have positive experimental data or trigger skin sensitisation alerts, the latter might be responsible for sensitisation. The simulated metabolites need a specific assessment and might require the generation of new experimental data. Other tools (e.g. Derek Nexus) incorporate the knowledge for metabolic transformation in developing alerts for skin sensitisation (e.g. hydroquinone and precursors).

It has been proposed that experimental data available from other endpoints (e.g. from *in vitro* mutagenicity) could provide additional information to support the conclusions on skin sensitisation obtained from non-animal test methods. The approach to use *in vitro* mutagenicity test results for assessing potential metabolite formation has not gone through an independent review and more experience has still to be gained. However, it may provide some useful insights into the potential electrophilic reactivity of the substance under metabolic conditions (Patlewicz *et al*., 2010). Information obtained from *in vitro* mutagenicity studies may provide useful information in the *Weight-of-Evidence* assessment when used in combination with computational tools, which could either support the results obtained from the *in chemico/in vitro* test methods or trigger further testing needs.

Use of non-animal data (e.g. *in vitro* methods) to support a category approach

In case a category approach is used to fulfil the REACH information requirements and data are available for some category members only, the generation of data by using e.g. *in chemico/in vitro* test methods could be used to support the category approach for this endpoint. This is especially the case when similar results on the skin sensitisation potential (or the lack thereof) are obtained from one (or more) non-animal testing method(s). In practice, it may be possible to perform only one or two *in chemico/in vitro* tests for the target substance of the read-across. In case of conflicting results, it is important to consider why they occurred: the reason might be that the specific substance does not belong to the category because of sensitising properties different from those of category members with good quality animal and/or human data, or that the substance does not fit into the applicability domain of the specific non-animal test. In those cases, *in vivo* testing may be required to assess the skin sensitisation potential of the substance.

Whenever a category approach is applied, it is essential to always justify why data can be read across from the category member substances to the target substance for which there is no good quality animal and/or human data. This justification also needs to be endpoint specific. Advice on how to build and report a category can be found on the ECHA website at: http://echa.europa.eu/support/grouping-of-substances-and-read-across.

Sub-categorisation

According to Annex VII, section 8.3, in addition to the assessment of whether the substance is a skin sensitiser or a non-sensitiser, the potency (Cat. 1A or not) of skin sensitising substances must be addressed. In case the substance is identified as skin sensitiser based on the results of *in vitro/in chemico* testing and these results allow a sufficiently reliable conclusion that the substance does not have the potential to produce significant sensitisation in humans, it can be presumed that the substance would be a moderate skin sensitiser (Cat. 1B). In case existing information from an *in vivo* study is available (i.e. initiated or generated before 10 May 2017), which does not enable potency assessment, this information can still be used to fulfil REACH information requirement for this endpoint. However, for skin sensitising substances, it is recommended to additionally consider existing information from other sources e.g. read-across and QSARs or generation of new non-animal test data to ensure adequate classification and risk assessment. Currently (at the time of publication of this Guidance), there is no widely accepted approach to integrate non-animal data into an adequate sensitisation potency classification. Some approaches have been proposed for potency prediction (see Appendix R.7.3–4). The LLNA (EU B.42/OECD TG 429), allows potency estimation and the setting of SCLs, however a retrospective analysis performed by ICCVAM (2011) of LLNA data compared to human revealed that for 27 strong sensitising substances analysed, approximately half of them were underclassified based on an EC3 cut-off value of <2%. Other *in vivo* test methods either have their limitations or are unable to predict potency (for further details see Section R.7.3.5.1, under "Animal data").

All available and newly generated information needs to be carefully considered for potency assessment within a *Weight-of-Evidence* approach[69]. This is particularly important for classification of a mixture containing a skin sensitising substance, since potency assessment is a basis for the concentration limits to be applied for classification of mixtures containing a sensitising substance. Therefore, depending on the skin sensitisation potency of a substance, different concentration limits are to be applied for classification of the mixture: i.e. for Cat. 1 and Cat. 1B the generic concentration limit (GCL) is 1%, for Cat. 1A (strong) the GCL is 0.1%, and for very strong (extreme) sensitisers an SCL of 0.001% (or lower) is recommended (see Section 3.4 of the *Guidance on the Application of the CLP criteria*). In short, if information on potency is lacking, this would lead to the situation where mixtures with potent sensitisers would not be classified in a way which reflects the hazard of the mixture. This would mean a lowering of the safety level as compared to current situation, which may lead to an increased incidence of human sensitisation to potent sensitisers. Currently, the non-animal test methods may not always provide sufficient information on potency estimation.

Therefore, in order to fulfil REACH information requirements for a substance, as laid down in Annex VII, section 8.3, the data from non-animal test methods (*in chemico/in vitro*) must allow the conclusion on whether the substance is a skin sensitiser or not and on whether it can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A). In case a conclusion cannot be drawn based on the results obtained from *in vitro/in chemico* methods, additional available information, e.g. from LLNA data for similar substances can be used in a *Weight-of-Evidence* approach and may help reach a conclusion on skin sensitisation potency (Cat. 1A or not) or conclude on no classification. In case no firm conclusion on skin sensitisation potency (Cat. 1A or not) can be drawn, while there is some evidence, e.g. from peptide reactivity, that the substance may be a strong sensitiser, a precautionary Cat. 1A classification may be considered.

A review of different approaches aiming to provide indications of skin sensitisation potency is provided in Appendix R.7.3–4. However, as work is still ongoing to try to address the lack of potency characterisation based on non-animal approaches (e.g. Reisinger *et al*., 2015), the registrant is advised to follow-up the recent and future developments in the field. The registrant is also advised to follow any updates to the ECHA webpage concerning testing methods and alternatives (see: http://echa.europa.eu/support/oecd-eu-test-guidelines).

When reliable and relevant human data are available, they can also be used for the classification of skin sensitising substances into sub-categories according to the CLP Regulation (for further information, see Section 3.4.2.2.3.1. of the *Guidance on the Application of the CLP criteria*).

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[69] For fulfilling the information requirement of Annex VII, 8.3.1, it is necessary to consider the information obtained from the three key events (unless data from fewer key events already allows classification and risk assessment, as specified in Annex VII, section 8.3) in a *Weight-of-Evidence* approach, even though no formal *Weight-of-Evidence* in the meaning of Annex XI, section 1.2 needs to be submitted.

How to perform and report a *Weight-of-Evidence* analysis based on non-animal approaches

For fulfilling the information requirement of Annex VII, 8.3.1, it is necessary to consider the information obtained from the three key events (unless data from fewer key events already allow classification and risk assessment, as specified in Annex VII, section 8.3) in a *Weight-of-Evidence* approach, even though no formal *Weight-of-Evidence* in the meaning of Annex XI, section 1.2 needs to be submitted in this case. The approach using *in chemico/in vitro* is based on the OECD AOP for skin sensitisation and its key events (OECD, 2012). It is recognised that, in the LLNA key events 1 to 4 are addressed since the biological response, i.e. induction of skin sensitisation, is caused by the cascade of these key events. Therefore, in the *Weight-of-Evidence* approach these key events must be covered to the extent possible. At present, three *in chemico*/*in vitro* tests, each addressing a specific key event of the AOP, have been adopted by the OECD and validated by EURL ECVAM.

As specified in Annex VII, section 8.3.1, three key events must be covered by an *in chemico*/*in vitro* test unless information obtained from test method(s) addressing one or two key events already allows classification and riks assessment, as specified in Annex VII, section 8.3. In case information on one or more key events is provided by e.g. (Q)SAR or read-across, Annex XI adaptations shoud be used (Annex XI, 1.2 – *Weight-of-Evidence*, Annex XI, 1.5 – read-across). It should be noted that, based on current knowledge, the information obtained from peptide reactivity, whether obtained from *in chemico* or *in silico* methods, seems to show the highest predictive power and may provide more weight to the overall assessement of skin sensitisation (Natsch *et al*., 2012, Urbisch *et al*., 2015). There is currently no scientifically valid or internationally adopted *in vitro* method to cover the fourth key event, i.e. lymphocyte proliferation. In case a suitable non-animal test method, e.g. *in vitro* method, becomes available, this could bring more weight to the overall *Weight-of-Evidence* approach. However, the available studies on the predictivity of different combinations of *in chemico/in vitro* methods/other information type seem to show that a good predictivity for hazard identification (Cat. 1 *vs.* non-sensitiser) can be achieved by covering the first three key events (Hirota *et al.*, 2015; Maxwell *et al*., 2014; Patlewicz *et al*., 2014; Takenouchi *et al*., 2015; Tsujita-Inoue *et al.,* 2014; Urbisch *et al.*, 2015; Van der Veen *et al*., 2014): the use of the non-animal test methods in combination seems to be comparable to that of the LLNA in the identification of human sensitisers and non-sensitisers (i.e. Cat. 1 *vs*. non-classified). However, the individual tests on their own were not as sensitive as the LLNA.

When *in chemico*/*in vitro* studies are used as specified in section 8.3.1 of Annex VII to fulfil the information requirement for skin sensitisation the registrant must provide a case-specific justification on why and how the *in chemico/in vitro* data used within a *Weight-of-Evidence* approach[70] can cover the information requirement. In that *Weight-of-Evidence* justification, e.g. coverage of the **key events** (see "Testing and assessment strategy for skin sensitisation" above), the quality and reliability of the data, scope and limitations of each test method used need to be considered. When all the evidence is taken together, the consistency of the evidence and completeness of the data need to be assessed. Further provisions on *Weight of Evidence* can be found in Section R.4.4 of Chapter R.4 of the *Guidance on IR&CSA* and in Art. 9(3) of the CLP Regulation.

It should be noted that the data used to cover the key events, whether they are *in chemico/in vitro* results or other data, can be inconsistent. For example it may happen that two tests/data

[70] For fulfilling the information requirement of Annex VII, 8.3.1, it is necessary to consider the information obtained from the three key events (unless information obtained from test method(s) addressing one or two key events already allows classification and risk assessment, as specified in Annex VII, section 8.3) in a *Weight-of-Evidence* approach, even though no formal *Weight-of-Evidence* in the meaning of Annex XI, section 1.2 needs to be submitted.

points are negative and one is positive for skin sensitisation. In case of inconsistent or conflicting data, a scientific explanation should be provided. The explanation may be, for example, that the substance needs metabolic activation to become a skin sensitiser and the test system misses the required metabolic competence. It may also be that the test substance does not fall into the applicability domain(s) of one or more of the *in chemico/in vitro* methods used. If the conflicting information/results cannot be explained, the registrant will need to generate/collect further information in order to support the prediction of the skin sensitisation potential of the substance. If in the end the registrant is not able to conclude on this endpoint due to inconsistent or inconclusive data, there may be a need to perform an LLNA.

As pointed out in elements 6 and 8 (*Weight-of-Evidence* analysis) of the testing and assessment strategy above, in case the skin sensitisation potential of a substance, including an assessement of whether the substance can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A or not), cannot be properly characterised based on the available data, generation of new data is necessary. This data can be e.g. (Q)SAR, data that are specific to a key event (e.g. *in chemico/in vitro*), read-across or, as a last resort, the *in vivo* study, i.e. the LLNA. The LLNA must be performed in any case if e.g.:

- The registrant may have some existing information from similar substances and/or QSAR(s) indicating that the substance may be a strong or extreme skin sensitiser and cannot conclude on adequate for classification and labelling and risk assessement, even by generating additional information by using *in chemico/in vitro* methods,
- The test substance does not fall into the applicability domain of any of the *in chemico*/*in vitro* tests for skin sensitisation (Note: assessment of the suitability of a test method for a substane should be performed before testing is initiated), or
- The results of the *in chemico*/*in vitro* tests are inconsistent and this inconsistency cannot be explained scientifically, or
- Determination of whether the substance can be presumed to have the potential to produce significant sensitisation in humans (Cat. 1A or not) is not possible based on non-animal testing approaches, as required in Annex VII, section 8.3.

At the end of the testing and assessment strategy, the data obtained, justification of the choice of the test methods, analysis of data consistency, conclusion made on hazard identification and on classification according to the CLP Regulation should be reported clearly and transparently. For the reporting of the approach applied according to a testing and assessment strategy it is recommended to use the template provided in Appendix R.7.3–3 of this Guidance and which is based on Annex I of the OECD Guidance Document On The Reporting Of Defined Approaches To Be Used Within Integrated Approaches to Testing And Assessment (OECD, 2016a).

Note that each individual information source shoud be reported in a separate endpoint study record in the IUCLID dossier. Then, the *Weight-of-Evidence* approach wrapping all of the information sources together can be attached to the skin sensitisation endpoint summary.

RESPIRATORY SENSITISATION

NOTE: Respiratory sensitisation is not a standard information requirement under REACH. However in case data are available they should be included in the technical dossier and used to support classification and labelling where relevant.

R.7.3.8 Mechanisms of respiratory sensitisation

For substances that sensitise via the respiratory tract, there is still uncertainty regarding the exact mechanisms leading to respiratory sensitisation. Based on the current knowledge the induction of respiratory sensitisation can occur *via* inhalation or dermal exposure to the sensitising substance (Redlich, 2010; Kimber *et al.*, 2015).

The current hypothesis is that the mechanism favours Th2-type immune responses (skin sensitisation favours Th1-type response), which is characterised by the production of cytokines, such as IL-4 and IL-5, and IgE antibodies. This is supported by studies performed in rodents and by human evidence (Adenuga *et al.,* 2012; Kimber *et al.*, 2014b; Helakoski *et al.*, 2015). Recently, it has been hypothesised that Th17 cells would also play a crucial role in respiratory sensitisation *via* secretion of IL-17 (Lambrecht and Hammad, 2013). The role of IgE may be the greatest reason for uncertainty, as there are patients who display serum IgE antibodies of the appropriate specificity, whereas in other instances (and particularly with respect to the diisocyanates) there are symptomatic subjects in whom it is not possible to detect these IgE antibodies. It has been hypothesised that either there may be a mechanism leading to respiratory sensitisation that is IgE-independent, or this is linked to technical difficulties in the accurate measurements of hapten-specific IgE-antibodies (Cochrane *et al.*, 2015).

In addition, an AOP for respiratory sensitisation to low molecular weight substances is currently under development at the OECD. The proposed Key Events for respiratory sensitisation are:

- Key Event 1: Covalent binding of substances to proteins (note: based on current knowledge, there seems to be a greater selectivity of respiratory sensitisers for lysine reactivity than for cysteine, whereas skin sensitisers bind both to cysteine and lysine (Lalko *et al.*, 2013a));

- Key Event 2: Cellular danger signals (activation of inflammatory cytokines and chemokines and cytoprotective gene pathways (Th2));

- Key Event 3: Dendritic cell activation and migration (Th2 skewed);

- Key Event 4: Activation and proliferation of T-cells (Th2) (Mekenyan *et al.*, 2014 and Sullivan *et al.*, 2015).

R.7.3.9 Information sources on respiratory sensitisation

R.7.3.9.1 Non-human data on respiratory sensitisation

Non-testing data on respiratory sensitisation

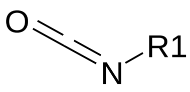
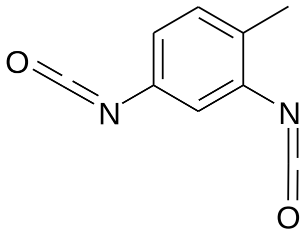
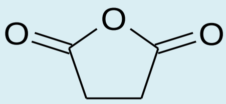
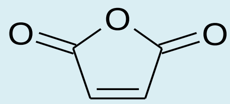
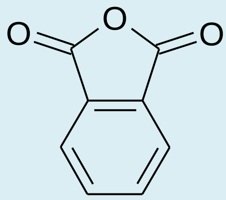
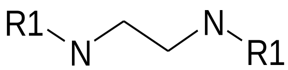
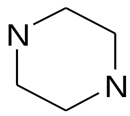
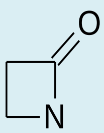
Attempts to model respiratory sensitisation have been hampered by the lack of a predictive test protocol for assessing chemical respiratory sensitisation. (Q)SAR models are available but these have largely been based on data for substances reported to cause respiratory hypersensitivity in humans. Examples of some structural alerts are shown in Table R.7.3–3.

Agius *et al.* (1991) made qualitative observations concerning the chemical structure of substances causing occupational asthma. This work drew attention to the large proportion of chemical asthmagens with at least two reactive groups, e.g. ethylene diamine and toluene diisocyanate. The earlier work was followed up by a simple statistical analysis of the occurrence of structural fragments associated with activity, with similar conclusions (Agius *et al.*, 1994 and 2000).

The MCASE group has developed three models for respiratory hypersensitivity (Karol *et al.*, 1996; Graham *et al.*, 1997, Cunningham *et al.*, 2005). The Danish (Q)SAR Database has an in-house model for respiratory hypersensitivity for which estimates can be extracted from the online database (available at http://qsar.food.dtu.dk/). Derek Nexus contains several alerts derived from a set of respiratory sensitisers/asthmagens (Payne *et al.*, 1995).

The structural alerts (SARs) are in principle transparent and easy to apply. Structural alerts related to respiratory sensitisation have been collected and described in the literature (Aigus *et al.*, 1991, 1994 and 2000; Payne *et al.*, 1995). It should be stressed however that these are derived from chemical asthmagens, i.e. substances causing asthma like symptoms with or without immunological mechanisms, and not specifically chemical respiratory allergens. Enoch *et al.* (2012) developed a set of mechanism-based structural alerts for low molecular weight organic substances with the potential to cause respiratory sensitisation. A need still remains to develop new (Q)SARs when a robust predictive test method becomes available.

Table R.7.3–3 Examples of structural alerts for respiratory sensitisation

Structural Alert Description	Examples of structures
 isocyanate	 Toluene-2,4-diisocyanate
 cyclic anhydride	 maleic anhydride  trimellitic anhydride
 diamine	 piperazine
 β-lactam	Some β-lactam antibiotics from penicillin and the cephalosporin groups

Recent work on the mechanism of respiratory sensitisation in humans and on the identification of structural alerts specific to respiratory sensitisation has been described in Enoch *et al*. (2009, 2010, 2012 and 2014). In these papers, the authors investigated a common molecular initiating event and mechanism for low molecular weight respiratory sensitisers (found to be the formation of a covalent bond in the lung) and applied their findings to predict respiratory sensitisation by read-across. The authors have proposed a set of 52 structural alerts which define the chemistry associated with covalent protein binding in the lung. Each structural alert is also characterised by a mechanistic domain ("mechanistic alert") and some data indicating presence of effect. Most of these alerts (a total of 41) have been encoded in the OECD QSAR Toolbox (ver. 3.3) profiler "Respiratory sensitisation". The full list of the encoded structural alerts for respiratory sensitisation is available under the OECD QSAR Toolbox feature "documentation", together with the description, applicability domain, mechanism, set of substances used for the profile training set, profile/alert analysis. Some examples of structural alerts are di-isocyanates, anhydrides and lactams (e.g. β-lactams). Dik *et al.* (2014) give some recent guidance to the reader on how to value the different models / alerts and how to improve the predictivity from SARs with different positive and negative predictivity in a tiered approach. The paper includes compiled datasets (QSAR training datasets, literature data, occupational asthma data) on substances which are considered respiratory sensitisers. Thus, it can provide a source of analogues for grouping and endpoint-specific read-across data.

Testing data for respiratory sensitisation

In vitro data

No validated or widely recognised *in vitro* test methods specific to respiratory sensitisation are available yet, owing to the complexity of the mechanisms of the sensitisation process. This is most likely due to the fact that there are still many uncertainties concerning the underlying immunological mechanisms, in particular with respect to the role of IgE antibody.

Some *in vitro* methods to assess respiratory sensitisation have been published and here are a few examples:

- The MucilAir model from Epithelix® that uses 3D human airway epithelial model to study multiple endpoints, e.g. cilia beating monitoring, trans-epithelial electrical resistance, cytotoxicity and cytokine/chemokine release (Huang *et al.*, 2013);
- The genomic Allergen Rapid Detection (GARD) assay that measures cellular expression markers common to myeloid and dendritic cells by using flow cytometry (Forreryd *et al*., 2015);
- The use of *in chemico* DPRA and PPRA for the identification of respiratory sensitisers has also been considered, as the peptide/protein binding step is similar for both skin and respiratory sensitisers; however based on current knowledge on respiratory sensitisers there seems to be a greater selectivity for lysine reactivity than for cysteine for the majority of the substances tested (Lalko *et al*., 2012; 2013a; 2013b).

Other test methods are currently under development. Therefore it is advised to follow the recent developments in the field.

Efforts are still needed to identify the most relevant endpoints in the optimisation of existing tests. However, a combination of several *in vitro* tests, covering the relevant mechanistic steps of respiratory sensitisation, into a test battery could eventually lead to the identification of respiratory sensitisers. There are efforts ongoing to develop an AOP for respiratory sensitisation.

Animal data

At present, although a number of *in vivo* test protocols have been published to detect respiratory allergens of low molecular weight, none of these are validated nor are these widely accepted. Some of the models are briefly discussed below, however the list is not comprehensive.

- One model is based on the LLNA, where mice are exposed *via* the inhalation route for 3 consecutive days, after which lymphocyte proliferation is measured in the draining (mandibular) lymph nodes. Known respiratory sensitisers, such as andhydrides and diisocyanates have been assessed in this model and were also shown to be positive in this assay. However many of the substances tested showed local toxicity in the lungs, therefore, due to local toxicity, lower doses can be applied in general when compared to skin exposure. Based on the readout, i.e. stimulation indices, the results could inform about the respiratory sensitising potency (Arts *et al.*, 2008).
- In other protocols, an LLNA-based method is used to assess and measure cytokine profiles relevant to respiratory sensitisation.
- In one protocol mice are exposed *via* the dermal route, after which lymphocytes are collected from the draining (auricular) lymph nodes and further cultured and prepared for cytokine determination (Dearman *et al.*, 2002).

- Another method to assess the cytokine profile uses inhalation exposure in mice, where the animals are exposed for 3 consecutive days *via* the inhalation route, after which lymphocytes are collected and following *ex vivo* proliferation are prepared for the cytokine profile determination (De Jong *et al.*, 2009; Johnson *et al.,* 2011). In addition to the assessment of cytokine profiles, the assessment of gene expression profiles in a similar mouse model could be useful in distinguishing respiratory sensitisers from respiratory irritants (Adenuga *et al.*, 2012).
- One model is based on Brown Norway rats, in which elicitation of respiratory sensitisation is assessed. This method has been used to assess known respiratory sensitisers such as diisocyanates. In this model rats are sensitised either *via* inhalation or dermal exposure, after which the animals are challenged *via* the inhalation route. The endpoints measured in this model are respiratory irritation, assessment of bronchoalveolar lavage, measurement of nitric oxide exhaled and delayed onset of respiratory response (Pauluhn, 2014).
- Another, relatively simple, approach may serve the purpose to specifically predict sensitisation of the respiratory tract: i.e. increases in total serum IgE antibodies after induction. This method is based on statistically significant increases in total serum IgE (Arts and Kuper, 2007; Kimber *et al.*, 2011; Vandebriel *et al.*, 2011).

There are currently no predictive methods to identify substances that induce asthma through non-immunological mechanisms. However, when performing challenge tests including non-sensitised but challenged controls, information can be obtained on non-immunological effects of these substances.

R.7.3.9.2 Human data on respiratory sensitisation

Human data on respiratory reactions (asthma, rhinitis, and extrinsic allergic alveolitis) may come from a variety of sources:

- consumer experience and comments, preferably followed up by professionals (e.g. bronchial provocation tests, skin prick tests and measurements of specific IgE serum levels);
- records of workers' experience, accidents, and exposure studies including medical surveillance;
- case reports in the general scientific and medical literature;
- consumer tests (monitoring by questionnaire and/or medical surveillance);
- epidemiological studies.

R.7.3.10 Evaluation of available information on respiratory sensitisation

R.7.3.10.1 Non-human data for respiratory sensitisation

Non-testing data for respiratory sensitisation

The freely downloadable OECD QSAR Toolbox software (http://www.qsartoolbox.org/) encodes a profiler (set of rules and structural domains) specific for respiratory sensitisation. The profiler offers support to the user in grouping substances which share common structural alerts and possibly predict the respiratory sensitisation potential *via* read-across. The current version of the profiler encodes 41 structural alerts for respiratory sensitisation.

This profiler is intended to be used for the assessment of the respiratory sensitisation potential of low molecular weight substances. The profiler has been developed based on the mechanistic knowledge of the elicitation phase of respiratory sensitisation, and thus identifies substances able to covalently bind to proteins in the lung. Presence of activity could be predicted from positive predictions. Absence of effect however cannot be predicted from the lack of alert because the lack of alert might be due to the lack of effect or lack of knowledge.

This profiler should also be used with caution due to the limited data available for the development of structural alerts. This is due to the lack of a standardised assay (*in vivo* or *in vitro*) suitable for identifying potential respiratory sensitisers. The available data are drawn from clinical reports of occupational asthma, which in a number of cases results in structural alerts defined based on a low number of substances. However, all structural alerts have a clear mechanistic rationale associated with them (in terms of covalent protein binding).

Experimental data on respiratory sensitisation can be found in two of the OECD QSAR Toolbox databases: "Skin sensitisation ECETOC" and ECHA Chem. The "Skin sensitisation ECETOC" database as named in the Toolbox contains data for both skin and respiratory sensitisers.

Testing data for respiratory sensitisation

In vitro data

Presently (at the time of publication of this Guidance) there are neither scientifically valid nor adopted *in vitro* tests available to assess respiratory sensitisation. Several *in vitro* test methods have been described in the literature; however more work is needed for wider acceptance of a given test method.

Some *in vitro* test methods are described in Section R.7.3.9.1. The list of *in vitro* methods is not complete and others exist. However, none of the current test methods have gone through a validation process and therefore expert judgement and care are needed when assessing information obtained from such methods and its relevance.

Animal data

Although generation of new information for respiratory sensitisation is not a standard information requirement under the REACH Regulation, existing information should be assessed. In case animal data are available on respiratory sensitisation those data should be assessed and included in the IUCLID dossier.

Information based on the LLNA model(s) as described in Section R.7.3.9.1 may provide valuable information on the possible respiratory sensitisation potential of the substance. Use of cytokine assessment in the studies could provide useful information by identifying Th2-type of cytokines. Also assessment of cytokines and gene expressions could be useful when trying to differentiate between respiratory sensitisation *vs.* respiratory irritation (Adenuga *et al.*, 2012).

Information based on the Brown Norway rat model that assesses the elicitation of respiratory sensitisation (see Section R.7.3.9.1) may provide relevant information for respiratory sensitisation, especially in case changes supporting sensitisation are noted in the endpoint measured. In addition, some indication for the NOAEL on the elicitation threshold in this test system may be obtained (Pauluhn and Poole, 2011; Pauluhn, 2014).

Moreover, measurement of serum IgE-levels in rodent models, even though variability has been observed in the animal models, can support the identification of respiratory sensitisers (Arts and Kuper, 2007; Kimber *et al.*, 2011, Vandebriel *et al.*, 2011; Kimber *et al.* 2014b).

R.7.3.10.2 Human data on respiratory sensitisation

Although predictive models are under validation, there is as yet no internationally recognised animal method for identification of respiratory sensitisation. Thus human data are usually evidence for hazard identification. In case existing human data are available on respiratory sensitisation, those data should be assessed and included in the IUCLID dossier.

Although human studies may provide some information on respiratory hypersensitivity, the data are frequently limited and subject to the same constraints as human skin sensitisation data.

For evaluation purposes, existing human experience data for respiratory sensitisation should contain sufficient information about:

- the test protocol used (study design, controls);
- the substance or preparation studied (should be the main, and ideally, the only substance or preparation present which may possess the hazard under investigation);
- the extent of exposure (magnitude, frequency and duration);
- the frequency of effects (versus number of persons exposed);
- the persistence or absence of health effects (objective description and evaluation);
- the presence of confounding factors (e.g. pre-existing respiratory health effects, medication; presence of other respiratory sensitisers);
- the relevance with respect to the group size, statistics, documentation;
- the healthy worker effect.

Evidence of respiratory sensitising activity derived from diagnostic testing may reflect the induction of respiratory sensitisation to that substance or cross-reaction with a chemically very similar substance. In both situations, the normal conclusion would be that this provides positive evidence for the respiratory sensitising activity of the substance used in the diagnostic test.

For respiratory sensitisation, no clinical test protocols for experimental studies exist but evidence can come from clinical history and data from appropriate lung function tests related to exposure to the substance, or data from one or more positive bronchial challenge tests with the substance (See section 3.4.2.1.2.3 of the CLP Regulation). The test should meet the above general criteria, e.g. be conducted according to a relevant design including appropriate controls, address confounding factors such as medication, smoking or exposure to other substances, etc. Furthermore, the differentiation between the symptoms of respiratory irritation and allergy can be very difficult. Thus, expert judgement is required to determine the usefulness of such data for the evaluation on a case-by-case basis.

Where there is evidence that significant occupational inhalation exposure to a substance has not resulted in the development of respiratory allergy, or related symptoms, then it may be possible to draw the conclusion that the substance lacks the potential for sensitisation of the respiratory tract. Thus, for instance, where there is reliable (e.g. supported by medical surveillance reports) evidence that a large cohort of subjects has had opportunity for regular significant inhalation exposure to a substance for a sustained period of time in the absence of respiratory symptoms, or related health complaints, then this will provide reassurance within a *Weight-of–Evidence* approach regarding the absence of a respiratory sensitisation hazard.

More information on how to apply human data for C&L purposes can be found in Section 3.4.2.1.3.1 of the *Guidance on the Application of the CLP criteria*.

R.7.3.11 Conclusions on respiratory sensitisation

R.7.3.11.1 Remaining uncertainty on respiratory sensitisation

Major uncertainties remain in our understanding of the factors that determine whether or not a substance is an allergen, and if so, what makes it a respiratory sensitiser. Evidence that a substance can lead to respiratory hypersensitivity will normally be based on human experience. Hypersensitivity is normally seen as asthma, but other hypersensitivity reactions such as rhinitis or extrinsic allergic alveolitis are also considered. The condition will have the clinical character of an allergic reaction. However, immunological mechanisms do not have to be demonstrated according to the CLP criteria. In case there is evidence available that the substance induces asthma-like symptoms by irritation only, these substances should not be considered as respiratory sensitisers.

Based on current knowledge, all low molecular weight respiratory sensitisers are also skin sensitisers (Kimber *et al.*, 2007), however this is not true in reverse. There are high molecular weight substances that do cause respiratory sensitisation, e.g. enzymes, but that, due to their size, are not able to penetrate the skin and are not skin sensitisers.

R.7.3.11.2 Concluding on suitability for Classification and Labelling

The CLP Regulation specifies that respiratory sensitisation should be allocated into sub-categories (i.e. 1A or 1B) whenever possible. In case the data are not sufficient for sub-categorisation, the substance must be classified in the general Category 1 (for further information, see Section 3.4 of the *Guidance on the Application of the CLP criteria*).

R.7.3.11.3 Concluding on suitability for chemical safety assessment: dose-response assessment and potency

There is evidence that for both skin sensitisation and respiratory hypersensitivity dose-response relationships exist although these are frequently less well defined in the case of respiratory hypersensitivity. The dose of agent required to induce sensitisation in a previously naïve subject or animal is usually greater than that required to elicit a reaction in a previously sensitised subject. Therefore the dose-response relationship for the two phases will differ. Little or nothing is known about dose-response relationships in the development of respiratory hypersensitivity by non-immunological mechanisms.

It is frequently difficult to obtain dose-response information from either existing human or animal data where only a single concentration of the test material has been examined. With human data, exposure measurements may not have been taken at the same time as the disease was evaluated, adding to the difficulty of determining a dose response.

Estimation of potency

The estimation of potency for respiratory sensitisation is currently (at the time of publication of this Guidance) solely based on human data (See Section 3.4.2.1 of the *Guidance on the Application of the CLP criteria)*.

Derivation of a DNEL

Even though respiratory sensitisation might be considered to be a threshold effect (induction and elicitation), currently available methods do not allow the determination of a threshold and establishment of a DNEL. Guidance on how to perform a qualitative safety assessment for respiratory sensitisers can be found in Section E.3.4.2 of Part E and Appendix R.8-10 of *Chapter R.8* of the *Guidance on IR&CSA*.

R.7.3.11.4 Additional considerations

Chemical allergy is commonly designated as being associated with sensitisation of the respiratory tract (asthma, rhinitis and extrinsic allergic alveolitis). According to current knowledge respiratory sensitisation can be induced via both dermal and inhalation routes. Therefore it is important for risk management purposes that exposures *via* both routes are prevented.

As the evidence for a substance leading to respiratory hypersensitivity is normally based on human data it may be difficult to distinguish respiratory sensitisation from respiratory irritation as the clinical symptoms for both are similar.

R.7.3.12 Assessment strategy for respiratory sensitisation

R.7.3.12.1 Objective / General principles

The objective of this assessment strategy is to give guidance on a stepwise approach to hazard identification with regard to the respiratory sensitisation endpoint. A key principle of the strategy is that the results of one study are evaluated before another is initiated. The strategy should seek to ensure that the data requirements are met in the most efficient and humane manner so that animal usage and costs are minimised.

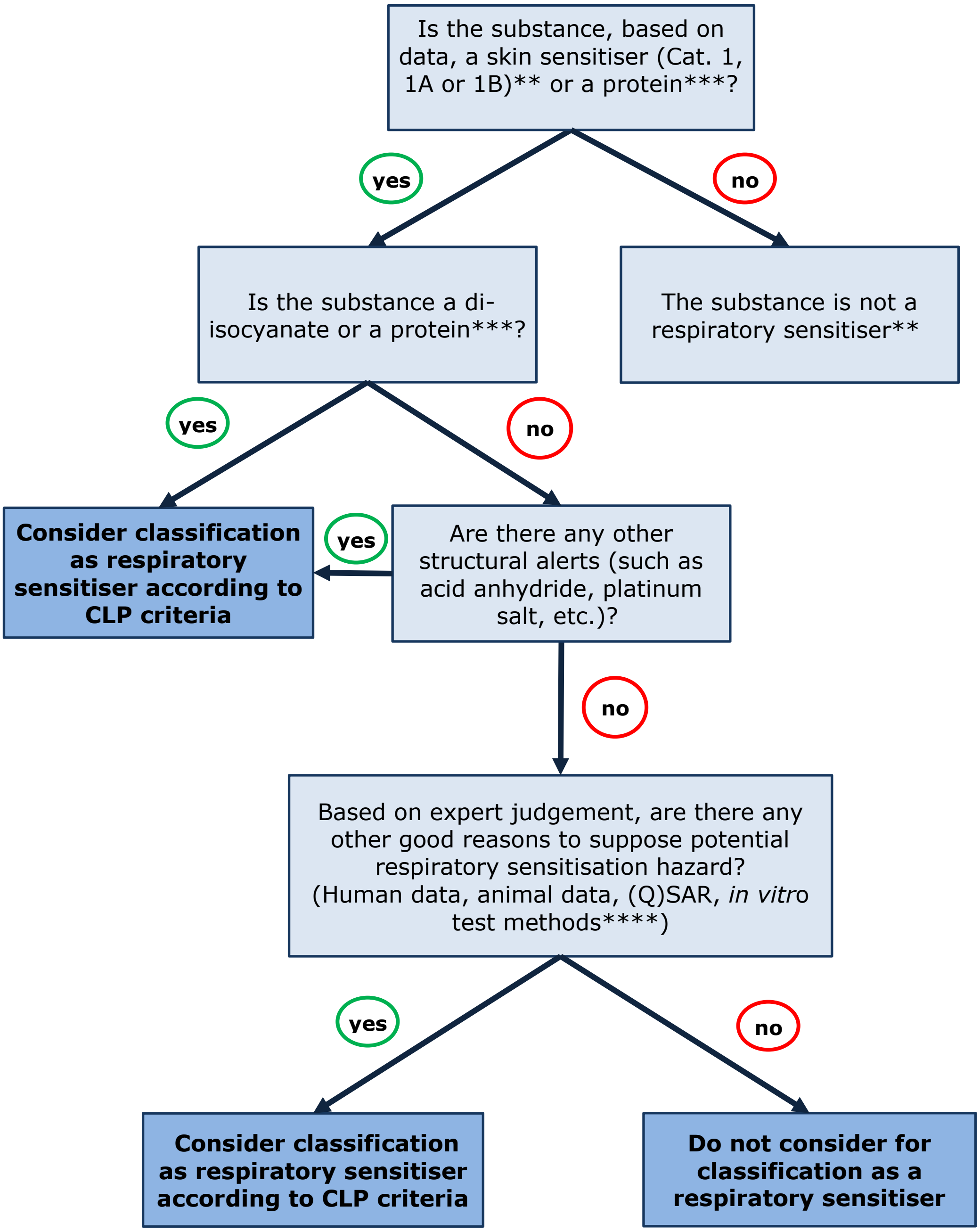
R.7.3.12.2 Preliminary considerations

Careful consideration of existing toxicological data, exposure characteristics and current risk management procedures is recommended to ascertain whether the fundamental objectives of the assessment strategy (see above) have already been met. Other factors that might mitigate data requirements for the endpoint of interest, e.g. possession of other toxic properties, characteristics that make testing technically not possible, should also be considered.

R.7.3.12.3 Recommended approach

The below strategy for respiratory sensitisation assessment (Figure R.7.3–4) can be followed:

Figure R.7.3–4 Assessment strategy for respiratory sensitisation data*



* In contrast to tests for skin sensitisation, the performance of tests for respiratory sensitisation is currently not required under REACH. Therefore the present strategy scheme depicts a strategy for evaluating existing data.
** Based on current knowledge there are no low molecular weight respiratory sensitisers that would not cause skin sensitisation (Kimber *et al*., 2007).
*** There is an indication that enzymes have the potential to cause respiratory sensitisation e.g. the Scientific Committee for Animal Nutrition (SCAN) states the following "Enzyme and microbial additives will be regarded as respiratory sensitisers unless convincing evidence to the contrary is provided" (http://ec.europa.eu/environment/archives/dansub/pdfs/enzymerepcomplete.pdf). Therefore, it is advised to consider respiratory sensitisation potential in the case of an enzyme, even though the CLP Regulation does not require to classify all enzymes as respiratory sensitisers.
**** No scientifically validated/independently reviewed or adopted test methods are yet available.

R.7.3.13 References on skin and respiratory sensitisation

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Appendices R.7.3-1 to 4 to Section R.7.3

Appendix R.7.3–1 Literature models and *in silico* tools for skin sensitisation

Content of Appendix R.7.3-1:

- Data and (Q)SAR models in scientific publications
- Data and models included in *in-silico* tools

DATA AND (Q)SAR MODELS IN SCIENTIFIC PUBLICATIONS

Data

Peer reviewed publications are a valuable source for skin sensitisation data. Cronin and Basketter (1994) published the results of over 270 *in vivo* skin sensitisation tests (mainly from the guinea pig maximisation test). All data were obtained in the same laboratory and represent one of the few occasions when large amounts of information from corporate databases were released into the open literature. A larger database of animal and human studies for 1034 substances is described by Graham *et al.* (1996), the MCASE database. Schlede *et al.* (2003) reports data on 244 substances. These data have been assessed and expert judgement on the potency ranking of 244 substances with contact allergenic properties based on data on humans and results of animal tests is provided in the paper.

A comparatively large number of data have been published for the local lymph node assay; examples include publications by Ashby *et al.* (1995), Gerberick *et al.* (2005) and Kern *et al*. (2010).

SARs

Some collections of structural alerts published in the literature have not (yet) been encoded into softwares (e.g. Gerner *et al.* (2004). However, experts in skin sensitisation can still manually use these structural alerts to make considerations on the substance of interest.

Local (Q)SAR models

Local models are (Q)SAR models developed for specific chemical classes or mode of actions. The majority of local models for skin sensitisation have been developed for direct-acting electrophiles using the relative alkylation index (RAI) approach. This is a mathematical model derived by Roberts and Williams (1982). It is based on the concept that the degree of sensitisation produced at induction, and the magnitude of the sensitisation response at challenge, depends on the degree of covalent binding (haptenation, alkylation) to carrier protein occurring at induction and challenge. The RAI is an index of the relative degree of carrier protein haptenation and was derived from differential equations modelling competition between the carrier haptenation reaction in a hydrophobic environment and removal of the sensitiser through partitioning into polar lymphatic fluids. In its most general form the RAI is expressed as:

$$\mathrm{RAI} = \log D + a \log k + b \log P \qquad (1)$$

Thus the degree of haptenation increases with increasing dose D of sensitiser, with increasing reactivity (as quantified by the rate constant or relative rate constant *k* for the reaction of the sensitiser with a model nucleophile) and with increasing hydrophobicity (as quantified by log P, P being the octanol/water partition coefficient). This RAI model has been used to evaluate a wide range of different datasets of skin sensitising substances. Examples include sulfonate esters (Roberts and Basketter, 2000), sulfones (Roberts and Williams, 1982), primary alkyl bromides (Basketter *et al.*, 1992), acrylates (Roberts, 1987), aldehydes and diketones (Patlewicz *et al*., 2001; Patlewicz *et al*., 2002; Patlewicz *et al*., 2004; Roberts *et al*., 1999; Roberts and Patlewicz, 2002; Patlewicz *et al*., 2003).

This approach has shown that local models tend to be transparent, simple and mechanistically derived but are labour-intensive to develop and restricted to local areas of chemistry (Cronin *et al.*, 2011).

The covalent hypothesis has served well and continues to be the most promising way of developing mechanistically based robust QSARs. These are local in that their scope is characterised by a mechanistic reactivity domain as outlined in Aptula *et al*. (2005), Aptula and Roberts (2006), and Roberts *et al*. (2007a). An example of this type of mechanistic model has been recently published (Roberts *et al.*, 2006). In the RAI model, log *k*, has been typically modelled by experimental rate constants, substituents' constants or molecular orbital parameters. More effort is needed to encode reactivity into descriptors, this could be achieved through the systematic generation of *in vitro* reactivity data as outlined by Aptula and Roberts (2006), Aptula *et al*. (2006), Schultz *et al*. (2006), Gerberick *et al*. (2004) and in the next section.

Global statistical models

Global Statistical models usually involve the development of empirical QSARs by application of statistical methods to sets of biological data and structural descriptors.

These are perceived to have the advantage of being able to make predictions for a wider range of substances. In some cases, the scope/domain of these models are well described, in most other cases a degree of judgement is required in determining whether the training set of the model is relevant for the substance of interest. Criticism often levied at these types of models is that they lack mechanistic interpretability. The descriptors might appear to lack physical meaning or are difficult to interpret from a chemistry perspective. The sorts of descriptors used may encode chemical reactivity/electrophilicity, e.g. LUMO (the energy of the lowest molecular orbital), and partitioning effects, e.g. Log P, but a more common case is that a large number of descriptors are calculated that encode structural, topological and/or geometrical information. A number of models have been reported in the recent literature; examples include those developed using LLNA data (Devillers, 2000; Estrada *et al*., 2003; Fedorowicz *et al*., 2004; Fedorowicz *et al*., 2005; Li *et al*., 2005; Miller *et al*., 2005; Ren *et al*., 2006; Li *et al*., 2007; Golla *et al*., 2009; Chaudhry *et al*., 2010).

DATA AND MODELS INCLUDED IN *IN-SILICO* TOOLS

Data

There is a number of computational tools and databases available to facilitate the search and retrieval of skin sensitisation data for the target substance or its analogues. Examples of such databases and tools are the OECD QSAR Toolbox (http://www.qsartoolbox.org/), Chemfinder (www.chemfinder.com), ChemID (http://chem.sis.nlm.nih.gov/chemidplus/), NICEATM LNA Database (http://ntp.niehs.nih.gov/pubhealth/evalatm/test-method-evaluations/immunotoxicity/nonanimal/index.html) and DssTox (http://www.epa.gov/nheerl/dsstox/) that are freely available to use on the internet, and Leadscope (http://www.leadscope.com) that is commercial.

Some of the available search engines are linked to databases (through hyperlinks and indexes) whereas others like DssTox provide a repository of available QSAR datasets which can be downloaded for subsequent use in appropriate QSAR/database software tools.

In-silico tools

OECD QSAR Toolbox

The OECD QSAR Toolbox software (http://www.qsartoolbox.org/, current version 3.3) encodes several mechanistic and skin sensitisation endpoint specific databases and profilers. They allow the user to group substances that share common structural alerts and to predict their skin sensitisation potential *via* read-across. ECHA has published illustrative examples on how to make skin sensitisation read-across predictions using the OECD QSAR Toolbox (https://echa.europa.eu/documents/10162/21655633/illustrative_example_qsar_part2_en.pdf).

The two dedicated databases for skin sensitisation are "Skin sensitisation", which includes 1 036 substances and 1 573 experimental data points (includes the OASIS skin sensitisation database and the Liverpool John Moores University skin sensitisation database) and "Skin sensitisation ECETOC", with 39 substances and 42 experimental data points. The ECHA Chem database, which collects the information found in REACH dossiers, also contains data on skin sensitisation.

There are four relevant profilers for skin sensitisation. They are all based on protein binding. Three of these profilers can be found under the general mechanistic profiler branch: Protein binding by OASIS v1.3, Protein binding by OECD, Protein binding potency. The fourth profiler is under the endpoint-specific branch: Protein binding alerts for skin sensitisation by OASIS v1.3.

Users can use profilers for the identification of analogues based on mechanistic commonalities and retrieve experimental information from the dedicated databases. Several data gap filling techniques can be used to predict skin sensitisation for the substance of interest: read-across, trend analysis and QSAR models.

The OECD QSAR Toolbox also encodes an Adverse Outcome Pathway (AOP) for skin sensitisation. This is the first attempt in the QSAR Toolbox to allow predictions through AOPs, and at this stage it is premature to advise the use of the AOP functionality within the OECD QSAR Toolbox for predicting skin sensitisation.

Expert systems

Software like VEGA or Toxtree are free to download and use. There are also several commercial (Q)SAR models for skin sensitisation available. Examples include TOPKAT, CASE, Derek Nexus (DN), TIMES (TIssue MEtabolism Simulator), Molcode, HazardExpert, and probably others.

- Statistical Models:

Toxtree allows the user to estimate toxic hazard by applying a decision tree approach. It includes the "Skin sensitisation reactivity domains" plug-in for the identification of mechanisms of toxic action using a SMARTS pattern based approach. It is important to note that the alerts are meant to provide grouping into reactivity mode of action and do not predict skin sensitisation potential.

TOPKAT (included in Discovery Studio package) marketed by BIOVIA Foundation (formerly Accelrys Enterprise Platform 'AEP') is a suite of two models: one for Non-sensitisers *vs.* Sensitisers and the other for Weak/Moderate *vs.* Strong sensitisers. The first model calculates the probability of a chemical structure being a sensitiser. If the probability is greater than or equal to 0.7, the substance is predicted to be a sensitiser, a non-sensitiser would have a probability of less than or equal to 0.30. The second model applies to structures predicted as sensitisers by the first model and resolves the potency: weak/moderate *vs.* strong, where a probability of 0.7 or more indicates a strong sensitiser and a probability below 0.30 indicates a weak or moderate sensitiser. Probability values between 0.30 and 0.70 are referred to as indeterminate. An optimum prediction space algorithm ensures that predictions are only made for substances within the applicability domain of the model. Please note that the models are all based on the guinea-pig maximization test (Enslein *et al*., 1997; http://accelrys.com/solutions/scientific-need/predictive-toxicology.html).

CASE methodology and all its variants were developed by Klopman and Rosenkranz. There is a multitude of models for a variety of endpoints and hardware platforms. The CASE approach uses a probability assessment to determine whether a structural fragment is associated with toxicity (Cronin *et al.*, 2003). The MCASE models (currently CASE Ultra) that have been developed for skin sensitisation are described further in primary articles (Gealy *et al.*, 1996; Graham *et al.*, 1996; Johnson *et al.*, 1997). There are two sensitisation modules available for purchase from MultiCase Inc (Ohio, USA) (http://www.multicase.com/case-ultra-models). In addition the (Q)SAR estimates for one MCASE skin sensitisation model are included in the Danish Environmental Protection Agency (EPA) (Q)SAR database (http://qsar.food.dtu.dk/).

VEGA platform, freely available for download (http://www.vega-qsar.eu/), incorporates a model (Chaudhry *et al*., 2010) developed using an Adaptive Fuzzy Partition (AFP) algorithm based on eight descriptors. The AFP assigns the substances to two classes, sensitisers and non-sensitisers. An in-depth assessment of the applicability domain of the prediction, mainly based on similarity with substances in the training set of the model, is also provided.

- Knowledge-based systems:

Derek Nexus (DN) is a knowledge-based expert system created with knowledge of structure-toxicity relationships and an emphasis on the need to understand mechanisms of action and metabolism. It is marketed and developed by LHASA Ltd (Leeds, UK) a not-for-profit company and educational charity (https://www.lhasalimited.org/).

Within DN (version 9), there are 361 alerts covering a wide range of toxicological endpoints. An alert consists of a toxicophore, a substructure known or thought to be responsible for the toxicity alongside associated literature references, comments and examples. The skin sensitisation knowledge base in DN was initially developed in collaboration with Unilever in 1993 using its historical database of guinea pig maximisation test (GPMT) data for 294 substances and contained approximately forty alerts (Barratt *et al.*, 1994). Since that time, the knowledge base has undergone extensive improvements as more data have become available (Payne and Walsh, 1994). The current version contains about seventy alerts for skin sensitisation and some alerts for photoallergenicity (Barratt *et al*., 2000; Langton *et al*., 2006). The predictivity of DN for skin sensitisation was recently assessed by Guesne *et al*. (2014). As a reminder, alert-based systems should not be assessed for their specificity and overall accuracy, contrary to discriminant models.

- Hybrids:

The **TIssue MEtabolism Simulator (TIMES)** software has been developed to integrate a Skin metabolism Simulator (SS) with 3D-QSARs for evaluating reactivity of substances in order to predict their skin sensitisation potency (Dimitrov *et al.,* 2005). The current version of the simulator (version 2.27.16) contains more than 200 hierarchically ordered spontaneous and enzyme-controlled reactions. Covalent interactions of substances/metabolites with skin proteins are described by 47 alerting groups. 3D-QSARs (COREPA) are applied for some of these alerting groups. Characterisation and evaluation of TIMES-SS can be found in Patlewicz *et al*. (2007) and Roberts *et al*. (2007b), respectively. New research with TIMES includes the work of Patlewicz *et al*. (2014).

The **Danish QSAR database** contains a collection of pre-calculated predictions for a range of hazard endpoints including allergic contact dermatitis (ACD) for over 600,000 discrete organic substances including more than 70,000 REACH pre-registered substances. The predictions were made in models developed or licensed by the Danish Technical University. The commercial CASE Ultra model for ACD is licensed from MultiCASE with special permission to remodel in Leadscope and SciQSAR. Included in the training set are 1,031 compounds with information from human epidemiological studies on ACD (Allergic Contact Dermatitis) and results from the Guinea Pig Maximization Test. The binary predictions of the models (positive/negative) are given together with information on whether the substance is within the defined model applicability domain. In addition, a fourth prediction based on "a majority vote algorithm" between the three other approaches is provided. The online database interface includes search functionalities on e.g. CAS RN, EC No, name, structure/sub-structure/chemical structure similarity, all the included prediction endpoints, as well as any complex AND/OR/NOT combinations of previous searches. The database including QMRFs for the three individual models is freely available at: http://qsar.food.dtu.dk/.

For expert systems such as DN, TOPKAT etc., the training sets and, to some extent, the algorithms or descriptors used are often kept latent within the software. Some supporting information is provided on the robustness and relevance for a given prediction. For example, within DN it is possible to see representative example substances and explanations of the mechanistic basis for the SAR developed.

TOPKAT supports users in assessing the reliability of the prediction by: 1) evaluating if the substance falls into the applicability domain of the model (based on structural fragments and descriptors), 2) checking if the substance is present in its database, and 3) identifying analogues of the target substance based on chemical similarity. Similar functionalities and features are present in many of the other commercial expert systems available.

Although the main factors driving skin sensitisation (and therefore the (Q)SARs) is the underlying premise of the electrophilicity of a substance, other factors such as hydrophobicity encoded in the octanol/water partition coefficient (log P) may also be considered as playing a role in the modification of the sensitisation response observed. Within DN, an assessment of the likely skin penetration ability is made using the algorithm by Potts and Guy. This relates the Kp value to log P and MW (Potts and Guy, 1992). It is then possible to rationalise the output in terms of bands of penetration potential. Some methods for assessing percutaneous absorption have been described in Howes *et al.* (1996).

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Appendix R.7.3–2 Template for the reporting of the individual information sources for a non-animal test method when using non-validated and/or non-adopted test methods

The following reporting format (Table R.7.3–4) should be considered especially when information is generated by non-animal test methods to fulfil the REACH information requirement for skin sensitisation. The use of this reporting template is very important in case (a) test method(s) is (are) used which has (have) not been considered scientifically valid in an international validation study and/or there is no internationally adopted test guideline available.

In case a test method has an internationally adopted test guideline available this template can also be used, however some of the points described below can already be included in the test guideline itself, hence detailed reporting of such (an) information source(s) is usually not needed. The reporting of each individual information source needs to be included in a separate endpoint study record (ESR) of the IUCLID dossier, i.e. one ESR per individual information source should be filled in.

Note: this reporting template is based on the OECD template for the reporting of individual information sources (OECD, 2016) and has been modified to be specific to the skin sensitisation endpoint and REACH information requirements.

Table R.7.3–4 Template for the reporting of the individual information sources describing a non-animal test method when using non-validated and/or non-adopted test methods to fulfil the REACH information requirement for skin sensitisation

Name of the information source	Provide the name of the information source and the acronym (if applicable)
Mechanistic basis including AOP coverage	Describe which key event of skin sensitisation AOP is addressed by the information source. A desription of the extent to which the mechanistic basis of the information source relates to the chemical/biological mechanism covered by the (key) event should be provided.
Description	Provide a short description of the information source including the experimental system used and any relevant aspect of the procedure (e.g. time of exposure of the experimental system with the test substance, number of doses/concentrations tested, number of replicates, concurrent testing of control(s) and vehicle(s), laboratory instruments/techniques used to quantify the response).
Response(s) measured	Specify the response(s) measured by the information source and its measure (e.g. *in chemico* binding to synthetic peptides, expressed as % of peptide depletion).
Prediction model	Indicate whether there is a prediction model associated to the information source and its purpose. Briefly describe the prediction model and provide a reference to a paper or document where the prediction model is described (if available).
Metabolic competence (if applicable)	Specify whether the information source encompasses any metabolically competent system/step and, to the extent possible, how this relates to the situation *in vivo*.
Status of development, standardisation, validation	Indicate whether the information source is: a) an officially adopted (standard) test method (e.g. a test method covered by an OECD Test Guideline); b) a validated but non-standard test method; c) a test method undergoing formal evaluation (e.g. prevalidation, validation, others); d) a non-validated test method widely in use; e) a non-validated test method implemented by a small number of users.
Technical limitations and limitations with regard to applicability domain	Indicate the substance(s) and/or chemical categories (e.g. based on physico-chemical properties or functional groups) for which the information source has been shown **not** to be applicable because of technical limitations, e.g. highly volatile substances, poorly water soluble substances, solid materials, interference of the substance with the detection system (e.g. coloured or autofluorescent substances interfering with spectrophotometric analysis). Indicate whether the information source is technically applicable to the testing of multi constituent-substances, UVCBs and mixtures. In addition indicate the substance(s) and/or chemical categories for which the information source has been experimentally shown to yield incorrect and/or unreliable predictions with respect to the reference classifications (e.g false negative predictions with substances requiring metabolic activation, high false positive rate for alcohols).
Strengths and Weaknesses	Provide an indication of the strengths and weaknesses of the information source, compared to existing similar non-testing or testing methods, considering among others the following aspects: a) extent of mechanistic information provided and relevance (i.e. measurement of various responses in the same experimental model, limited or good coverage of the mechanisms at the basis of the effect being investigated, predictive of responses in humans);

	b) level of information provided (single-point estimate or dose-response information); c) level of performance (e.g. higher or lower reproducibility, predictive capacity); d) extent of domain of applicability; e) number of substances with published information.
Reliability (within and between laboratories) (if applicable)	Describe the level of reliability of the information source (i.e. the degree of agreement among results obtained from testing the same substances over time using the same protocol in one or multiple laboratories) and to what extent this has been characterised including the number of substances used for the assessment.
Predictive capacity (if applicable)	Describe the extent to which the information source predicts the key event of interest (as reported in scientific publications and as determined in validation studies). Express the predictive capacity in terms of sensitivity, specificity and accuracy if applicable or by other goodness-of-fit statistics (e.g. linear correlation analysis). Include the number of substances used in this assessment and their predictions using the reference method.
Proposed regulatory use	Indicate the proposed regulatory use of the information source (e.g. stand-alone full replacement method, partial replacement method, screening method, others).
Potential role within a Testing and Assessment Strategy	Indicate the potential weight the information source is expected to carry within a structured approach to data integration (if applicable) and/or within a Testing and Assessment Strategy, and for which specific purpose the information source can potentially be used on its own.

Reference

OECD (2016) Guidance Document On The Reporting Of Defined Approaches To Be Used Within Integrated Approaches to Testing And Assessment

Available at: http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono(2016)28&doclanguage=en

Appendix R.7.3–3 Reporting format for defined approaches to testing and assessment based on multiple information sources

This template aims to provide advice on the reporting of defined approaches to testing and assessment to be used within IATA and for the integration of the individual information sources used to fulfil the REACH information requirement for skin sensitisation. The reporting of the defined approaches to testing and assessment and the conclusions obtained from them should be included in the dossier, i.e. as an attachment to the endpoint summary record of skin sensitisation of the IUCLID dossier.

Note: the reporting template is based on the OECD reporting format for reporting defined approaches as described in Annex I of the OECD Guidance Document On The Reporting Of Defined Approaches To Be Used Within Integrated Approaches to Testing And Assessment (OECD, 2016a), however the template has been adapted to REACH-specific purposes.

1 Summary

Summarise the information in the reporting format in order to provide a concise overview of the proposed defined approach.

2 General information

2.1 Identifier: *Provide a short and informative title for the structured approach.*

2.2 Reference to main scientific papers: *List the main bibliographic references (if any).*

3 Endpoint addressed

Specify the endpoint (here skin sensitisation). Also specify related properties that have been measured or predicted by the proposed approach and indicate whether these address (or partially address) an endpoint, or key event being predicted by an existing test guideline.

4 Definition of the purpose

Default: meeting the REACH information requirement for skin sensitisation (Annex VII, 8.3) and the relevant classification and/or risk assessment obligations.

5 Rationale underlying the construction of the defined approach

Describe the rationale used to construct the defined approach. This should include an assessment of the linkage of the individual information sources used within the approach to the known substance and the key events being predicted. The reason for the choice of (a) specific information source(s)/test(s) addressing (a) specific key event(s) possibly in the light of other existing similar information sources should be provided. In case a non-guideline information source for a key event is used, for which an existing test guideline is available (e.g. EU or OECD), this should be justified.

6 Description of the individual information sources used within the approach (see OECD Guidance Documents (OECD, 2016a to d) and Appendix R.7.3–2 **of this Guidance)**

List the information sources employed within the proposed defined approach (e.g.

physico-chemical properties, non-testing (in silico) methods and testing (in chemico, in vitro, in vivo) methods, including the response(s) measured and the respective measure(s) (e.g. in chemico binding to synthetic peptides, expressed as % peptide depletion). Detailed descriptions of each in chemico, in vitro, and in vivo method should be provided using the endpoint study records (ESRs) in IUCLID (i.e. one ESR per individual information source).

In addition, when QSAR models are used the QSAR Model Reporting Format (QMRF) should be provided and individual predictions, if applicable, should be reported using the QSAR Prediction Reporting Format (QPRF) and included in the ESR of the IUCLID. Both reporting formats are accessible at: https://eurl-ecvam.jrc.ec.europa.eu/laboratories-research/predictive_toxicology/qsar_tools/QRF.

7 Data interpretation procedure applied

Describe the data interpretation procedure (DIP) used. Indicate whether the DIP output is qualitative or quantitative. If possible, provide a workflow to illustrate the manner in which the DIP should be applied.

8 Substances used to develop and test the DIP

8.1 Availability of training and test sets: *Indicate whether a training set (i.e. chemical data used in the development of the DIP) and test set (i.e. chemical data used to evaluate the DIP) are available (e.g. published in a paper, stored in a database) or appended to this Reporting format. If they are not available, explain why. Example: "It is available and attached"; "It is available and referenced"; "It is not available because the data set is proprietary"; "The data set could not be retrieved".*

8.2 Selection of the training set and test set used to assess the DIP: *If the training set and test set are available please describe the rationale for their selection (e.g. availability of high quality in vivo data for the endpoint being predicted, coverage of the range of effects observed in vivo, coverage of diverse physico-chemical properties, coverage of structural diversity, others).*

8.3 Supporting information on the training and test sets: *If the training and/or the test sets are available, append them as supporting information, preferably in the form of an Excel table. The following information on both sets should be reported where available and to the extent possible: a) chemical name (common and/or IUPAC); b) CAS and/or EC numbers; c) in case of multi-constituent or UVCBs, report the composition to the extent possible; d) reference data or classifications(s) for each substance (e.g. in vivo data); e) data from the individual information sources used in the defined approach; f) final result/prediction for each substance.*

8.4 Other information on training and test sets: *If the training and/or the test sets are not available for inclusion as supporting information, indicate any other relevant information about the training and/or test sets (e.g. number and type of substances). This will be useful to gain an appreciation of e.g. the chemical coverage.*

9 Limitations in the application of the defined approach

Indicate the type(s) of substances, in terms of their physico-chemical properties, structures and functional groups, for which the approach is considered ***not*** *to be applicable because of technical constraints in the testing of those substances (e.g. poor solubility, interference with detection system etc.) or because such substances have been found to give unreliable results (e.g. non-reproducible results when the defined approach is applied multiple times or because of wrong predictions with respect to reference classification).*

10 Predictive capacity of the approach

Provide an indication of the extent to which the defined approach predicts the skin sensitisation potential by considering the associated information sources and by excluding chemical types identified in the limitations above. Express the predictive capacity in terms of sensitivity, specificity and concordance, if applicable, or by other goodness-of-fit statistics (e.g. linear correlation analysis). Rationalise to the extent possible potential misclassifications (i.e. substances under-predicted or over-predicted with respect to the reference classification) or unreliable predictions for substances that are considered to be covered by the applicability domain of the approach.

11 Consideration of uncertainties associated with the application of the defined approach

11. 1 Sources of uncertainty

Describe the uncertainty(ies) which is/are considered to be associated with the application of the defined approach by capturing the source(s) of uncertainty that result(s) from:

1. The DIP's structure

- *What are the uncertainties related to the chosen DIP's structure?*
- *How does the DIP's coverage or weighing of the AOP key events affect your confidence in the overall prediction?*
- *How does your confidence in the DIP's prediction vary between different substances?*

2. Information sources used within the defined approach

- *How does the variability in approach information source's data for a given substance (i.e. reproducibility) affect your confidence in the DIP's prediction?*

3. Benchmark data used

- *How does the inherent variability of the reference data (e.g. LLNA, human) affect your confidence in the DIP's prediction?*

4. Others sources

11.2 Impact of uncertainty on the DIPS's prediction

Consider how these sources of uncertainty affect the overall uncertainty in the final prediction in the context of the defined approach application.

12 References

List relevant references, weblinks etc., including those describing the structured approach itself (also provided under Section 2 on General Information).

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OECD (2016c) Annex I: Case Studies To The Guidance Document On The Reporting Of Defined Approaches and Individual Information Sources To Be Used Within Integrated Approaches To Testing And Assessment (IATA) For Skin Sensitisation (ENV/JM/MONO(2016)29/ANN1). Available at: http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono(2016)29/ann1&doclanguage=en

OECD (2016d) Annex II: Information Sources Used Within The Case Studies To The Guidance Document On The Reporting Of Defined Approaches and Individual Information Sources To Be Used Within Integrated Approaches To Testing And Assessment (IATA) For Skin Sensitisation (ENV/JM/MONO(2016)29/ANN2). Available at: http://www.oecd.org/officialdocuments/publicdisplaydocumentpdf/?cote=env/jm/mono(2016)29/ann2&doclanguage=en

Appendix R.7.3–4 Potency estimation for skin sensitisation

Background

The estimation of potency for skin sensitisers is important for protecting workers and consumers. According to the CLP Regulation skin sensitisers can be divided in two classes i.e. Extreme and strong sensitisers (Cat. 1A) and moderate sensitisers (Cat. 1B).

Where non-animal testing methods, e.g. *in chemico/in vitro* test methods, are used to fulfil the information requirements of the REACH Regulation, it should be noted that no widely accepted approach is currently available to assess the potency leading to sub-categories according to the CLP Regulation. In contrast, the current standard requirement under REACH for fulfilling the requirements of AnnexesVII is an *in vivo* test method, i.e. the LLNA, which allows the assessment of skin sensitisation potency and subsequent sub-categorisation of substances (Cat. 1A *vs.* Cat. 1B) when EU method B.42/OECD TG 429 is used.

Identification of strong and extreme skin sensitisers and subsequent classification into sub-category 1A according to CLP is important for the protection of human health. This is due to the fact that, depending on the skin sensitisation potency, different concentration limits are to be applied, i.e. for Cat. 1 and Cat. 1B the generic concentration limit (GCL) is 1%, for Cat. 1A (extreme or strong) the GCL is 0.1% and for extreme sensitisers a specific concentration limit (SCL) of 0.001% is recommended according to the Guidance on the Application of the CLP criteria (for further information, see Section 3.4 of the *Guidance on the Application of the CLP criteria*). In short, this may lead to the situation that mixtures containing potent sensitisers are not correctly classified, if general Cat. 1 is used and the GCL of 1% is applied instead of 0.1% or 0.001%. This would lead in lowering of the safety levels, which may lead to an increased incidence of human sensitisation to more potent sensitisers.

In this appendix different approaches for assessing skin sensitisation potency are reviewed, in the context of the capabilities and limitations of the current standard *in vivo* method.

Uncertainty of the LLNA

One of the challenges for alternative (replacement) approaches when trying to predict an apical endpoint is the uncertainty associated with the inherent variability of the animal-based reference data. In the case of skin sensitisation, the variability of the LLNA defines an upper limit for the predictivity of alternative methods, especially when trying to predict potency classes which can be significantly affected by solvent effects (Basketter *et al.*, 2001). Recently this was confirmed by Hoffmann (2015) who analysed potential vehicle-related variability with respect to the five category classification system used by ECETOC (2003). A retrospective analysis by ICCVAM (2011) comparing LLNA data with human data showed that based on a limited number of cases investigated it seems that the cut-off values used in the CLP criteria for identifying strong sensitisers (Cat. 1A) approximately half of Cat. 1A sensitisers were underclassified according to LLNA-based (EU method B.42/OECD TG 429) CLP criteria, whereas approximately a third of Cat. 1B sensitisers were missclassified, and approximately 60% of non-sensitisers were overclassified[71].

In a more recent study of LLNA variability by EURL ECVAM (Dumont *et al.*, 2016), the number of studies predicting skin sensitisation potency (UN GHS / EU CLP hazard sub-categorisation) was considered, analysing the variability per substance (irrespective of the solvent) and per substance-solvent combination. Consistent with the ICCVAM analysis, the results showed that the inherent variability of LLNA is less significant for Cat. 1A substances, but more significant for Cat. 1B and non-sensitising substances[72].

Review of approaches for predicting skin sensitisation potency (classes and EC3)

At this point in time, the results portrayed in the subsequent paragraphs, while published in the peer-reviewed literature, have not been endorsed or validated by bodies such as EURL ECVAM or the OECD. It is the purpose of this section to provide an overview of the available literature, but ECHA cannot take responsibility for the correctness of the reported results, which may or may not be used at the registrant's own risk. In any case, the reader is advised to screen the recent literature for the latest developments.

Several attempts have been made to combine the EURL-ECVAM validated methods Direct Peptide Reactivity Assay (DPRA; (Gerberick *et al.*, 2004), KeratinoSens^{TM} (Natsch and Emter 2008; Emter *et al.*, 2010), h-CLAT (Ashikaga *et al.*, 2006; Nukada *et al.*, 2011, 2012), and other methods (Ade *et al.*, 2006; Piroird *et al.*, 2015; Python *et al.*, 2007, Ramirez *et al.*, 2014), have been made (Natsch *et al.*, 2009, 2015; Bauch *et al.*, 2012; Hirota *et al.*, 2013, 2015; Jaworska *et al.*, 2013, 2015; Tsujita-Inoue *et al.*, 2014; Urbisch *et al.*, 2015) to predict skin sensitisation potency. A summary of the most promising approaches follows.

Natsch *et al.* made one of the first attempts (Natsch *et al.*, 2009) to predict skin sensitisation potency of a dataset of 116 substances with a combination of non-animal methods. The model incorporates four descriptors accounting for different key events of the adverse outcome pathway (AOP). They used a system based on scores (Jowsey *et al.*, 2006) from single methods, i.e. peptide reactivity as a surrogate for protein binding, the induction of antioxidant/electrophile responsive element dependent luciferase activity as a cell-based assay, and an in silico prediction (Dimitrov *et al.*, 2005) for skin sensitisation potential (TIMES-SS). The relationship between scores and potency was not sufficient (R^2=0.423) to properly distinguish between potency classes. However, extreme sensitisers (not strong) could be easily distinguished from weak sensitisers and non-sensitisers, as 57 out of 59 weak/NS had average predicted EC3 values <3. Due to the large overlap between strong, moderate and weak

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71 More specifically, the ICCVAM analysis revealed that 48% of known strong human sensitisers (showing positive responses at an induction dose per skin area of ≤ 500 µg/cm2, i.e., Cat. 1A according to CLP criteria for human data) showed an EC3 > 2% (41%) or were negative in the LLNA (7%), therefore being underclassified according to LLNA CLP criteria. Furthermore, of the non-strong human sensitisers (Cat. 1B according to CLP criteria for human data), 6% were overpredicted as Cat. 1A and 22% were underpredicted as non-sensitisers by the LLNA. Finally, 7% and 52% of the NS in humans were overclassified as Cat. 1A and Cat. 1B, respectively, by the LLNA.

72 More specifically, the EURL ECVAM analysis revealed that for substances having at least one LLNA study giving a non-sensitiser result, only 66% of all available LLNA studies (performed with the same solvent) identified these substances consistently as non-sensitiser. The rest of the studies classified them as either Cat. 1B (23%) or Cat. 1A (11%). For substances having at least one LLNA study giving a Cat. 1B result, only 68% of all available LLNA studies (performed with the same solvent) classified them consistently as 1B. The remaining studies (32%) classified the substances in equal proportion to non-sensitisers or Cat. 1A sensitisers. The classification of substances having at least one LLNA study giving a Cat. 1A result was found to be less variable, with 79% of the studies (performed with the same solvent) classifying these substances as 1A (15% Cat. 1B and 6% non-sensitisers).

sensitisers, the published model is not recommended for distinguishing between Cat. 1A and 1B substances.

Other approaches (also leading to over-prediction of moderates as compared to the LLNA) have been based on different ways of integrating data, i.e. artificial neural networks (Hirota *et al.*, 2013, 2015; Tsujita-Inoue *et al.*, 2014), decision trees, score-based models (Nukada *et al.*, 2013; Takenouchi *et al.*, 2015), and mechanistic domain based regression models (Natsch *et al.*, 2015). Consistent with the variability in the LLNA, these models, which integrate validated *in vitro* methods or similar cell based assays with or without *in silico* descriptors, have overall accuracies in predicting skin sensitisation potency categories ranging from 70% to 85%. It is worth noting, though, that these methods usually try to predict a large number of substances, 244 substances in the most extreme case (Natsch *et al.*, 2015), which is more difficult than predicting smaller numbers (≤50 substances) because the applicability domain is much smaller and the models are more local.

One of the conclusions that can be drawn from these studies is that some mechanistic domains are easier to predict than others. For instance, Natsch *et al.* found better predictivity for epoxides and nucleophile substitution domains, with R^2>0.80. Aldehydes were the worst predicted group, with R^2=0.21. The parameters with most prediction power also vary across domains and this is valuable information for the further development of models: kinetic rate constants were found to be the most prominent predictor for the S_N2/S_NAr domain; KeratinoSens™ EC3 was the best predictor for Michael acceptors, and cytotoxicity and vapour pressure for epoxides. Given the large number of substances predicted by the Natsch *et al.* method and the fact that it can distinguish Cat. 1A substances from the rest (sensitivity=0.70 if Cat. 1A are distinguished from 1B & NS, results not shown), the method could be used in a *Weight-of-Evidence* approach or as the basis for grouping and read-across.

Important information on mechanistic domains was also provided by Urbisch and colleagues (Urbisch *et al.,* 2015). They observed that ARE based assays, like KeratinoSens™ and LuSens, did not perform well at predicting the skin sensitisation potential of acylating agents, and Schiff base could not be well classified with any of the methods investigated (DPRA, KeratinoSens™, LuSens, h-CLAT, mMUSST). This information can be of high value for those methods (discussed below) that need further testing.

In contrast to *in vitro* based models, Dearden et al. developed QSARs (Dearden *et al.*, 2015) to predict LLNA EC3 values from purely computational descriptors (CODESSA, MOE, and winMolconn41 software). They divided a dataset of 204 sensitisers with LLNA EC3 into 10 mechanistic domains and derived QSARs for each domain. They obtained good predictivities[73] (R^2>0.83) for 7 out of 10 domains and Q^2>0.79 for 6 out of 10 domains – Q^2 is the R^2 equivalent for the test set. The domains with best predictions were in this order: oxidation potential (R^2=0.91), acyl transfer (R^2=0.90), Michael acceptor (R^2=0.83), pro-Michael acceptor (R^2=0.83), S_N2 (R^2=0.82), and Schiff base + pro-Schiff base formers (R^2=0.82). The S_N1, pro-S_N2, and S_NAr domains contained too few substances to develop meaningful QSARs. The publication does not use the predictions to classify into potency classes, but given that the correlation with LLNA-EC3 is so high, a good performance is expected. While this model might not be adequate as a standalone prediction method, it has high potential to be used in a *Weight-of-Evidence* approach, especially given its computational and, therefore, fast and reproducible character. It could also be used as a way of grouping substances for read-across.

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[73] Dearden et al. transformed the EC3 (g/ml) into an equivalent parameter with molar units named SSP. In principle, this transformation is convenient from the point of view that if EC3 is expressed in g/ml, two substances that are equally sensitisers and have different molecular weight would have significantly different EC3 values and might fall into different potency classes. The SSP corrects for this as it is not dependent on the molecular weight of the substances. In practice, the transformation from EC3 (g/ml) to SSP (M) had no effect on the potency class definitions for the substances studied and both parameters were strongly correlated (R^2=0.96).

Another partially *in silico* based model with high predictivity is the Bayesian network (BN) proposed by Jaworska *et al.* (Jaworska, 2011; Jaworska *et al.*, 2013, 2015). The BN model integrates several sources of information and is capable of guiding the tests that need to be conducted to obtain a higher confidence in the prediction. The second version of the model (ITS-2; Jaworska *et al.*, 2013) was trained on 124 substances (training set) and tested on 21 substances, predicting correctly 95% and 86% of the substances in the test set for hazard and LLNA potency classes, respectively. The LLNA classes were reduced to four as strong and extreme sensitisers were merged under strong sensitisers. However, the model could be used to predict CLP categories if weak and moderate are considered as Cat. 1B substances. The model uses different *in silico* (TIMES-SS), *in vitro* (KeratinoSens™, mMUSST U937, skin permeation model readouts), *in chemico* (DPRA readouts), and octanol-water partition coefficient (K_{ow}) parameters. The model uses all the provided parameters to derive a probability value of belonging to each of the LLNA potency groups. However, the model also works with data gaps, allowing one to estimate how much certainty in the prediction would be gained if specific test data were obtained before performing such tests. In general the model performed very well, although better for non-sensitisers and weak substances. The performances were: non-sensitisers (93, 100), weak (89, 93), moderate (75, 83), and strong/extreme (81, 73). The values in parenthesis correspond to the % area under the curve (AUC) for the training set and test set, respectively (see original publication for further information on the scoring). The model includes a correction factor for Michael acceptors which were systematically over-predicted.

A third improved version of the model (ITS-3) has been recently published (Jaworska *et al.*, 2015). The system was constructed with an aim to improve precision and accuracy for predicting LLNA potency beyond ITS-2, by improving representation of chemistry and biology. Among novel elements are corrections for bioavailability both *in vivo* and *in vitro* as well as consideration of the individual assays' applicability domains in the prediction process. The three validated alternative assays, DPRA, KeratinoSens and h-CLAT, representing the first three key events of the adverse outcome pathway for skin sensitisation, are all integrated in ITS-3. In this model, skin sensitisation potency prediction is provided as a probability distribution over four potency classes. The probability distribution is converted to Bayes factors to: 1) remove prediction bias introduced by the training set potency distribution and 2) express uncertainty in a quantitative manner, allowing transparent and consistent criteria to accept a prediction. The novel ITS-3 database includes 207 substances with a full set of *in vivo* and *in vitro* data. The accuracy for predicting LLNA outcomes on the external test set (n = 60) was assessed in three levels, and was found to be high (>90%) following the order: hazard (two classes) > GHS potency classification (three classes) > potency (four classes).

Another *in vitro* model is the epidermal-equivalent (EE) potency assay (Gibbs *et al.*, 2013). The model uses 3D reconstituted human epidermis, cytotoxicity, and IL-1α (IL-1α2x) fold increase in order to predict potency (2 key event of the AOP). Its reproducibility and predictive capacities were evaluated in an international ring trial for 13 substances (Teunis, 2014). The model appears to be able to separate strong/extreme from weak/moderate sensitisers, with sensitivity=69% and specificity=84%. One of the advantages of this method is that it does not have solubility and stability issues in water like most *in vitro* methods since the substances can be tested neat. This model could offer a means of differentiating between Cat. 1A and Cat. 1B substances, but cannot on its own be used to distinguish non-sensitisers from sensitisers.

The "classic" *in vitro* and *in chemico* assays to predict skin sensitisation do not always predict LLNA potency with high accuracy even when combined with *in silico* models. Newer versions of methods that are gene-based seem more promising. One of these methods (McKim *et al.*, 2010) is the *in vitro* toxicity index (IVTI) developed by Cyprotex in 2010, also known as SenCeeTox (McKim *et al.*, 2012). It is based on a combination of cell viability, direct and indirect chemical (peptide) reactivity, and ARE/EpRE- mediated gene expression and was developed for use with human keratinocyte (HaCaT) cells and human 3D skin models. It predicts four potency classes: extreme/strong, moderate, weak, and non sensitisers. The model showed high specificity (92%) and sensitivity (81%) in discriminating between extreme, strong, and moderate sensitisers from weak and non-sensitisers and it was reported that 4 out of 4 extreme/strong, 70% of the moderates, 79% of the weaks, and 73% of the non-sensitisers were correctly classified. The model was tested on 97 substances (39 in the training set and 58 in the test set) and seems capable of identifying extreme/strong sensitisers (Cat. 1A) substances and distinguishing them from non-Cat. 1A substances, even though it is not capable of distinguishing weak from non-sensitisers.

Another method for predicting LLNA EC3 values is VITOSENS, which is based on a linear combination of cell cytotoxicity (IC20) caused by the test substance and the fold change in the expression of CCR2 (C–C chemokine receptor type 2) and the transcription factor cAMP responsive element modulator (CREM) (Lambrechts *et al.*, 2010). Cell cultures from two different cord blood donors (three in case of discordant results) are used. The authors showed a very high correlation (Pearson R^2=0.79, Spearman rank correlation coefficient=0.91) between the predicted and the EC3 values for 15 substances. The method properly predicts some pro-haptens showing some metabolic capabilities but might not be adequate for extreme cytotoxicants as two substances were considered outliers due to too high cytotoxicity.

The gene allergen rapid detection (GARD) method uses differentially regulated transcripts of 200 genes in MUTZ-3 cells (as surrogate of primary human dendritic cells) after exposure to predict skin sensitisation (Johansson *et al.*, 2011, 2013). A support vector machine model trained on 38 substances performs the final skin sensitisation prediction. The authors did not provide statistics on potency classification performance, but PCA plots showed clusters of substances belonging to the same potency groups. The model has recently been reported (Forreryd *et al.*, 2015) to predict respiratory sensitisation hazard for 30 substances. However it should be pointed out that the regulated transcripts of the 200 genes in MUTZ-3 cells are not public, similarly to a proteomic assay (with approximately 110 candidate proteins to reliably identify human skin sensitisers) also using MUTZ-3 and human keratinocytes (Reisinger *et al.*, 2015; Thierse *et al.,* 2011; Roggen *et al*., 2011). Note: currently (at the time of publication of this Guidance), only one laboratory in EU is conducting the assay.

It seems that the evolution of the two methods just described is the so-called SENS-IS assay (Cottrez *et al.*, 2015). SENS-IS makes use of a series of over 60 genes that are modulated during sensitisation either in mice or humans, and uses the modulation of these genes in reconstructed human epidermis models (Episkin) to predict the skin sensitisation potential and potency of substances. This set of sensitisation biomarker genes was selected based on a thorough analysis of the genes modulated during the sensitisation process in mice (LLNA), humans (blisters) or reconstructed human epidermis, starting from a panel of over 900 target genes identified through data mining. Fine analysis of their expression pattern indicated that it was the number of modulated genes rather than the intensity of up-regulation that correlated best with sensitization potential (Cottrez *et al.*, 2015). Thus, the model simply consists of determining the number of genes that are up-regulated after exposure. A threshold determines whether the substance is predicted as sensitiser or non-sensitiser. A test substance is considered to be a skin sensitiser if it increases the expression of at least seven genes in either the so-called "SENS-IS" gene set (consisting of 17 genes) or the "ARE" gene set (consisting of 21 genes). A third set of 23 genes is used to identify if the test substance is irritant (if at least 15 out of these 23 genes are induced). The skin sensitisation potency of a test substance is determined based on the lowest concentration at which it becomes positive in the SENS-IS assay. The test substance is therefore considered to be an extreme sensitiser if it is positive at 0.1%, strong if it is positive at 1%, moderate if it is positive at 10% or weak if it is positive at 50%.

The SENS-IS is capable of distinguishing skin sensitisers from non-sensitisers with accuracies of 97% based on LLNA data for 150 substances (9 extreme, 17 strong, 27 moderate, 36 weak and 61 non-sensitisers) and of 96% based on human data for 130 substances (52 non-sensitisers and 78 sensitisers). The performance of the test method in predicting skin sensitising potency reaches an accuracy of 93% when used to distinguish between five different LLNA classes (non-sensitiser, weak, moderate, strong and extreme sensitiser) for the same 150 substances (Cottrez *et al.*, 2016). A manuscript reporting the data for these 150 substances has been submitted for publication and was provided as an attachment to a recent

SPSF proposal submitted to the OECD for the development of a new Test Guideline. These data were also presented to the OECD expert group on skin sensitisation in October 2015.

Finally, it should be mentioned that test methods using reconstructed human epidermis models also have the advantage that their test systems possess a metabolic capacity closer to that of native human skin (Hewitt *et al*., 2013), which is the target organ of interest when assessing the skin sensitisation potential/potency of substances.

Conclusions

The prediction of skin sensitisation potency by alternative methods is currently an important objective, and efforts to develop approaches have to be judged in the context of the inherent variability of the LLNA.

While firm recommendations for alternative testing strategies cannot be made, this mini-review shows that some *in vitro* and *in silico* methods show promise either for the identification of Cat. 1A substances or even discrimination between potency classes. In particular, promising results have been reported for several approaches combining different methodologies (e.g. Dearden *et al.*, 2015; Natsch *et al.*, 2015; Takenouchi *et al.*, 2015; Hirota *et al*., 2015; Jaworska et al. 2015), and some of *in vitro* gene based methods (e.g. Cottrez *et al.*, 2015, 2016; Lambrechts *et al.*, 2010).

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R.7.4 Acute toxicity

R.7.4.1 Introduction

Assessment of the acute toxic potential of a substance is necessary to determine the adverse health effects that might occur following accidental or deliberate short-term exposure. The nature and severity of the acute toxic effects are dependent upon various factors, such as the mechanism of toxicity and bioavailability of the substance, the route of exposure and the total amount of substance to which the person or animal is exposed.

R.7.4.1.1 Definition of acute toxicity

The term *acute toxicity* is used to describe the adverse effects occurring following oral or dermal administration of a single dose of a substance or multiple doses given within 24 hours, or an inhalation exposure of 4 hours (see Section 3.1.1.1 of Annex I to the CLP Regulation).

The adverse effects can be seen as mortality, clinical signs of toxicity (for animals, refer to OECD Guidance Document 19 (OECD, 2000)), abnormal body weight changes, and/or pathological changes in organs and tissues. In addition to acute systemic effects, some substances may have the potential to cause local irritation or corrosion of the gastro-intestinal tract, skin or respiratory tract following a single exposure. Acute irritant or corrosive effects due to the direct action of the substance on the exposed tissue are not specifically covered by this document, although their occurrence may contribute to the acute toxicity of the substance and must be reported. The endpoints of skin corrosion/irritation, serious eye damage/eye irritation and respiratory tract corrosion/irritation are addressed in Section R.7.2 of this Guidance.

At the cellular level acute toxicity can be related to three main types of toxic effect, i.e. (i) general basal cytotoxicity, (ii) selective cytotoxicity and (iii) cell-specific function toxicity. Acute toxicity may also result from substances interfering with extracellular processes (Seibert, 1996). Toxicity to the whole organism also depends on the degree of dependence of the organism on the specific function affected.

R.7.4.1.2 Objective of the guidance on acute toxicity

A substance may induce systemic and/or local effects. This document is concerned with assessment of systemic effects following acute exposure.

The objectives of an acute toxicity study are to establish:

- whether a single exposure (or multiple exposures within 24 hours) to the substance of interest (when administered up to the limit dose of 2000 mg/kg bw (oral or dermal route), or equivalent concentration (inhalation route)) could be associated with adverse effects on human health; and/or
- what types of toxic effects are induced, their time of onset, duration and severity (all to be related to dose); and/or
- the dose-response relationship to determine the Acute Toxicity Estimate or ATE[74] (LD_{50}, LC_{50}), the discriminating dose, or the acute toxicity category; and/or
- when possible, the slope of the dose-response curve; and/or

[74] See Table 3.1.1 of Annex I to the CLP Regulation.

- when possible, whether there are marked sex differences in response to the substance.

Consequently this information enables to correctly decide on the classification and labelling of the substance for acute toxicity.

The indices of LD_{50} and LC_{50} are derived values, which relate to the dose that is expected to cause death in 50% of treated animals. These indices do not provide information on all aspects of acute toxicity. Other parameters and observations and their type of dose-response may yield valuable information.

Also, according to REACH Article 13(1) and Article 25(1), "*in order to avoid animal testing, testing on vertebrate animals for the purpose of [REACH] shall be undertaken only as a last resort. It is also necessary to take measures limiting duplication of other tests."*

Consequently the objectives of this Guidance are to address the REACH information requirements related to acute toxicity testing as well as to inform registrants on alternatives to animal testing.

The potential to avoid acute toxicity testing should be carefully exploited by application of read-across or other non-testing means.

To this end, Appendix R.7.4–1 on a Weight-of-Evidence (WoE) adaptation of the standard information requirement for an acute oral toxicity study should be considered, as it can help the registrant determine whether any non-animal or non-testing approach could be used instead of *in vivo* testing The WoE adaptation proposed primarily applies to low toxicity substances.

Background information on how this WoE approach was developed is provided in Appendix R.7.4–2.

Other approaches not explicitly outlined in Appendix R.7.4–1 may also be appropriate. Some generic alternative approaches, mostly referring to read-across and physico- chemical properties, can also be found in the draft OECD "Guidance Document on bridging or Waiving Acute Mammalian Toxicity Studies" (OECD, 2016). However, it should be noted that those alternative approaches may not all be applicable in the context of the REACH Regulation.

For risk assessment, further considerations on the nature and reversibility of the toxic effects are necessary.

R.7.4.2 Information requirements for acute toxicity

The standard information requirements for acute toxicity under the REACH Regulation are as follows:

Annex VII (≥1 t/y): acute toxicity study(ies) via the oral route of exposure is(are) required (Section 8.5.1);

Column 2 of Section 8.5 of Annex VII details specific rules for adaptation of the information requirement, notably allowing for the waiving of acute oral toxicity testing if the substance is corrosive to the skin or if a study on acute toxicity by the inhalation route is available.

Annexes VIII -X (≥ 10 t/y): acute toxicity study(ies) *via* the oral and dermal or inhalation route(s) of exposure is(are) required (Sections 8.5.2 and 8.5.3).

Column 2 of Section 8.5 of Annex VIII details specific rules for adaptation, notably requiring information on at least one other route of exposure depending on the nature of the substance and the likely route of human exposure. In addition allowance is made for the waiving of acute toxicity testing if the substance is corrosive to the skin.

Column 2 of Section 8.5.3 of Annex VIII further allows for the waiving of acute dermal toxicity testing if (i) the substance does not meet the criteria for classification for acute toxicity or STOT SE by the oral route and (ii) no systemic effects have been observed in *in vivo* studies with dermal exposure (e.g. skin irritation, skin sensitisation) or, in the absence of an *in vivo* study by the oral route, no systemic effects after dermal exposure are predicted on the basis of non-testing approaches (e.g. read across, QSAR studies).

R.7.4.3 Information sources on acute toxicity

Information on acute toxicity, as detailed below, can be obtained from a variety of sources including unpublished studies, databases and publications such as books, scientific journals, criteria documents, monographs and other publications (see Chapter R.3 of the *Guidance on IR&CSA* for further general guidance).

R.7.4.3.1 Non-human data on acute toxicity

R.7.4.3.1.1 Non-testing data on acute toxicity

Non-testing data can be provided by the following approaches:

a) structure-activity relationships (SARs) and quantitative structure-activity relationships (QSARs), collectively called (Q)SARs, and expert systems;

b) read-across and grouping.

Note that other types of data may also be proposed by the registrants.

(Q)SAR models

Compared with some other endpoints, there are relatively few (Q)SAR models and expert systems[75] capable of predicting acute toxicity. Available approaches have been reviewed in the literature (Cronin *et al*., 1995, 2003; Lessigiarska *et al*., 2005; Lapenna *et al.*, 2010; Fuart Gatnik and Worth, 2010; Diaza *et al.*, 2015; Kleandrova *et al.*, 2015).

(Q)SAR software packages (free and commercial) that contain models for the prediction of acute toxicity include: the OECD QSAR Toolbox, HazardExpert, Topkat, CASE Ultra, T.E.S.T, Derek Nexus and ACD/Percepta. Some of the models available from the scientific literature and the aforementioned software are described in detail in Appendix R.7.4–3.

On the basis of these reviews, the following conclusions can be made:

i) the relatively small number of models for *in vivo* toxicity is related to the nature of the endpoint – acute toxicity measurements are usually related to whole body phenomena and are

[75] In this context we mean by "(Q)SAR" global or local models relating the structure or properties of chemical substances to a specific property, in this case usually the 48h LD_{50} in the rat after exposure *via* the oral route. Expert systems comprise groups or packages of several local (Q)SAR models, and then apply some reasoning (based on expert knowledge) to decide which one of them (if any) is best suited to generate a prediction.

therefore very complex. The complexity of the mechanisms involved leads to difficulties in the QSAR modelling process;

ii) most QSAR models identify hydrophobicity as a parameter of high importance for the modelled toxicity. In addition, many models indicate the role of the electronic and steric effects;

iii) most literature-based models are restricted to single classes of substances, such as phenols, alcohols, anilines. Models based on more heterogeneous data sets are those incorporated in the expert systems.

Read-across and grouping

Read-across/chemical categories are described in Sections R.6.1 and R.6.2 of Chapter R.6 of the *Guidance on IR&CSA*. The scientific basis for building grouping arguments and read-across cases were revisited in the second version of the OECD Guidance on grouping of chemicals (OECD, 2014).

More detailed advice on the assessment of read-across can be found in ECHA's Read-Across Assessment Framework – RAAF (see http://echa.europa.eu/en/support/grouping-of-substances-and-read-across). Software such as the OECD QSAR Toolbox can be used to find data for analogues and support read-across cases. The OECD eChemPortal (http://www.echemportal.org/echemportal/index?pageID=0&request_locale=en) can be used to collect further data on suitable analogues.

R.7.4.3.1.2 Testing data on acute toxicity

In vitro data

There are currently no *in vitro* tests that have been officially adopted by the EU or OECD for the (regulatory) assessment of acute toxicity.

- *In vitro* Neutral Red Uptake (NRU) Cytotoxicity Assay

Based on the validation study to assess the predictive capacity of the *in vitro* NRU cytotoxicity assay to identify substances not requiring classification for acute oral toxicity (DB-ALM Protocol n°139, see http://ecvam-dbalm.jrc.ec.europa.eu/beta/), EURL ECVAM issued a recommendation concerning the validity and limitations of this *in vitro* test (EURL ECVAM, 2013). This recommendation is based on the views expressed by the EURL ECVAM Scientific Advisory Committee (ESAC) (see https://eurl-ecvam.jrc.ec.europa.eu/eurl-ecvam-recommendations/3t3-nru-recommendation).

According to the validation study, the *in vitro* NRU cytotoxicity assay shows a high sensitivity (ca. 95%) and, consequently, a low false negative rate (ca. 5%) when employed in conjunction with a prediction model to distinguish potentially toxic versus non-toxic (i.e. classified versus non-classified) substances. However, substances inducing acute toxicity by mechanisms specific only to certain cell types or tissues or requiring metabolic activation may not be correctly predicted. Moreover, the *in vitro* NRU cytotoxicity assay has a high false positive rate and, therefore, positive results cannot be readily used in a meaningful way in characterising the acutely toxic substances.

Following the provisions of the REACH Regulation, and in particular those contained in Annex XI, data from the *in vitro* NRU cytotoxicity assay could be used within a WoE approach to adapt the standard information requirements for acute oral toxicity, but this assay cannot be used as a stand-alone test.

A recommended application and the limitations of the *in vitro* NRU cytotoxicity assay are described in Appendix R.7.4–1.

Animal data

Data may be available, particularly for phase-in substances, generated from a wide variety of animal test guideline studies, which give various direct or indirect information on the acute toxicity of a registered substance, e.g.:

- EU B.1 / OECD TG 401 "Acute Oral Toxicity" (method deleted from the OECD Guidelines for testing of chemicals and from Annex V to Directive 67/548/EEC[76]);
- EU B.1 bis / OECD TG 420 "Acute oral toxicity – Fixed dose procedure";
- EU B.1 tris / OECD TG 423 "Acute oral toxicity – Acute toxic class method";
- OECD TG 425 "Acute oral toxicity – Up-and-down procedure" (updated in 2008);
- EU B.3 / OECD TG 402 "Acute dermal toxicity";
- EU B.2 / OECD TG 403 "Acute inhalation toxicity" (updated in 2009);
- Draft OECD TG 433 "Acute Inhalation Toxicity, Fixed Dose Procedure" (under drafting);
- EU B.52 / OECD TG 436 "Acute Inhalation Toxicity, Acute Toxic Class Method" (adopted in 2009);
- Draft OECD TG 434 "Acute Dermal Toxicity, Fixed Dose Procedure" (under drafting);
- ICH compliant studies;
- Mechanistic and toxicokinetic studies;
- Studies in non-rodent species.

Some repeated dose toxicity (RDT) studies can also give useful information. Guidance on how to use information from a sub-acute oral toxicity study is given in Appendix R.7.4–1.

Traditionally, acute toxicity tests on vertebrate animals have used mortality as the main observational endpoint, usually in order to determine the LD_{50} or LC_{50} values. These values were regarded as key information for hazard assessment and as supportive information for risk assessment.

However, derivation of a precise LD_{50} or LC_{50} value is no longer considered essential. Indeed, some of the current standard acute toxicity test guidelines, such as the fixed dose procedures (EU B.1 bis / OECD TG 420 and draft OECD TG 433), use signs of non-lethal toxicity. These test methods should be preferred as they present advantages over the other guidelines in terms of animal welfare.

Generic definitions of "Evident toxicity" and clinical signs indicative of "predictable death" can be found in Annex 1 of the EU B.1 bis / OECD TG 420.

Published and unpublished toxicological or general data

In addition to the current regulatory *in vivo* methods, acute toxicity data on animals may be obtained by conducting a literature search and reviewing all available published and unpublished toxicological or general data, and the official/existing acute toxicological reference values. Table R.7.4-1 lists a number of databases from where acute toxicity data may be retrieved. For more extensive general guidance see Section R.3.1 of Chapter 3 of the *Guidance on IR&CSA*.

Based on all the available information from sources such as those above, a WoE approach should be undertaken to maximise the use of existing data and minimise the commissioning of

[76] Existing EU B.1 / OECD TG 401 data would normally be acceptable but testing using this deleted method must no longer be performed.

new *in vivo* testing. A WoE adaptation, specific to substances of low acute oral toxicity is described (and instructed for) in Appendix R.7.4–1.

Table R.7.4-1 List of databases containing data on acute toxicity (adapted and expanded from Lapenna *et al.*, 2010[77]).

Database[78]	Availability	Information
ChemIDplus, developed by the US National Library of Medicine (NLM) http://chem.sis.nlm.nih.gov/chemidplus/	Freely available through the Internet	Toxicity data for over 139,000 records, retrieved from TOXNET (TOXicology Data NETwork; http://toxnet.nlm.nih.gov) which includes HSDB (Hazardous Substances Data Bank). The HSDB is an older subset of the RTECS database. A search for rat and mouse oral LD_{50} values found 13,548 and 28,033 records, respectively.
Chemical Effects in Biological Systems (CEBS), developed by the US National Institutes of Health (NIH) http://www.niehs.nih.gov/research/resources/databases/cebs/index.cfm	Freely available through the Internet	*In vivo* study data and acute dose of a small number of known hepatotoxicants to rat.
Registry of Toxic Effects of Chemical Substances (RTECS), originally compiled and maintained (until 2001) by the US NIOSH and currently maintained by Symyx Technologies. Structure searchable through the Symyx Toxicity Database: http://www.symyx.com/products/databases/bioactivity/rtecs/index.jsp Also searchable via the Leadscope Toxicity Database (http://www.leadscope.com/databases/)	Commercial	Rat acute oral toxicity (LD_{50}) and acute inhalation toxicity (LC_{50}) data compiled from the open scientific literature for approximately 7,000 compounds (organic, inorganic and mixtures), including approxmately 4,000 organic compounds.
TerraBase databases http://www.terrabase-inc.com/	Commercial	Several databases containnig rat and mouse LD50 values for different product types (natural compounds, drugs, pesticides).
ZEBET, compiled by BfR ZEBET; http://www.dimdi.de	Freely searchable through the DIMDI website	Includes rat or mouse LD_{50} values (from the RTECS database) and cytotoxicity (IC_{50}) data for 347 compounds compiled from the open literature.

77 http://publications.jrc.ec.europa.eu/repository/bitstream/JRC61930/eur_24639_en.pdf

78 The databases in the table are mentioned for information only, and their inclusion in the table does not represent any endorsement by ECHA on the quality or adequacy of the data. Ultimately it is up to the registrant to decide whether data found in these sources are suitable for REACH purposes.

ACToR http://actor.epa.gov/actor/faces/ACToRHome.jsp	Freely available through the Internet	The EPA Aggregated Computational Toxicology Resource (ACToR) includes acute-toxicity data that are compiled from the Integrated Risk Information System (IRIS), Organisation for Economic Co-operation and Development (OECD) Summary reports, and Agency for Toxic Substances and Disease Registry documents.
Hazardous Substances Data Bank (HSDB) http://toxnet.nlm.nih.gov/cgi-bin/sis/htmlgen?HSDB	Freely available through the Internet	The National Library of Medicine (NLM) manages a network of databases called TOXNET®, which makes it possible to search for acute-toxicity information that is available in the Hazardous Substances Data Bank (HSDB).
Priority-based Assessment of Food Additives (PAFA) available in Leadscope http://www.leadscope.com/toxicity_databases/	Commercial	Leadscope, Inc. markets a toxicity database that contains nearly 180,000 chemical structures and over 400,000 toxicity-study results derived from the US Food and Drug Administration Priority-based Assessment of Food Additives (PAFA) Database, the National Toxicology Program Chronic Database, the Registry of Toxic Effects of Chemical Substances (RTECS), and the DSSTox Carcinogenicity Potency Database (CPDB) (Leadscope 2012). Acute-toxicity data related to multiple exposure routes are available in the PAFA database and RTECS.
eChem Portal http://www.echemportal.org/echemportal/substancesearch/substancesearchlink.action	Freely available through the Internet	eChemPortal, is a no-cost publicly available acute-toxicity database that can be searched by using a variety of chemical identifiers.
ECHA dissemination http://echa.europa.eu/ also available in the OECD QSAR Toolbox http://www.qsartoolbox.org/	Freely available through the Internet	Database containing endpoint study records from REACH registration dossiers.
Toxicity Japan MHLW database http://dra4.nihs.go.jp/mhlw_data/jsp/SearchPageENG.jsp and Rodent Inhalation Toxicity Database http://www.qsari.org/index.php/databases both available in the OECD QSAR Toolbox.	Freely available through the Internet	The Toxicity Japan MHLW database contains experimental results from single dose toxicity test and mutagenicity test results performed under Japan's Existing Chemicals Programme. The Rodent Inhalation Toxicity Database is a compilation of high quality data from rat inhalation studies reported in the literature. The collection effort focused on a primary file of approximately 500 scientific papers and reports for comprehensive review. Of the 500 scientific papers, only 79 papers passed the minimum quality assurance reviews based on verification that the paper was the primary reference for the test, verification that the paper used experimental methods that would produce reliable observations, and verifications that the reporting of the toxicity endpoints were unambiguous.

R.7.4.3.2 Human data on acute toxicity

Acute toxicity data on humans may be available from:

- Epidemiological data identifying hazardous properties and dose-response relationships;
- Routine data collection, poisons data, adverse event notification schemes, coroner's reports;
- Biological monitoring/personal sampling;
- Human kinetic studies – observational clinical studies;
- Published and unpublished studies from e.g. industry, occupational safety authorities, academia;
- National poison centres.

The main obstacles to the use of human data are their limited availability and often limited information on levels of exposure (ECETOC, 2004).

For further information, see Section 3.1 of the *Guidance on the application of the CLP criteria*.

R.7.4.3.3 Exposure considerations for acute toxicity

With regard to acute toxicity, exposure considerations are detailed in column 2 of Annex VIII to the REACH Regulation, but not in Annex XI.

Where the potential for human exposure exists, the most likely route(s) of exposure should be determined so that the potential for acute toxicity by this (these) route(s) can be assessed. If there is only one demonstrated route of exposure, acute toxicity by this route must be addressed. Determination of the most likely route of exposure will have to take into account not only how the substance is manufactured and handled, including engineering controls that are in place to limit exposure, but also the physico-chemical properties of the substance, for instance, whether the substance is a solid or liquid, the particle size and proportion of respirable and inhalable particles, vapour pressure and log K_{ow}.

R.7.4.4 Evaluation of available information on acute toxicity

The detailed generic guidance provided in Chapter R.4 of the *Guidance on IR&CSA* on the process of judging and ranking the available data for its adequacy (reliability and relevance), completeness and remaining uncertainty is relevant to information on acute toxicity.

R.7.4.4.1 Non-human data on acute toxicity

R.7.4.4.1.1 Non-testing data on acute toxicity

Physico-chemical properties[79]

It may be possible to infer from the physico-chemical characteristics of a substance whether it is likely to be corrosive or absorbed following exposure by a particular route, which needs to be taken into account when deciding the route of administration for testing for acute toxicity. Physico-chemical properties may be important in the case of exposure *via* the inhalation route

[79] Refer also to Appendix R.7.4–1 and to Tables R.7.12-1 to R.7.12-6 in Section R.7.12 of Chapter R.7c of the *Guidance on IR&CSA*.

(vapour pressure, mean mass aerodynamic diameter (MMAD)[80], log K_{ow}), not only to determine whether this route is relevant, but also to determine the technical feasibility of the testing and act upon the distribution of the substance in the airways, in particular for *local-acting substances*. Indeed, some physico-chemical properties of the substance or mixture could be the basis for waiving of testing. In particular, waiving should be considered for low volatility substances, which are defined as having vapour pressures <1 x 10^{-5} kPa (7.5 x 10^{-5} mmHg) for indoor uses, and <1 x 10^{-4} kPa (7.5 x 10^{-4} mmHg) for outdoor uses. Furthermore, inhalable particles are capable of entering the respiratory tract via the nose and/or mouth, and are generally smaller than 100 µm in diameter. Particles larger than 100 µm are less likely to be inhalable. For that reason, particular attention should be given to the results of aerosol particle size determination.

In particular, for substances in powder form, particle size of the material decisively influences the deposition behaviour in the respiratory tract and potential toxic effects. For mists, particle size is less determinative since the substance or the solvent may evaporate after mist formation, resulting in smaller particles more likely to reach the respiratory tract. Particle size considerations (determined by e.g. granulometry testing, OECD TG 110) can be useful for:

- selecting a representative sample for acute inhalation toxicity testing;
- assessing the respirable and inhalable fractions, preferably based on aerodynamic particle size;
- justifying derogations from testing, for instance, when read-cross (or chemical grouping approach) data can be associated with results from particle size distribution analyses (see Section R.6.2 of Chapter R.6 of the *Guidance on IR&CSA*).

Physico-chemical properties are also important for determining the potential for exposure through the skin, for example, log K_{ow}, molecular weight and volume, molar refraction, degree of hydrogen bonding, melting point (Hostýnek, 1998). Further information on dermal absorption can be found in the guidance documents from OECD (2011) and EFSA (2012).

(Q)SAR

Several (Q)SAR systems are available that can be used to make predictions about, for example, dermal penetration or metabolic pathways. However, these systems have not been extensively validated against appropriate experimental data and it has not been yet verified whether the results genuinely reflect the situation *in vivo*. That is why the modelled data can be used for hazard identification and risk assessment purposes only as part of a WoE approach.

These approaches can be used to assess acute toxicity if they provide relevant and reliable (adequate) data for the substance of interest. (Q)SARs can also be used to provide adequate data on single components of multi-constituent substances or UVCBs for defining ATEs. Guidance on how to assess the relevance and reliability of non-testing data is provided in the general guidance on (Q)SARs in Section R.6.1 and on grouping approaches in Section R.6.2 of Chapter R.6 of the *Guidance on IR&CSA*. Non-testing methods should be documented according to the appropriate reporting formats (see Sections R.6.1.9 and R.6.2.6). In the case of (Q)SARs and expert systems, a detailed description of available models is provided in the JRC QSAR Model Database (http://qsardb.jrc.it/).

The complexity of the acute toxicity endpoint (possibility of multiple mechanisms) is one of the reasons for limited availability and predictivity of QSAR models for this endpoint. In the

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80 Forms or physical states in which the substance or mixture is placed on the market and in which it can reasonably be expected to be used must be taken into consideration for classification.

absence of complete validation information, available models could be used as a part of the WoE approach for hazard identification and risk assessment purposes after precise evaluation of the information derived from the model.

Evaluation of the validity of the method

An evaluation of model validity according to the OECD principles should be available, as described in Section R.6.1 in Chapter R.6 of the *Guidance on IR&CSA*, using the QSAR Model Reporting Format (QMRF).

Evaluation of the reliability of the individual prediction

The reliability of individual (Q)SAR predictions should be evaluated, as described in Section R.6.1 in Chapter R.6 of the *Guidance on IR&CSA*, using the QSAR Prediction Reporting Format (QPRF).

Read-across and grouping

Generic guidance on the application of grouping approaches is provided in Section R.6.2 of Chapter R.6 of the *Guidance on IR&CSA* and in the RAAF document. The RAAF document describes the assessment of the suitability of the analogues distinguishing six possible scenarios to build a read-across argumentation (see http://echa.europa.eu/en/support/grouping-of-substances-and-read-across).

R.7.4.4.1.2 Testing data on acute toxicity

In vitro data

The NRU cytotoxicity assay (see Section R.7.4.3.1.2) may provide supplementary information, which may be used e.g. to determine starting doses for *in vivo* studies (OECD, 2010; Schrage *et al.*, 2011), and to assist in the evaluation of data from animal studies. The NRU cytotoxicity assay cannot replace testing in animals completely, and should rather be used in a WoE context.

Generic guidance is given in Chapter R.4 of the *Guidance on IR&CSA* for judging the applicability and validity of the outcome of various study methods, assessing the quality of the conduct of a study (including how to establish whether the substance falls within the applicability domain of the method and the validation status for the given domain) and aspects such as vehicle, number of duplicates, exposure/ incubation time, GLP-compliance or comparable quality description.

Animal data

Acute toxicity tests on animals have primarily used mortality as the main observational endpoint, usually in order to determine LD_{50} or LC_{50} values, although current standard protocols, such as the fixed dose procedure (EU B.1 bis / OECD TG 420), use evident signs of toxicity in place of mortality. In most cases, there will be no information on the cause of death or mechanism underlying the toxicity, and only limited information on pathological changes in specific tissues or clinical signs, such as behavioural or activity changes.

Many acute toxicity studies on substances of low toxicity are performed as limit tests. For more harmful substances the choice of an optimum starting dose will minimise use of animals. When multiple dose levels are assessed, characterisation of the dose-response relationship may be possible and signs of toxicity identified at lower dose levels may be useful in estimating LOAELs or NOAELs for acute toxicity. The use of sub-acute oral toxicity studies for the characterisation of acute oral toxicity is described in Appendix R.7.4–1. For local acting substances, mortality after inhalation may occur due to tissue damage in the respiratory tract. In these cases, the severity of local effects may be related to the dose or concentration level and, therefore, it might be possible to identify a LOAEL or NOAEL. For systemic toxicity, there could be some evidence of target organ toxicity (pathological findings have to be documented) or signs of toxicity based on clinical observations.

Whichever approach is used in determining acute toxicity critical information needs to be derived from the data to be used in risk assessment. It is important to identify those dose levels which produce signs of toxicity, the relationship of the severity of these with dose and the level at which toxicity is not observed (i.e. the acute NOAEL).

In addition to currently available OECD or EU test methods (see Section R.7.4.3), alternative *in vivo* test methods for assessing acute dermal and inhalation toxicity are in the process of adoption or revision and use for regulatory purposes. Whichever test is used to evaluate acute toxicity in animals, the evaluation of studies takes into account the reliability based on the approach of Klimisch *et al*. (1997) (standardised methods, GLP, detailed description of the publication), the relevance, and the adequacy of the data for the purposes of evaluating the given hazard from acute exposure (for more guidance see Section R.4.2 of the *Guidance on IR&CSA*). The preferred studies are those that give a precise description of the mechanism and reversibility of the toxic effect, the number of subjects, gender, the number of animals affected by the observed effects and the exposure conditions (atmosphere generation for inhalation, duration and concentration or dose). The relevance of the data should be determined in describing the lethal or non-lethal endpoint being measured or estimated.

In addition, when several studies results are available for one substance, the most relevant one should be selected; data from other studies that have been evaluated should be considered as supportive data for the full evaluation of the substance.

The classification criteria for acute inhalation toxicity relate to a 4-hour experimental exposure period. If data for a 4-hour period are not available then extrapolation of the results to 4 hours are often achieved using Haber's Law (C x t = k). However, there are limits to the validity of such extrapolations, and it is recommended that the Haber's Law approach should not be applied to experimental exposure durations of less than 30 minutes or greater than 8 hours in order to determine the 4-hour LC_{50} for C&L purposes. CLP criteria also include criteria for conversion of existing inhalation toxicity data which have been generated using a 1-hour exposure (for further details see footnote c to Table 3.1.1 of Annex I to the CLP Regulation and Section 3.1.2.2 of the *Guidance on the application of the CLP criteria*).

Nowadays a modification of Haber's Law is used (C^nx t = k) as for many substances it has been shown that *n*, which is specific to individual substances, is not equal to 1 (Haber's Law). In case extrapolation of exposure duration is required, the *n* value should be considered. If this *n* value is not available from the literature, a default value may be used. It is recommended to set *n* = 3 for extrapolation to shorter duration than the duration for which the LC_{50} or EC_{50} was observed and to set *n* = 1 for extrapolation to longer duration (ACUTEX project, 2006), also taking the range of approximately 30 minutes to 8 hours into account.

Experimentally, when concentration-response data are needed for specific purposes, the EU B.2 / OECD TG 403 could be taken into consideration. The EU B.2 / OECD TG 403 will result in a concentration-response curve at a single exposure duration, the C x t approach will result in a concentration-time-response curve, taking different exposure durations into account. The C x t approach uses two animals per C x t combination and exposure durations may vary from about 15 minutes up to approximately 6 hours. This approach may provide detailed information on the concentration-time-response relationship in particular useful for risk assessment and determination of NOAEC/LOAEC.

R.7.4.4.2 Human data on acute toxicity

When available, epidemiological studies, poisoning case reports or information from occupational surveillance may be crucial for acute toxicity and can provide evidence of effects that are undetectable in animal studies (e.g. symptoms like nausea or headache). However, the conduct of human studies is not allowed for the purpose of the REACH and CLP Regulations.

Such human data could also be useful to identify particular sensitive sub-populations like new born, children, patients with diseases (in particular with chronic respiratory diseases, e.g. asthma, chronic obstructive pulmonary disease (COPD)).

Additional guidance is provided on the reliability and the relevance of human data because there are no standardised guidelines for such studies (except for odour threshold determination) and these are not usually conducted according to GLP. Such guidance is provided in Section R.4.3.3 of Chapter R.4 of the *Guidance on IR&CSA*.

R.7.4.4.3 Exposure considerations on acute toxicity

Particular attention should be given to the potential routes of exposure in humans to select the appropriate testing strategy. The oral route is the primary route of choice based on practical considerations, e.g. on the likelihood of achieving the maximal systemic uptake of the test substance in most cases.

R.7.4.4.4 Remaining uncertainty on acute toxicity

In most cases, remaining uncertainties will exist due to the absence of valid human acute toxicity data, and so appropriate assessment factors should be applied. Toxicokinetic data could help in deriving substance-specific interspecies assessment factors. As acute toxicity testing does not usually include clinical chemistry, haematology and detailed histopathology and functional observations, an additional assessment factor may need to be applied when a NOAEL or LOAEL from these studies is used to derive DNELs (for more guidance on the setting of DNELs for acute toxicity, see Appendix R.8-8 of Chapter R.8 of the *Guidance on IR&CSA*).

R.7.4.5 Conclusions on acute toxicity

R.7.4.5.1 Concluding on suitability for Classification and Labelling

In order to determine correct classification and labelling for acute toxicity, the criteria set forth in the CLP Regulation (Annex I, section 3.1) must be applied. The criteria for classification are based on specific "cut off values" (acute toxicity estimates) based on the LD_{50} or LC_{50} (For further details, see Section 3.1 of the *Guidance on the application of the CLP criteria*).

Information from acute toxicity testing can also be used to assess specific target organ toxicity after single exposure (STOT SE) as other (non-lethal) effects may be relevant for STOT SE classification in Cat 1, 2 or 3 (with respect to narcotic effects) (see sections 3.8.2.1.5 and 3.8.2.1.7.3 of Annex I to the CLP Regulation and Section 3.8.2.1.2 of the *Guidance on the application of the CLP criteria*).

Ideally, classification and labelling should be achieved using data generated from studies conducted in accordance with officially adopted test methods incorporated into the EU Test

Methods Regulation (Council Regulation (EC) No 440/2008)[81] or OECD TGs. Such studies will permit identification of the LD_{50}, LC_{50}, the discriminating dose (fixed dose procedures), or a range of exposure where lethality and/or severe toxicity is expected (acute toxic class methods). For substances of low expected toxicity (no mortalities expected at the upper dose limit) testing may be limited to this dose level (the limit test) and if the absence of mortalities is confirmed, classification of the substance with respect to acute toxicity is not required. This option/approach is described in detail in Appendix R.7.4–1.

In the Up-and-Down Procedure (OECD TG 425), where individual animals are dosed sequentially, estimation of the oral LD_{50} with a confidence interval is possible and this can be used for classification purposes. Data generated in the fixed dose/concentration procedures (EU B.1 bis / OECD TG 420, draft OECD TG 433 (under drafting) and draft OECD TG 434 (under drafting)) and the acute toxic class methods (EU B.1 tris / OECD TG 423 and EU B.52 / TG 436) are equally sufficient for classification purposes. In the fixed dose/concentration procedures, the discriminating dose is identified as the dose causing evident toxicity but not mortality, and must be one of the four dose levels specified in the test method. Evident toxicity is a general term describing clear signs of toxicity such that at the next highest dose level, either severe pain and enduring signs of severe distress, moribund status or probable mortality can be expected in most animals. In the acute toxic class methods, the range of exposure where death is expected is determined by testing at one or more of the four fixed doses. The OECD and EU guidelines for fixed dose procedure and acute toxic class methods include flow charts that allow conclusions to be drawn with respect to GHS classification. In addition the flow charts in the acute toxic class methods allow identification of LD_{50} or LC_{50} cut offs. In the absence of GLP compliant data generated in accordance with OECD or EU methods, all other available information should be considered. Each individual set of data (e.g. a non-GLP study) must be assessed for reliability and relevance as stated in Section R.7.4.4 and any unsuitable data (i.e. that are considered unreliable or not relevant) should be disregarded. When experimental data for acute toxicity are available in several animal species, scientific judgement should be used in selecting the most relevant data from among the valid, well-performed tests. When equally reliable data from several species are available, priority should be given to the data relating to the most sensitive species, unless there are reasons to believe that this species is not an appropriate model for humans (for further details on the preferred test species for evaluation of acute toxicity see Section 3.1.2.2.1 of Annex I to the CLP Regulation and Sections 3.1.2.1.2 and 3.1.2.3.2 of the *Guidance on the application of the CLP criteria*). If definitive classification and labelling cannot be achieved from any individual source, but multiple sets of data all lead to the same conclusion, then, the WoE approach might be sufficient to classify and a robust proposal detailing this should be put forward (see Appendix R.7.4–1).

Where evidence is available from both humans and animals and there is a conflict between the findings, the quality and reliability of the evidence from both sources should be evaluated in order to resolve the question of classification. Generally, data of good quality and reliability in humans should have precedence over other data. However, even well designed and conducted epidemiological studies may lack the sufficient number of subjects to detect relatively rare, but nevertheless important, effects. Also, the interpretation of many studies is hampered by difficulties in identifying and taking account of confounding factors. Positive results from well-conducted animal studies are not necessarily negated by the lack of positive human experience but require an assessment of the robustness and quality of both the human and animal data.

[81] The Test Methods Regulation is regularly updated to follow the approval of new OECD Test Guidelines and was last amended in January 2014 by Commission Regulation (EU) N° 260/2014. Please note that the latest version of an adopted test guideline should always be used when generating new data, independently from whether it is published by the EU or OECD.

If the existing data are contradictory, not concordant or insufficient to reliably determine the appropriate classification and labelling of the substance, additional *in vitro* studies, QSARs, read-across should be considered before conducting any OECD or EU compliant *in vivo* study. In that way such non-animal data could have a supporting role in a read-across or chemical grouping approach. Study data, which permit an assessment of dose-response relationship, should be considered for risk assessment and classification and labelling.

According to Section 3.1.2.3.2 of Annex I to the CLP Regulation, of particular importance in classifying for inhalation toxicity is the use of well-articulated values in the high toxicity categories for dusts and mists. Inhaled particles with an MMAD between 1 and 4 microns will deposit in all regions of the rat respiratory tract. In order to achieve applicability of animal experiments to human exposure, dusts and mists would ideally be tested in this range in rats. The cut-off values indicated in table 3.1.1 of Annex I to the CLP Regulation for dusts and mists allow clear distinctions to be made for materials with a wide range of toxicities measured under varying test conditions.

R.7.4.5.2 Concluding on suitability for Chemical Safety Assessment

For chemical safety assessment, standard EU test method / OECD TG data, as well as all applicable data considered both reliable and relevant, should be used. A quantitative rather than qualitative assessment is preferred to conclude on the risk posed by a substance with regards to acute toxicity dependent on the data available and the potential exposure to the substance during the use pattern/lifecycle of the substance. If quantitative data are not available, the nature and the severity of the specific acute toxic effects can be used to make specific recommendations with respect to handling and use of the substance.

Information on acute toxicity is not normally limited to availability of an LD_{50} or LC_{50} value. Additional information which is important for chemical safety assessment will be both qualitative and quantitative and will include parameters such as the nature and severity of the clinical signs of toxicity, local corrosive or irritant effects, the time of onset and reversibility of the toxic effects, the occurrence of delayed signs of toxicity, body weight effects, dose-response relationships (the slope of the dose-response curve), sex-related effects, specific organs and tissues affected, the highest non-toxic and lowest lethal dose (adapted from ECETOC Monograph No. 6, 1985).

If human data on acute toxicity are available, it is unlikely that they are derived from carefully controlled studies or from a significant number of individuals. In this situation, it may not be appropriate to determine a DNEL from these data alone, but the information should certainly be considered in a WoE assessment and may be used to confirm the validity of animal data. In addition, human data should be used in the risk assessment process to determine (a) DNEL value(s) for particularly sensitive sub-populations like new-born, children or patients with diseases.

For more extensive guidance on the setting of DNELs for acute toxicity, see Appendix R.8-8 of Chapter R.8 of the *Guidance on IR&CSA*.

The effects anticipated from physico-chemical properties and bioavailability data on the acute toxicity profile of the substance must also be considered in the Chemical Safety Assessment.

R.7.4.5.3 Information not adequate

A WoE approach, comparing available adequate information with the tonnage-triggered information requirements of the REACH Regulation, may result in the conclusion that the requirements are not fulfilled.

In the absence of data obtained using approved test guidelines or equivalent methods, data from other endpoints could be helpful for the determination of acute toxicity potential. For example, data could be provided by subchronic toxicity or neurotoxicity studies, as in general the design of these studies includes a pilot study to determine a dose of departure for the main test. In order to proceed with further information gathering the following testing and assessment strategy can be adopted.

R.7.4.6 Testing and assessment strategy for acute toxicity

R.7.4.6.1 Objective / General principles

The main objective of this testing and assessment strategy is to provide advice on how the REACH Annexes VII and VIII information requirements for acute toxicity can be met using the most humane methods. If the strategy is followed, the information generated will be sufficient to make a classification decision with respect to acute toxicity hazard and may provide data for the risk assessment and DNEL derivation. In addition, assessment of acute toxicity may provide information that is valuable for the conduct of repeated dose toxicity studies, such as identification of target organ toxicity and dose selection.

By adhering to the criteria outlined in the previous sections, informed decisions may be made on whether sufficient data already exist to cover the objectives, or whether further testing is required.

If further testing is deemed necessary, the use of the most appropriate study in accordance with the REACH Regulation is considered rather than a *one study fits all* approach. An overarching principle is that all data requirements are met in the most efficient and humane manner so that animal usage is avoided or minimised, whenever feasible, and that costs are minimised.

R.7.4.6.2 Preliminary considerations

The standard information requirements for acute toxicity under the REACH Regulation are given in Section R.7.4.2.

According to column 2 of Section 8.5 in REACH Annexes VII and VIII, information requirements for acute toxicity studies can be adapted if the substance is classified as corrosive to the skin, so as to avoid unnecessary testing and suffering of animals. However, if there are health concerns regarding exposure to non-corrosive concentrations, i.e. if there is a suspicion of systemic toxicity e.g. from structural alerts indicating that the substance exhibits both corrosivity and systemic toxicity, then acute toxicity assessment may be considered appropriate. In such cases, a specific protocol should be developed as standard LC_{50} or any other *in vivo* acute toxicity testing cannot be performed. For example, *in vitro* data on basal cytotoxicity could be used to establish the most appropriate range of concentrations to be tested.

Regardless of the tonnage level, before any testing is triggered, careful consideration of existing toxicological data and current risk management procedures is recommended to ascertain whether the fundamental objectives of the strategy have already been met. This consideration should take account of discussions that have taken place under other regulatory schemes, such as CLP, BPR, including earlier regulatory schemes such as the Existing Substances Regulation (EEC) No 793/93, and the EU hazard classification scheme. If it is concluded that further testing is required, then a series of decision points are defined to help shape the scope of an appropriate testing programme.

The following four-stage process has been developed for clear decision-making:

- **Stage 1**: gather existing information according to Annex VI;
- **Stage 2**: consider information needs according to the relevant Annex(es) VII to X;
- **Stage 3**: identify data gaps (and adequacy of all available data for classification and labelling and/or risk assessment, or to fulfil the criteria for adaptation of REACH information requirements);
- **Stage 4**: generate new data.

R.7.4.6.3 Testing and assessment strategy for acute toxicity (see Figure R.7.4–1)

Stage 1. Gathering of existing information

The starting point of the strategy is the review of existing data (e.g. human or animal data, physico-chemical properties, (Q)SARs, *in vitro* test data, read-across). For non-corrosive substances, the results of skin and eye irritation and skin sensitisation studies (Annex VII) may provide useful information on the potential for systemic toxicity. However, new *in vivo* tests on these endpoints should not be carried out **solely** for the purpose of obtaining information on the acute toxicity potential of a substance.

All existing human and test data (e.g. from clinical reports, poisoning cases, animal studies, corrosivity, physico-chemical properties) should be considered. Some information from the existing data e.g. *in vitro* studies (*de novo in vitro* basal cytotoxicity and dermal penetration studies), systemic effects observed in other studies, route of human exposure, physico-chemical properties, dermal or respiratory toxicity of structurally-related substances, might primarily be used for the selection of either an acute *in vivo* inhalation test or an acute *in vivo* dermal test.

Section R.7.4.3 presents a detailed discussion of the sources that may provide relevant information for the assessment of acute toxicity.

Stage 2. Considerations on information needs

A detailed evaluation of the existing information collated in Stage 1 is conducted to allow an informed decision on the testing needs to fulfil the REACH requirements. It is important to ensure that the available data are relevant and reliable to fulfil these requirements.

It should be noted that if a substance is predicted to be corrosive then further consideration should be given as to whether or not an acute test can be justified (in particular in relation with animal welfare considerations). Justifications for conducting a study must be provided in order to minimise the animal use. If the substance is considered to be corrosive, no acute toxicity testing should normally be conducted (see Section R.7.4.6.2). Where information on corrosivity is not available then *in vitro* corrosivity tests should be conducted first.

The standard information requirements for acute toxicity under the REACH Regulation are given in Section R.7.4.2.

When acute toxicity *via* a second route is required (i.e. at Annex VIII and above), the choice of the second route (dermal or inhalation) depends on the nature of the substance and the likely route of human exposure. However, information on only one route of exposure may be sufficient and justified based on physico-chemical, toxicokinetic or human data and review of all possible exposure scenarios. For example, with gases only the inhalation route could be evaluated as no relevant human exposure may occur by the oral or dermal route. For liquid substances with high viscosity, no testing by the inhalation route should be conducted.

If human exposure is possible *via* inhalation, or if physico-chemical properties indicate that such an exposure may occur, then testing for acute toxicity *via* this route should be conducted. Data from skin/eye irritation, skin sensitisation and acute oral toxicity should be used as indicators to help testing *via* inhalation (for example whenever possible, exposure concentrations should be chosen so that respiratory irritation is avoided). If no systemic effects are shown during acute oral testing, then the requirement to conduct inhalation testing should be considered on a case-by-case basis.

Consideration of the need for assessment of acute dermal toxicity should be given if the inhalation route is not considered appropriate and the conditions described in Column 2 of Section 8.5.3 of Annex VIII for waiving acute dermal toxicity testing are not met. In some cases, it may be possible to draw conclusions about the potential for acute dermal toxicity without further testing, on the basis of the data available from acute oral toxicity and/or dermal absorption studies. Evidence for the potential of high dermal absorption should be considered on a case-by-case basis taking into account physico-chemical properties e.g. Log K_{ow}, water solubility, molecular weight and melting point of the substance. Testing for acute dermal toxicity is indicated if:

- Systemic toxicity is observed in skin/eye irritation and/or skin sensitisation studies;
- Death is observed in an acute oral toxicity test and there is potential for dermal absorption;
- Systemic toxicity is observed in an acute oral toxicity test and there is a potential for high dermal absorption (determined following e.g. EU B.45 / OECD TG 428);
- There is a potential for high dermal exposure (case-by-case basis).

Conversely, testing for acute dermal toxicity should not be conducted if:

- the substance does not meet the criteria for classification for acute toxicity or STOT SE by the oral route, and
- no systemic toxicity is observed in *in vivo* studies with dermal exposure (e.g. skin irritation, skin sensitisation) or, in the absence of an *in vivo* study by the oral route, no systemic effects after dermal exposure are predicted on the basis of non-testing approaches.

Stage 3. Identification of data gaps / adequacy of data

The purpose of this step is to identify what additional information is required in order to classify the substance and to perform a risk assessment. For those substances for which the available data suggest low toxicity, the WoE-based adaptation described in Appendix R.7.4–1 should be considered.

The available information may include data generated using study protocols that differ from the standard regulatory test methods. The evaluation should include whether the available information meets or exceeds the data requirements from standard regulatory study protocols. Therefore it may be possible that the tonnage-driven minimum information needs can be met through combined data obtained from several sources.

At this stage, it is also necessary to verify if the available information is adequate for hazard characterisation. For this process, all relevant information should be taken into account in a WoE assessment. Quantitative data on the dose-response relationship for the critical toxicological effects and/or estimations of either the LC_{50}/LD_{50} values or the Discriminating Dose will be important for assessing the hazard classification and can be used in risk assessment. Information from the testing for other toxicological endpoints (e.g. repeated dose toxicity) may also be useful for risk assessment (see also Appendix R.8-8 of Chapter R.8 of the *Guidance on IR&CSA*). Mathematical modelling should be considered for estimating a threshold exposure level (e.g. benchmark dose), as an alternative to generating additional *in vivo* data.

For the inhalation route, standard protocols involve a 4-hour exposure. If data for other time periods are available (e.g. for 30 min to 8 hours), extrapolation to a 4-hour exposure period can be achieved using a modification of Haber's Law ($C^n \times t = k$). If this n value is not available from the literature, a default value may be used; it is recommended to set $n = 3$ for extrapolation to shorter duration than the duration for which the LC_{50} or EC_{50} was observed and to set $n = 1$ for extrapolation to longer duration (ACUTEX project, 2006). Experimentally, the value of n can be determined using the C x t approach (OECD TG 403).

CLP criteria also include criteria for conversion of existing inhalation toxicity data which have been generated using a 1-hour exposure (for further details see Note (c) to Table 3.1.1 of Annex I to the CLP Regulation and Section 3.1.2.2 of the *Guidance on the application of the CLP criteria*).

If the data and subsequent decisions are deemed consistent with an adequate hazard characterisation and are sufficient to classify the substance or to conduct a risk assessment, then no further testing for acute toxicity is necessary.

In some cases, the substance may be excluded from acute toxicity testing if it does not appear as scientifically necessary. This might be the case for example if:

- A WoE analysis demonstrates that the available information is sufficient for an adequate hazard characterisation and the exposure to the substance is adequately controlled;
- The substance is not bioavailable via a specific route and possible local effects of the substance are adequately characterised (example, no dermal absorption for dermal route);
- For the inhalation route, no testing is required if it is not technically possible to generate an atmosphere suitable for testing, e.g. because the vapour pressure is very low.

Finally, the conclusion that no further testing is required may be reached when the data meet the requirements for classification for toxic effects or if the substance has already been classified for acute toxic effects.

Where evidence is available from both existing human data and animal tests and there is a conflict between the findings, the evidence should be evaluated in order to understand the toxicological basis for these diverging findings. Issues relating to the quality and reliability of the data should also be taken into account. Generally, data of good quality and reliability in humans should take precedence over other data. However, well-designed and conducted epidemiological studies may lack a sufficient number of subjects to detect relatively rare but still significant effects or to assess potentially confounding factors. Positive results from well-conducted animal studies are not necessarily negated by the lack of positive human experience but require an assessment of the robustness and quality of both the human and animal data.

Stage 4. Generation of new data

If the data considered at stage 3 are contradictory, not concordant or insufficient to determine reliably the appropriate classification and labelling of the substance, additional *in vitro* studies, QSARs and/or read-across should be considered before conducting any OECD compliant *in vivo* study. Study data that allow an assessment of the dose-response relationship should be considered particularly valuable for risk assessment purposes.

If data gaps still need to be filled, new data must be generated (Annexes VII and VIII to the REACH Regulation). Due to animal welfare considerations, new tests on animals should only be performed as a last resort, when all other sources of information have been exhausted.

Internationally adopted test methods for acute toxicity are described in the Annex to the EU Test Methods Regulation (Council Regulation (EC) No 440/2008)[82] and in OECD TGs (available at http://www.oecd.org/env/ehs/testing/oecdguidelinesforthetestingofchemicals.htm). These standard guidelines should normally be used as they provide the necessary information on acute toxicity hazard in a way that balances the need to protect human health with animal welfare concerns (see Section R.7.4.3 and the above guidance for Stage 3).

The route(s) of exposure to be used for acute toxicity evaluation depend(s) on the nature (e.g. gas or not, molecular weight, log K_{ow}) and use of the substance and should reflect the most likely route(s) of human exposure. If any specific human exposure may be identified, further testing for risk assessment should be considered as proposed in REACH Annex VIII, Section 8.5. If exposure by inhalation is likely, then the testing strategy by inhalation should be proposed (Figure R.7.4–2).

The first considerations should aim at defining the potential of the substance for acute toxicity. In that respect, information may be provided by existing data from SARs, QSARs, chemical categories approaches and available *in vitro* and *in vivo* data. If no potential for acute toxicity is shown, then no further testing is required and a decision on classification can be taken. Such information may also provide relevant information for risk assessment considerations. This approach, which is based on evidence of low/no acute oral toxicity (without performing the relevant *in vivo* test according to REACH Annex VII, 8.5), should be documented in a WoE analysis as explained in Appendix R.7.4–1. For this specific WoE case, the sub-acute oral toxicity study is crucial and should usually be available in order to reach a definitive conclusion.

Following the general testing strategy, dose selection, including a decision to perform only a limit test, appears to be an important aspect in order to select the most appropriate starting point. When validated *in vitro* tests are available, they may provide relevant results and help in the dose selection for oral route testing (see Section R.7.4.4.1).

For substances in the ≥10 t/y tonnage band, testing by the dermal route should be considered if (i) a human exposure is identified, (ii) results from physico-chemical properties and in particular skin irritation/sensitisation tests show any dermal absorption or any systemic toxicity, and (iii) the conditions described in Column 2 of Section 8.5.3 of Annex VIII for waiving acute dermal toxicity testing are not met. Depending on such information, dermal testing should be conducted or not following standard protocols (see Section R.7.4.3).

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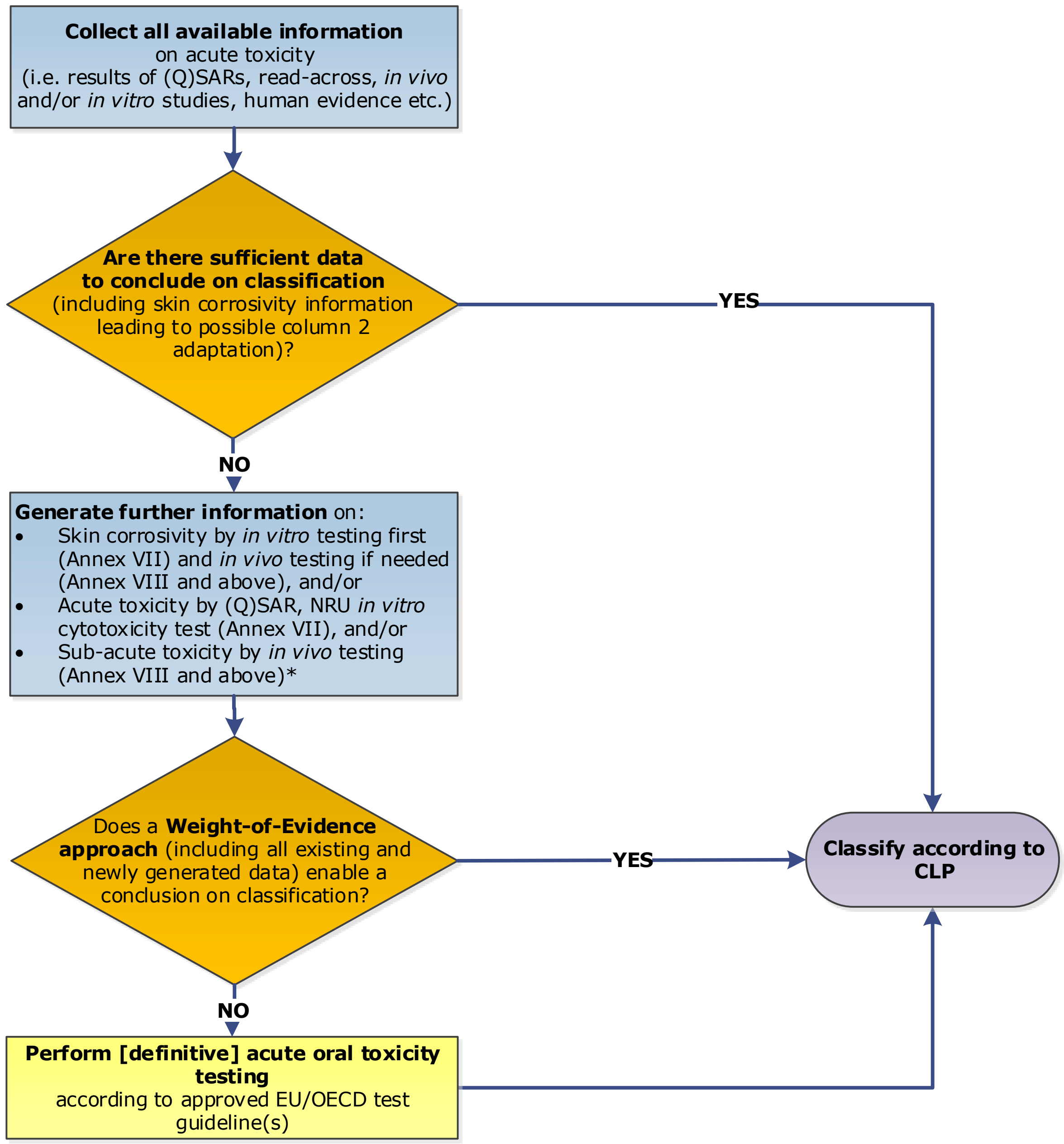
82 The Test Methods Regulation is regularly updated to follow the approval of new OECD Test Guidelines and was last amended in January 2014 by Commission Regulation (EU) N° 260/2014. Please note that the latest version of an adopted test guideline should always be used when generating new data, independently from whether it is published by the EU or OECD.



* A sub-acute toxicity study is only required at REACH Annex VIII and above. Generation of further information is not a requirement, but can be done on a voluntary basis in case the registrant decides to use a WoE approach.

Figure R.7.4–1 Testing and assessment strategy for acute oral toxicity (REACH Annexes VII and VIII).

Section 1 of Annex XI to the REACH Regulation is the basis for the proposed adaptations of the standard information requirement for an acute oral toxicity study and should be consulted for further details on the conditions of application of the general adaptation rules to the different steps of this strategy.

A specific testing and assessment strategy is proposed for the **selection of additional routes of exposure** for acute toxicity testing (Figure R.7.4–2).

Regarding the **inhalation route**, primary considerations should be based on the (in)ability to generate a suitable testing atmosphere, depending on the physico-chemical properties of the substance (for example, low volatility, solid, particle size >100 μm (see also Section R.7.4.4.1)). In case an inhalable testing atmosphere cannot be generated, no human exposure may be identified and no further testing is required.

Wherever possible, assessment of acute inhalation toxicity should be conducted in accordance with the draft OECD TG 433 (under drafting) and EU B.52 / TG 436 since they have been designed to use less animals than the EU B.2 / OECD TG 403. In addition, the draft OECD TG 433 does not require mortality as endpoint. However, in some circumstances, i.e. if a dose-response curve is needed for risk assessment purposes, testing according to EU B.2 / OECD TG 403 may be considered appropriate (see also the OECD Guidance Document 39 (OECD, 2009)).

Regarding the **dermal route**, acute toxicity *via* this route should be assessed if the inhalation route is not considered relevant, human dermal exposure is likely and dermal absorption or systemic toxicity *via* this route can be predicted or demonstrated.

Wherever possible, assessment of acute dermal toxicity should be conducted in accordance with the EU B.3 / OECD TG 402. However, before any *in vivo* study is envisaged, the registrant must check whether the conditions described in Column 2 of Section 8.5.3 of Annex VIII for waiving acute dermal toxicity testing are met, i.e. a column 2 adaptation can be justified if:

- The substance does not meet the criteria for classification for acute toxicity or STOT SE by the oral route, and
- No systemic effects have been observed in *in vivo* studies with dermal exposure or, in the absence of an *in vivo* study by the oral route, no systemic effects after dermal exposure are predicted on the basis of non-testing approaches.

If based on the above the dermal route is not considered relevant or the acute dermal toxicity study can be waived, no further testing nor classification for acute dermal toxicity is required.

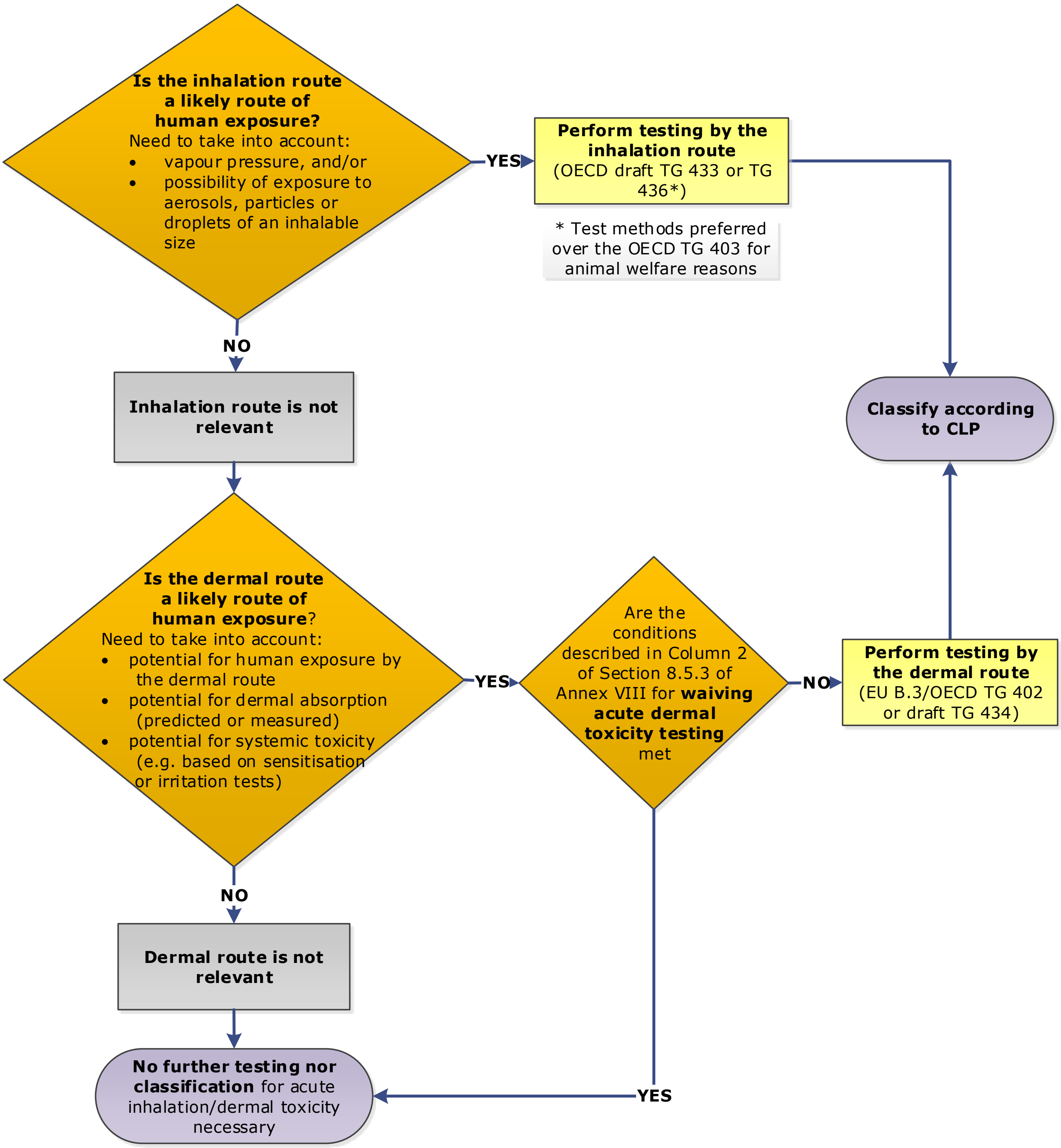


Figure R.7.4–2 Selection of (an) additional route(s) of exposure for acute toxicity testing (REACH Annex VIII) (see also the OECD GD 39 (OECD, 2009)).

Please note that draft test guidelines are also included in this figure (for further details on the status of development of these draft test guidelines, see Section R.7.4.3.1.2 "Testing data on acute toxicity").

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Appendices R.7.4-1 to 2 to Section R.7.4

Appendix R.7.4–1 Weight-of-Evidence based adaptation to the standard information requirement for an acute oral toxicity study

The aim of this Appendix is to advise the registrants on how they can perform an *in vivo* acute toxicity study only as a last resort. An *in vivo* acute oral toxicity study can potentially be avoided, if a registrant has relevant data, which are used in a Weight-of-Evidence (WoE) approach. In cases where the WoE adaptation leads to the assumption of low/no expected acute oral toxicity (> 2000 mg/kg bw/d), the registrant can avoid unnecessary animal testing pursuant to REACH Articles 13(1) and 25(1). The description of the "elements of evidence" that can be included in a WoE case, is the main scope of this Appendix.

To use the WoE approach described below, the registrant should perform a sub-acute toxicity study **before** acute toxicity testing, and in case the test substance is shown to be of low toxicity, he should eventually use the results of the 28-day study to waive the acute oral toxicity study.

1. Scope of the WoE adaptation

Acute oral toxicity is one of the standard information requirements in Annexes VII-X.

An alternative to performing the acute oral *in vivo* acute toxicity test is outlined in this Appendix. Its aim is to reduce the number of animal studies needed and the cost of testing by proposing a WoE adaptation, according to REACH Annex XI, section 1.2.

Furthermore, information on repeated dose toxicity (RDT), e.g. no mortality during days 0–3 of an RDT study, may be relevant for acute toxicity and can be useful in supporting classification and labelling for acute systemic toxicity (see Section R.9.2.5.2 of the *Guidance on the application of the CLP criteria*).

The scope of the WoE based adaptation outlined below is the following:

- The WoE approach is mainly meant for substances to be registered at **Annex VIII** tonnage level and above (i.e. registrations at >10 tpa), for which an oral sub-acute toxicity study (OECD TG 407) or the combined RDT study with the reproduction/developmental toxicity screening test (OECD TG 422) is required;
- The type of adaptation described below could be used, independently of the tonnage band, in case a sub-acute toxicity study is available;
- The WoE approach is intended for substances of low acute toxicity, i.e. for substances with an $LD_{50,\ oral}$ expected to be greater than 2000 mg/kg bw, or where other data that may be available indicate low toxicity.

These and other limitations are described in the specific sections of this Appendix in more detail.

The background and rationale of this guidance for a WoE-based adaptation for the acute oral toxicity study are provided in Appendix R.7.4–2.

There are several types of studies and information that can be used in the characterisation of the acute oral toxicity of a substance. The types of information, which are presumably of high value in the prediction of the acute oral toxicity, have been included in this Appendix.

The WoE approach outlined below is one of many adaptation possibilities, which are available to registrants under REACH. If this approach is used, it should consist of more than one of the following elements of evidence[83] and it has to include in all cases a 28-day RDT study, as the most valuable and essential part of the WoE approach proposed[84]:

1. Results of a 28-day RDT study *via* the oral route (i.e. a sub-acute study)[84]; and
2. Results of (a) dose range finding (DRF) study/ies, which can be supplemented with relevant clinical observations during the first day of dose administration, which would provide valuable information; or
3. Data from an NRU (Neutral Red Uptake) *in vitro* study for cytotoxicity (or equivalent); according to the ECVAM recommendation (EURL ECVAM, 2013). The NRU cytotoxicity assay predicts well substances of low acute oral toxicity (i.e. not classified for acute toxicity); or
4. (Q)SAR results which may provide information on the acute oral toxicity; or
5. Data on such physico-chemical properties of the substance, which inform on the bioavailability or the reactivity of the substance, and/or which can contribute to the assessment scheme and/or to the grouping approach; or
6. Other supporting evidence, such as justified read-across information, results from mechanistic and/or tissue-based *in vitro* studies, e.g. addressing neurotoxicity or human data.

These elements of evidence, which are addressed in detail in the next sections, can be examined and considered by the registrants to adapt the standard information requirement for an oral *in vivo* acute toxicity test for their substances.

This Appendix also provides guidance on how to obtain and assess these different elements of evidence.

Finally, two "decision-trees" for the WoE assessment, with different starting elements are outlined in Figure R.7.4–3 and Figure R.7.4–4 of this Appendix.

2. Prediction of acute oral toxicity based on the results of a sub-acute oral toxicity study

2.1. Introduction

The WoE approach for the Annex VIII substances with tonnage > 10 tpa has to include data on oral sub-acute toxicity. An analysis initiated by JRC (Graepel *et al.*, 2016) and then continued by ECHA (see Appendix R.7.4–2) has shown that, for **substances of low toxicity**, the prediction of **acute oral toxicity** classification can be based on the data from oral **sub-acute studies** in most cases. In particular, the non-classification for oral acute toxicity (i.e. the substance is not to be classified if the LD_{50} is above 2000 mg/kg bw) can be correctly predicted

[83] The requirement of obtaining and reporting more than one piece of evidence within the WoE follows from the provisions of REACH Annex XI, 1.2.

[84] Also a 90-day study, when it provides a NOAEL at or above 1000 mg/kg bw, can be used as an element of the WoE approach.

based on the results of oral sub-acute studies, when the NOAEL[85] is at or above 1000 mg/kg bw.

In this Appendix, the term "low toxicity" is used for substances which have an $LD50_{acute,oral}$ greater than 2000 mg/kg bw and a $NOAEL_{sub\text{-}acute,oral}$ of 1000 mg/kg bw or greater, derived from an RDT study with a duration of at least 28 days.

A quantitative correlation between acute oral toxicity and sub-acute oral toxicity across the whole range of toxicity (i.e. from low toxic to severely toxic substances) was also examined, but the results have not been promising.

Therefore the scope of the present WoE approach is explicitly for the substances of **low toxicity**, and relies on a "limit test" dose for repeated dose toxicity studies (i.e. NOAEL ≥ 1000 mg/kg bw) and the classification threshold applied for the acute oral toxicity in the EU (i.e. > 2000 mg/kg bw).

2.2. Conclusion on the use of an existing sub-acute oral toxicity study to adapt the acute oral toxicity study requirement

Where registrants hold an existing sub-acute oral toxicity study, the results of which indicate that the substance falls within the scope of this WoE approach, the prediction of the acute oral toxicity potential may be used as an element of a WoE adaptation (pursuant to the REACH Annex XI, 1.2). This approach supports registrants in fulfilling their obligation under REACH Article 13(1).

Based on this prediction, and other pieces of evidence, registrants may also conclude that the classification and labelling for acute toxicity is not warranted.

Since this prediction focuses on substances of low toxicity, it is important to note the following limitations:

- The WoE cannot be used for any substance for which the results of a sub-acute oral toxicity study resulted in a NOAEL below 1000 mg/kg bw. A quantitative analysis made by JRC has shown that the correlation between the sub-acute and acute toxicity across the whole range of NOAELs and LD_{50} values is poor (Bulgheroni *et al.*, 2009).
- The WoE approach cannot be proposed if no sub-acute oral toxicity study (OECD TG 407 or TG 422) has been performed.
- The WoE cannot be used for any substance that requires the GHS classification as "acute toxicity category 5"[86] (i.e. where the $LD50_{acute,oral}$ is higher than 2000 mg/kg bw and lower than 5000 mg/kg bw).

[85] A Maximum Tolerated Dose (MTD) could also in principle be used as a measure of toxicity. However, an MTD is not regularly provided for the sub-acute toxicity studies in the REACH registration dossier, whereas a NOAEL is provided. It should be noted that the present WoE approach was developed using NOAEL values (see Appendix R.7.4–2). Therefore the prediction model described in this Appendix is based on the use of a NOAEL.

[86] The GHS "Acute toxicity category 5" classification may be needed for some countries outside the EU. In the EU, category 5 classification is not required.

2.3. Use of a novel dose range finding study and of a novel sub-acute toxicity study

When registrants do not hold a (valid) sub-acute oral toxicity study for substances manufactured or imported at tonnage > 10 tpa, they will need to perform a novel study to fulfil the legal requirements at Annex VIII (Section 8.6.1).

2.3.1. Dose-range-finding (DRF) studies

Before a novel sub-acute oral toxicity study (OECD TG 407 or OECD TG 422) is conducted, appropriate doses must be identified. For this purpose, the registrant should use existing data (e.g. screening studies, acute toxicity studies, literature data) and relevant results from validated *in vitro* tests, and only if all those data are insufficient will he need to perform one or more dose-range-finding studie(s) (DRF(s)). Under the section on "Dosage", the OECD TG 407 stipulates that "*If there are no suitable data available, a range finding study (animals of the same strain and source) may be performed to aid the determination of the doses to be used.*" Furthermore, DRFs are standard practice followed by contract research organisations (CROs).

DRF1

If virtually nothing is known about the substance, the first part of the DRF study (pilot study) may consist of a single administration of one dose to 2 animals (1 male and 1 female) and subsequently, depending on the reaction of the animals, with single administrations of lower or higher doses to additional animals. Thus one gets some preliminary information on the acute toxicity of the substance.[87]

Investigations are normally restricted to cage-side observations for signs of toxicity and gross necropsy in an attempt to identify target organs. Normally, the frequency of observations is several times on the first day, then once or twice a day. The observation period is typically limited to 7 days after administration.

DRF2

Having found the highest dose which will most probably not lead to the death of the animals after repeated administration of the test substance, a second DRF study is usually performed by administering 3 or 4 different doses to groups of 3-5 animals per sex, once daily, for one week (7 days). Investigations include body weight development, cage-side observations and possibly also clinical observations. The frequency of cage-side observations is normally twice to four times on the first day, then twice a day for 7 days. At the end of the administration period, gross necropsy is performed, but no histopathology or clinical chemistry or haematology is undertaken.

Based on these findings the doses for the main study are selected.

Please note that data generated in DRF2 is not considered as valuable as the results of an enhanced DRF1 and therefore, DRF2 is not a recommended element in the WoE described in this Appendix.

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[87] This WoE approach is primarily meant for cases where no acute toxicity study, nor repeated dose toxicity (28-day) study are available. In those cases, one possible starting point is to perform an "enhanced" DRF1, to find out whether the substance is of low toxicity and to identify appropriate doses for the 28-day study. There can be cases where the acute toxicity study could be an indicator for the appropriate dose range to use in the 28-day study. It is anticipated that these cases will not be many, because DRFs normally have a longer duration of exposure than the acute oral study and thus, usually LD_{50} cannot replace the DRF. Furthermore, if the NOAEL is below 1000 mg/kg bw in the DRF, the substance is considered to not fall within the scope of the proposed WoE approach. Consequently the substance is likely to be acutely toxic. Therefore, in most cases, using the DRF1 will not lead to using more animals, as compared to the conventional way of testing (i.e. first LD_{50}, then DRFs as necessary and finally the 28-day study).

Contribution of DRFs to the WoE

The advantages of using **DRF1** as one element in this WoE approach are that (i) only a low number of test animals is needed and (ii) high doses up to 2000 mg/kg bw may be administered (in most EU member states). Furthermore, more frequent observations of the signs of toxicity can be relatively easily arranged for, to obtain valuable information on whether animals dosed with up to 2000 mg/kg bw survive without showing signs of toxicity.

DRF2, on the other hand, provides data on toxicity after repeated exposure. As the doses may be higher than in the main study, some additional information on acute toxicity may be gained.

The information will be most valuable if animals which are dosed up to 2000 mg/kg bw survive without showing signs of toxicity. However it should be noted that the NOAEL derived from a 7-day toxicity study (DRF2) cannot be used as stand-alone for the sub-acute oral toxicity requirement.

2.3.2. Enhanced DRF1

To enhance the information provided by the DRF1 tests, the frequency of the clinical observation needs to be adjusted for the first day of DRF1 to the scheme of the acute oral toxicity test guidelines.

The observation period may be prolonged to a total of 14 days after the administration of the test substance, so that "*animals are observed individually after dosing at least once during the first 30 minutes, periodically during the first 24 hours, with special attention given during the first 4 hours, and daily thereafter, for a total of 14 days*" (EU B.1 bis / OECD TG 420, Acute Oral Toxicity – Fixed Dose Procedure, Adopted in 2001).

The clinical observations during the enhanced DRF1 (type and level of details) should follow the ones specified in the OECD acute oral toxicity test guidelines (Table R.7.4-2).

Notes: Registrants are reminded that the DRF1 observations (mostly from the first day) should be reported separately as an endpoint study record under the Acute Toxicity Endpoint in the IUCLID dossier (section 7.2.1). The observations and the findings made in the enhanced DRF1 should be submitted with the registration dossier, as part of the WoE documentation.

It is acknowledged that, in some CROs, the practice of performing the DRF1 may be different from the one recommended in this Appendix. Furthermore, in some EU Member States, for animal welfare considerations, a dose of 1000 mg/kg bw cannot be exceeded. If a short duration of observation (omitting the 14-day recovery period) and the low dose limit (i.e. 1000 mg/kg bw) are used in the DRF1, the information obtained from it will be of less value in the context of the WoE adaptation presented here. Where the registrant can choose to perform the DRF1 following the recommendations given below, the results obtained are of higher value in the WoE analysis. Note that Figure R.7.4–4 may also be applied when a non-enhanced DRF1 is available. In that case, DRF1 is one of the "Additional elements of evidence within the WoE".

Costs of additional steps

Together with CROs' experts, ECHA has estimated that, compared to a typical DRF1 study, additional costs would be generated from (i) approximately 1-3 hours of extra time for observations and recording, and (ii) the housing of the animals for 14 days after administration (as opposed to 7 days). It is therefore anticipated that the cost increase of the enhanced DRF1 study will be limited.

2.3.3. Main study: the sub-acute oral toxicity study

The main study, i.e. the sub-acute oral toxicity study, and the screening studies (see Table R.7.4-2 below) provide data on toxicity after repeated exposure. However, information on acute oral toxicity may also be gained from that study.

The obvious advantage of the main study is that its results will be valuable for the WoE approach, in case the NOAEL is at or above 1000 mg/kg bw/day (see Section 1 of this Appendix).[88]

The schedule of observations and the scope of clinical observations in the acute and sub-acute oral toxicity studies are summarised in Table R.7.4-2, according to the relevant paragraphs of the relevant OECD test guidelines.

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[88] It is noteworthy that the 1000 mg/kg bw dose is not the definite upper threshold for a 28-day repeated dose study and that higher doses can be applied, if deemed useful, e.g. for deriving DNELs. The main study (sub-acute oral toxicity study) is understood as resulting from performing the test under the OECD TG 407 or OECD TG 422. Regarding the results of an OECD TG 422 study, it is important to note that the NOAEL used refers to the maternal/paternal toxicity, and not to the NOEL for developmental effects.

Table R.7.4-2 Comparison of the general clinical observations as required by the OECD test guidelines for acute oral toxicity and sub-acute oral toxicity and the proposed schedule of observations in the enhanced DRF1 study.

OECD Test Guideline	**Day 1**		**Days 2-14 (acute and enhanced DRF1)** **Days 2-28 (RDT)** **Days 2-7 (DRF2)**
	At 30 min	**At 4 hour + periodically until 24 hrs**	**Daily**
TG 420 (Fixed Dose Procedure), **TG 423 (Acute Toxic Class method),** **TG 425 (Up-and-Down-Procedure)**	Animals are observed individually, at least once	Animals are observed individually, with special attention given during the first 4 hours	Animals are observed individually
	Additional observations* necessary if animals continue to show signs of toxicity		
Enhanced DRF1 for TG 407[89]	Animals are observed individually, at least once	Animals are observed individually, with special attention given during the first 4 hours	Animals are observed individually
TG 407 or TG 422 (Repeated dose oral toxicity study)[90]	• General clinical observations at least once a day • Morbidity/ mortality at least twice daily		
DRF2 for TG 407	• General clinical observations at least once a day • Morbidity/mortality at least twice daily		• General clinical observations at least once a day • Morbidity/ mortality at least twice daily

* (At least) changes in skin and fur, eyes and mucous membranes, and also respiratory, circulatory, autonomic and central nervous systems, and somatomotor activity and behaviour pattern. Attention should be directed to observations of tremors, convulsions, salivation, diarrhoea, lethargy, sleep and coma.

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[89] Enhancement of the DRF1 means that the observation schedule is identical to the one in the acute toxicity test and that observation lasts for 14 days.

[90] There are parameters/observations that are common to both sub-acute and acute toxicity tests, e.g. signs of toxicity, body weight/body weight changes, necropsy. However, there are other parameters/observations that are routinely recorded in a sub-acute toxicity test, but not in an acute toxicity test, such as clinical biochemistry and haematology. In some cases, the effect(s) of a substance on these "common" parameters/observations may be used to determine the NOAEL, whereas the "non-common" parameters/observations would be affected at a higher dose level. In these cases, a NOAEL lower than 1000 mg/kg bw might allow prediction of acute toxicity (although this has not been looked at in the current IUCLID-based analysis used to develop the present WoE approach). However, these cases are only likely to be a few, since the parameters/observations recorded only in a sub-acute toxicity study are usually more sensitive than the "common" parameters, i.e. likely to be affected at dose levels lower than 1000 mg/kg bw. Moreover, a prediction model that is based on the NOAEL of the sub-acute toxicity study as such is simpler to apply than a model that would require/advise the registrants to consider all the parameters/observations at each dose level and make their prediction of acute toxicity based on these.

As part of the OECD TGs 420, 423, 425 and enhanced DRF1 (i.e. during the general clinical observations), *"the duration of observation should not be fixed rigidly. It should be determined by the toxic reactions, time of onset and length of recovery period, and may thus be extended when considered necessary" […] "The times at which signs of toxicity appear and disappear are important, especially if there is a tendency for toxic signs to be delayed […]. All observations are systematically recorded, with individual records being maintained for each animal."* In addition *"[T]he principles and criteria summarised in the Humane Endpoints Guidance Document should be taken into consideration […] Animals found in a moribund condition and animals showing severe pain or enduring signs of severe distress should be humanely killed. When animals are killed for humane reasons or found dead, the time of death should be recorded as precisely as possible."* (extracted from the OECD TG 420: Acute oral toxicity study, Fixed Dose procedure, paragraphs 27 and 28, as an example for OECD TGs 420, 423 and 425).

When internal exposure information is available (i.e. ADME studies), kinetic parameters (such as C_{max}, T_{max}, $AUC_{0\text{-}t}$, non linearity ranges, etc…) can be taken into account in determining the dosing and intervals for clinical observations.

2.4. Conclusion on the use of the novel DRFs and sub-acute oral toxicity study to adapt the acute oral toxicity requirement

When a sub-acute oral toxicity study is not available and the registrant generates a novel study, it is recommended that the registrant performs an enhanced DRF1 study as proposed in Table R.7.4-2. If no signs of toxicity are seen in the enhanced DRF1 and if the main sub-acute toxicity study falls within the scope of this WoE approach (i.e. NOAEL ≥ 1000 mg/kg bw), this prediction may be used to justify that the performance of a novel acute oral toxicity test is not scientifically necessary (pursuant to REACH Annex XI, 1.2). In this case, the two main elements of the WoE are the enhanced DRF1 and the main sub-acute toxicity study. Consequently the registrant can propose to not classify the registered substance for acute oral toxicity (Figure R.7.4–3). This approach also supports registrants in fulfilling their obligations under Article 13(1).

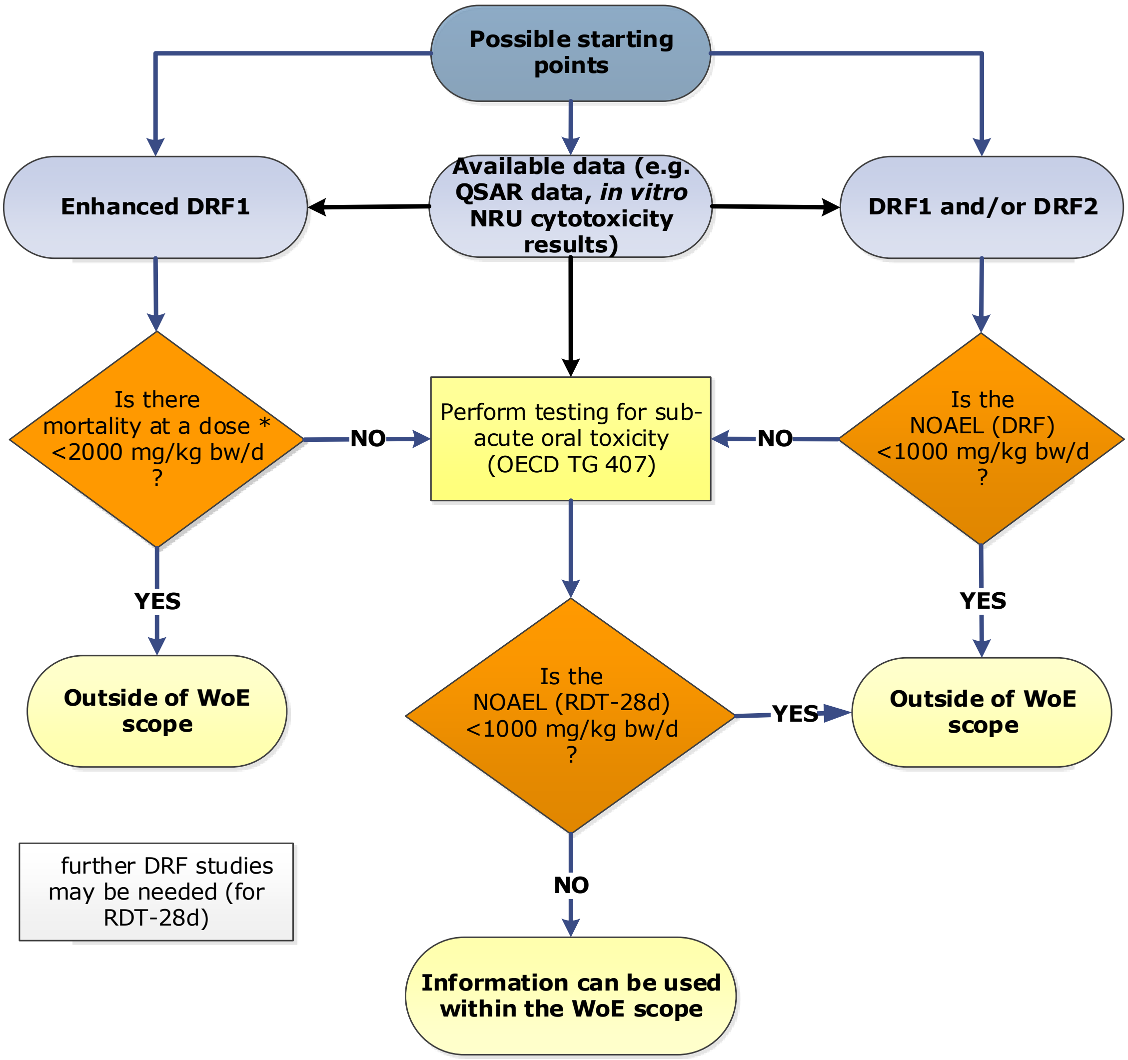

Figure R.7.4–3 Decision tree to assess whether an *in vivo* acute toxicity test is required, when the registrant has to generate a novel repeated dose sub-acute oral toxicity study.

Figure R.7.4–3 illustrates cases where data are available from the DRF study(ies) and from a sub-acute oral study and where these data confirm that the substance is of low acute oral toxicity. The Figure also illustrates the situations where the registered substance would fall outside of the scope defined for this WoE approach and where an *in vivo* acute oral toxicity test will therefore be required.

It is acknowledged that registrants may have other data, such as data from a DRF1 study, data from other *in vivo* studies in rats where single doses higher than 2000 mg/kg bw or doses higher than 1000 mg/kg bw for several days have been administered, an NRU cytotoxicity assay (which is currently the only validated *in vitro* cytotoxicity test), a QSAR model or data from human evidence, which provide a conclusion consistent with the one obtained from a 28-day sub-acute study. Registrants may then use these elements of evidence together with the 28-day sub-acute study, instead of using the enhanced DRF1 in their WoE approach (see Figure R.7.4–4).

It is noteworthy that, currently, the observations made in the DRF studies are not standardised, and therefore ECHA provides relevant instructions in Table 1. Furthermore, the 14-day observation period that is included in an acute oral toxicity test is usually not followed in a DRF study for a sub-acute oral toxicity study. This concurs with the need to generate the "Enhanced DRF1" where the observation period is prolonged.

2.5. Conclusion on the regulatory use of the DRF studies for the WoE approach

If an enhanced DRF1 study (with a limited number of animals, typically 2) is used as part of the WoE approach, at least one of the doses applied should be up to 2000 mg/kg bw (or above in case of old studies). The observations should be made according to the scheme outlined in Table R.7.4-2. The enhanced DRF1 provides information which resembles that obtained from an OECD test guideline for acute oral toxicity, but obtained with less animals than recommended in the test guideline. It can therefore not be a replacement for an OECD guideline study, but may be a part of the WoE approach. The (enhanced) DRF1 should be used in the registration dossier with an adequate justification of how this information, when taken together with other WoE elements, meets the specified REACH information requirement.

When an (enhanced) DRF1 study is used within the WoE, two scenarios may occur:

1. There are **no or only transient** signs of toxicity at a dose level up to 2000 mg/kg bw (or above). This evidence could be considered as one element of the WoE to address acute toxicity.

2. There is **mortality or signs of severe toxicity**, leading to interim kills of the test animals, in DRF1 at 2000 mg/kg bw. Therefore the $LD_{50,\ oral}$ of the substance is most probably below 2000 mg/kg bw and the substance does not fit in the scope of this adaptation.

A DRF2 (typically using 3-5 male and 3-5 female animals per dose and an administration period of 7 days) can also be used as a valuable element of the WoE approach, if the highest dose is 1000 mg/kg bw or higher, and if no mortality or signs of severe toxicity leading to interim kills of test animals for humane reasons are observed. No data are available to confirm a correlation between an acute $LD_{50,\ oral}$ > 2000 mg/kg bw and a $NOAEL_{oral}$ ≥1000 mg/kg bw, obtained after only 7 days of administration. Therefore a DRF2 as described above can only be used as one element of evidence in the WoE approach.

In summary, the DRF studies, in particular DRF1, will provide very valuable element(s) of evidence for the WoE approach. Furthermore, there would be no or only limited cost implications, as both DRF1 and DRF2 are usually performed ahead of the 28-day study.

The enhanced DRF1 should be reported as a separate study record under the acute oral toxicity section 7.2.1 in IUCLID.

3. Use of an *in vitro* cytotoxicity assay (Neutral Red Uptake) within the WoE approach

3.1 Introduction

ECHA can accept *in vitro* studies as standalone key studies only if conducted in line with validated and internationally accepted methodologies. Non validated *in vitro* methods can still be used according to the adaptation possibilites described in REACH Annex XI. At the time of drafting of this Appendix, only the *in vitro* NRU cytotoxicity assay can be accepted as part of the proposed WoE approach.

The *in vitro* NRU basal cytotoxicity assay is based on the ability of viable cells to incorporate and bind neutral red (NR), a supravital dye (Borenfreund and Puerner, 1985). NR is a weak cationic dye that readily diffuses through the plasma membrane and concentrates in lysosomes where it electrostatically binds to the anionic lysosomal matrix (OECD, 2010). Toxicants can alter the cell surface or the lysosomal membrane to cause lysosomal fragility and other adverse changes that gradually become irreversible. Such adverse changes cause cell death and/or inhibition of cell growth, which then decrease the amount of NR retained by the culture. Since the concentration of NR dye desorbed from the cultured cells is directly proportional to the number of living cells, cytotoxicity is expressed as a concentration-dependent reduction of the uptake of NR after exposure to the test substance. The amount of NR in the cells (fibroblast cell line, BALB/c 3T3) is measured with a spectrophotometer.

Based on the EURL ECVAM validation study to assess the predictive capacity of the NRU cytotoxicity assay to identify substances not requiring classification for acute oral toxicity (Prieto *et al.*, 2013), EURL ECVAM has issued recommendations concerning the validity and limitations of this *in vitro* test (EURL ECVAM, 2013). Considering the results of that validation study, the NRU cytotoxicity assay shows a high sensitivity (ca. 95%) and, consequently, a low rate (ca. 5%) of false negative results, when employed in conjunction with a prediction model to distinguish potentially toxic versus non-toxic (i.e. classified versus non-classified) substances.

The validated NRU cytotoxicity assay appears to be particularly relevant for the assessment of industrial substances since they are not designed to act on specific biological targets and, in general, tend not to be acutely very toxic. Following the provisions of the REACH Regulation and in particular its Annex XI, data from the NRU cytotoxicity assay could be used within a WoE approach to adapt the standard information requirements.

3.2. Limitations

The NRU cytotoxicity assay is sensitive to hazardous substances acting through general mechanisms of toxicity common to most cell types, often referred to as "basal cytotoxicity". Consequently, substances not exhibiting significant cytotoxicity but acting through:

(i) **mechanisms specific only to certain cell types and tissues** (e.g. of the heart or central nervous system) may not be identified as potentially acutely toxic by this method;

(ii) **metabolic activation** to induce toxicity may go undetected since the cell model lacks significant metabolic capacity.

Therefore, care must be taken in interpreting negative results derived from this assay.

The NRU cytotoxicity assay has a high false positive rate. Therefore, positive results cannot be readily used in a meaningful way in characterising acutely toxic substances (i.e. acute toxicity classifications Cat. 1 – Cat. 4). A likely reason is that the test method does not capture important biokinetic processes such as absorption, distribution, metabolism and excretion. Thus, certain substances, despite having cytotoxic potential, may not actually be acutely toxic *via* the oral route.

3.3. Regulatory use of the in vitro NRU cytotoxicity assay within the WoE approach

Considering the above limitations, results derived from the NRU cytotoxicity assay should **always be used in combination with other information sources** (with the data from a sub-acute study) to build confidence in the decision not to classify a substance for acute oral toxicity. Possible information sources complementary to a sub-acute toxicity study include physico-chemical properties and results of QSAR modelling (structural alerts, structure–activity relationships). The *in vitro* NRU cytotoxicity assay therefore fits within a WoE approach or as a component of a testing and assessment strategy (e.g. Norlén *et al.*, 2012).

Even in cases where the information resulting from the NRU cytotoxicity assay and QSAR models is available, the WoE should also include a sub-acute oral study (Table R.7.4-3) which fits within the scope of this adaptation (i.e. NOAEL ≥ 1000 mg/kg bw), as classification requirements must be fulfilled.

In line with the provisions of Directive 2010/63/EU on the protection of animals used for scientific purposes, and the provisions of Article 13 and Annex XI, 1.2 of the REACH Regulation, the *in vitro* NRU cytotoxicity assay should be used in combination with other data, in particular the results of a sub-acute oral toxicity test. Due to its limitations the *in vitro* NRU cytotoxicity assay should primarily be used to correctly identify and classify substances of low toxicity. The *in vitro* NRU cytotoxicity assay may be a valuable component of a WoE approach for supporting hazard identification and safety assessment in agreement with the EU CLP Regulation implementing the upper threshold of UN GHS Category 4 as the cut-off for non-classification of substances.

The *in vitro* NRU cytotoxicity assay is not considered to be the only confirmatory element in the WoE approach that is primarily based on the results of the sub-acute oral test or its DRF studies. Sub-acute oral toxicity studies have higher biological relevance and better predictivity than the *in vitro* NRU cytotoxicity assay. Therefore, while the *in vitro* NRU cytotoxicity assay is seen as a useful element of the WoE approach, it is not regarded as an obligatory element of it. Other types of information, such as convincing data on bioavailability or data from well documented (Q)SAR modelling, may also be used to build confidence in the prediction, as explained in Sections 4, 5 and 6 of this Appendix.

4. Prediction of acute oral toxicity based on the results of (Q)SAR

The use and restrictions of using (Q)SAR in order to provide information for acute oral toxicity are explained in Section R.7.4.3.1.1. Some physico-chemical parameters have been proposed as possible predictors of acute toxicity and it may be possible to generate relevant information with (Q)SAR methodologies, e.g. on systemically acting volatile compounds causing narcosis (Weed, 2005; Veith *et al.,* 2009). Furthermore since other methodologies (in particular the NRU cytotoxicity assay described in section 3 of this Appendix) are not appropriate for the identification of substances with specific toxic mechanisms, QSAR modelling should be applied to find if structurally related substances act *via* a specific mechanism. If there are indications that a substance may have a neurotoxic mechanism of action, QSAR modelling should be applied to find if structurally related substances are neurotoxic. This indication could be based on structural similarity with a known neurotoxicant (supported by adequate read-across justification) or on mechanistic *in vivo* or *in vitro* studies. If that is the case, the substance would not fit under this WoE adaptation, since neurotoxic substances often have a high acute toxicity.

Within the adaptation possibility considered in this Appendix, the core question concerning the use of (Q)SARs is whether the substance to be registered under REACH fits in the domain of a well-documented (Q)SAR model, including an open training set. If that is the case, the (Q)SAR modelling is a potential element within the WoE approach.

ECHA's Practical Guide 5 (How to report (Q)SARs)[91], illustrates the general aspects to take into account when using (Q)SAR models for regulatory purposes. It is important to distinguish between the proposed validity of the (Q)SAR model *per se*, the reliability and adequacy of an individual (Q)SAR estimate (i.e. the application of the (Q)SAR model to a specific substance), and the appropriateness of the documentation associated with models and their predictions. The appropriate documentation consists normally in a QSAR Model Reporting Format (QMRF), which documents transparently that the model is scientifically valid, and a QSAR Prediction Reporting Format (QPRF), which justifies that the prediction generated with a model for a specific substance is reliable and appropriate. Guidance on how to characterise (Q)SARs according to the OECD (Q)SAR validation principles is provided in the OECD GD 69 (OECD, 2007a).

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91 http://echa.europa.eu/documents/10162/13655/pg_report_qsars_en.pdf

The decision on whether to accept a (Q)SAR prediction is to be taken on a case-by-case basis.

(Q)SAR predictions may be gathered from databases (in which the predictions have already been generated and documented) or generated *de novo* through the available models. Data obtained by grouping approaches can also be used to generate local QSARs and derive a predicted toxicity value.

Programs such as the OECD QSAR Toolbox[92] serve this purpose. This software can be used to find analogue substances that have a toxicological profile similar to the substance with a data gap, which can be filled with a prediction of the relevant endpoint generated via read across or trend analysis. Furthermore, certain structures indicative of higher acute toxicity can be identified thanks to the Toolbox profilers[93].

Within this WoE adaption, it is not anticipated that QSAR prediction alone could be used to meet the information requirement. WoE by default has to consist of more than one "data element". Therefore, QSAR modelling may be useful e.g. in case it supports or confirms the evidence of low toxicity that has been obtained from the sub-acute study and, if applicable, from other elements of evidence such as a DRF study (see Figure R.7.4–4).

5. Use of physico-chemical data within the WoE approach

Certain physico-chemical properties are regarded as indicative for low bioavailability and low toxicity. However, it is noteworthy that these parameters cannot be used as standalone evidence to justify the adaptation of a systemic toxicity test, including the acute oral toxicity study. Therefore, whenever physico-chemical data are provided for the purpose of an adaptation they have to be accompanied by additional types of evidence, including a sub-acute oral toxicity study with a NOAEL equal to, or greater than, 1000 mg/kg bw, as specified below.

5.1. Low reactivity

Low reactivity, chemical and biological inertness or very low solubility are examples of physico-chemical properties of a substance that usually suggest that the bioavailability of the substance will be low. In the REACH registration dossiers, relevant data on low bioavailability have been provided for some substances, which have e.g. a crystalline structure and extremely low solubility even in aggressive media (hydrogen chloride solution mimicking the gastro intestinal tract). In order to contribute to the WoE, this type of data would normally need to be given as results of bioaccessibility or bioelution tests. Simulated gastric fluid and other relevant biological media need to be used in these tests to be convincing. While the bioelution method has not been accepted as an OECD TG, there is a standard protocol available as ASTM (American Society for Testing and Materials) D-5517[94] (US EPA, 2008), and BARGE (Bioaccessibility Research Group of Europe). By the initiative of Eurometaux, test method development is under consideration, aiming at an OECD TG project. In some read-across and trend analysis cases, bioelution studies have been found useful under REACH.

The rationale of "unreactivity" and lack of bioavailability as indicators of low toxicity is referred to in the column 2 adaptation in Annex IX, 8.6.2, fourth indent, according to which *"the sub-*

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92 www.qsartoolbox.org

93 For instance, quinones are known to be able to form covalent binding with proteins via a Michael addition reaction. Aliphatic secondary amines are associated with enhanced toxicity. Pyrethroids are known to cause neurotoxicity, and therefore an increased toxicity can be expected.

94 ASTM D-5517: extractability of metals from art materials (gastric fluid)

chronic toxicity study (90 days) does not need to be conducted if: […] the substance is unreactive, insoluble and not inhalable and there is no evidence of absorption and no evidence of toxicity in a 28-day 'limit test', particularly if such a pattern is coupled with limited human exposure."

Non-reactive substances with very high molecular weight may also have a low bioavailability *via* the relevant routes of exposure. However, a high molecular weight alone is not considered to be useful data in the WoE approach addressed in this part of the Guidance. It also has to be considered that metabolism *in vivo* may influence reactivity.

If the registrant uses physico-chemical data as an element of a WoE adaptation, reliable and good quality data have to be provided with a justification of how and why a given physico-chemical property is supportive of low toxicity.

6. Use of other information within the WoE approach

The WoE elements described above are the most relevant ones for the purpose of adaptation of the acute oral toxicity study. They should normally be considered when data are collected and generated. Beside information on mechanistic and/or tissue-based *in vitro* studies (e.g. addressing neurotoxicity), there are also other useful information sources, which are outlined below.

6.1. Read-across

The basic prerequisite to justify a read-across approach is that the source and target substances of the read-across are chemically and structurally similar and, therefore, they are expected to exhibit similar properties. The target substance should not have any such functional or chemical difference, which potentially makes its properties or reactivity and its toxicity different from that of the source substance. Also a mechanistic hypothesis has to be formulated in case a registrant proposes to use a read-across argumentation. For example, very low bioavailability or lack of reactivity associated with low toxicity, or dissociation/hydrolysis to normal constituents of biological media, are hypotheses that may be associated with the read-across in support of low acute toxicity. In order to be relevant in the regulatory context, the mechanistic hypothesis needs to be supported by reliable data. Also, if low toxicity and low biological activity are observed for both the source and the target substances of the read-across in toxicity studies, this can be used to build confidence in the read-across justification.

Furthermore, the data that are available on the source substance and target substance must enable the prediction of the acute toxicity potential or rather the lack of it. Within the present WoE adaptation, the registrant must be able to predict, with sufficient certainty and confidence, that the LD_{50} of the target substance of the read-across will be above 2000 mg/kg bw. While the paragraph above illustrates some principles of read-across when applied for the purpose of this specific WoE adaptation, more detailed guidance on read-across can be found in Chapter R.6 of the *Guidance on IR&CSA* and in the illustrative examples available on ECHA's web-site at: http://echa.europa.eu/support/grouping-of-substances-and-read-across.

The same considerations apply to a sub-acute oral toxicity study: where properly justified and documented, a sub-acute oral toxicity study with an analogue substance may be proposed, according to Annex XI, section 1.2, and then be used as an element of the specific WoE adaptation.

The conclusions about the likely properties of a substance can also be based on knowledge of the properties of one or more similar substances, by applying grouping methods. The corresponding OECD guidance provides information on the use of grouping of chemicals and read-across approaches (OECD, 2014).

6.2. Existing human data

The strength of the epidemiological evidence for specific health effects depends, among other things, on the type of analyses and on the magnitude and specificity of the response. Relevant human data may be available e.g. in reports of the poison control centres or from published case studies. Confidence in the findings is increased when comparable results are obtained in several independent studies on populations exposed to the same agent under different conditions. Other characteristics that support a causal association are the presence of a dose-response association, a consistent relationship in time and (biological) plausibility, i.e. aspects covered by epidemiological criteria such as those of Hill (1965).

A comprehensive guidance on both the evaluation and use of epidemiological and human evidence for risk assessment purposes is provided by Kryzanowski *et al.* (WHO, 2000).

High quality human data may also be obtained from historical data from individual clinics or collated clinic data and/or from dose-response studies (Mowry *et al.*, 2012; Dolgin *et al.*, 2014; Cassidy *et al*,. 2015). High quality human data may be considered as a strong basis for C&L decision making (subject to the ethical considerations relevant for the respective regulatory programme). It is acknowledged that new human studies are not allowed for the purpose of CLP and REACH, but existing data may be used.

The usefulness of human data in the context of this WoE adaptation is limited, since the scope of this adaptation is limited to substances of low toxicity, whereas the most definitive human data are usually available on substances that are toxic.

7. Weight-of-Evidence analysis

When applying the WoE approach proposed in this Appendix, the registrant should aim at obtaining adequate and reliable data for hazard identification and classification purposes for substances of low acute toxicity. Within the WoE approach, different types of data can be obtained and assessed, in order to find out whether the information requirement for the acute oral toxicity can be met, or whether further information needs to be generated.

The objective of this WoE approach is to correctly identify substances that are not acutely toxic, i.e. with an $LD_{50\ acute,\ oral}$ higher than 2000 mg/kg bw and which, therefore, do not need to be classified under the CLP Regulation.

7.1. Introduction

The term "weight of evidence" is widely used in scientific publications and government agency guidelines in the context of risk assessment. The term has been used with reference to a specific body of evidence without reference to an interpretative method, but also methodologically, with prescribed methods addressing specific purposes such as confidence in causation (Weed, 2005).

A WoE determination means that all available and scientifically justified information bearing on the determination of hazard are considered together. In the case of acute oral toxicity, this includes animal data on sub-acute oral toxicity (including DRF studies), physico-chemical parameters, information from category approaches (e.g. grouping, read-across), (Q)SAR results, the results of suitable *in vitro* tests (e.g. validated NRU cytotoxicity assay), and possibly human data. The quality and consistency of the data should be taken into account when weighing each piece of the available information. In this context, the highest weight should be given to the sub-acute oral toxicity study and its related DRF studies, as described in Table R.7.4-3.

A WoE approach involves an assessment of the relative values/weights of different pieces of the available information that has been gathered and generated. These weights/values can be assigned either in a more structured (even quantitative) way by applying a formalised procedure (e.g. based on Bayesian logic, as in Rorije *et al.*, 2013) or by using expert judgement. The weight given to the available evidence will be influenced by the quality of the data, consistency of results/data, and relevance of the information for the given regulatory endpoint. A matrix for the WoE analysis is provided below (Table R.7.4-3).

Examples of tools available to evaluate the quality of data include the Klimisch scores (Klimisch *et al.*, 1997), Hill's criteria for evaluation of epidemiological data (Hill, 1965) as well as the JRC's ToxRTool for scoring *in vivo* and *in vitro* data (Schneider *et al.*, 2009). The ToxRTool[95] provides an assessment system which allows the evaluator of a given study to derive an appropriate Klimisch score.

Under Article 9(3)of the CLP Regulation, a WoE approach should be used when the specific criteria for classification cannot be directly applied. According to that provision, all available information that can contribute to the determination of classification for an endpoint are considered together.

7.2. Role of WoE in the assessment of acute oral toxicity

After the necessary testing has been performed and non-testing data have been generated and assessed, the WoE approach is applied in order to consider whether the hazard characterisation and the classification can be achieved without performing the legally required acute oral toxicity test.

As explained above and described in Table R.7.4-3, the most relevant *in vivo* test is the sub-acute oral toxicity test (OECD TG 407 or 422 screening test), and then the enhanced DRF1, whereas the most useful *in vitro* test is the NRU cytotoxicity assay.

However, in case other relevant and good quality data can be obtained, e.g. from open literature and/or from the registrant's own databases, a WoE analysis could actually be performed, but not necessarily completed, even before performing new *in vitro* or *in vivo* tests. In case a WoE analysis is based on available data, there are two possible conclusions: either the data are considered sufficient and a WoE adaptation is submitted in the registration dossier without new testing, or the WoE based on the available data remains insufficient or inconclusive and generation of further data is necessary.

Considering human evidence, several types of existing information can be used, provided that these are of sufficient quality. In the WoE analysis, the availability of the specified types of data should be checked. The sources of those data may vary, ranging from clinical study reports, scientific publications, data from poison information centres, guideline tests, to worker surveillance data from the chemical industry.

7.4. Assessment of data quality

The quality of the data obtained for a WoE approach needs to be assessed, since the quality will contribute to the weight of each data element. In case the quality of a certain study is deemed to be inappropriate, those data should not be included in the WoE. Instead it is recommended to focus on other elements of information that are of sufficient quality. Quality might be inappropriate, e.g. due to the missing validation of a methodology, the "non-adherence" to the relevant test guideline/method, the lack of adequate controls, and/or the deficiencies in data reporting, etc.

The quality of toxicological studies is usually described by assigning Klimisch scores. Epidemiological data can be evaluated using Hill's criteria (Hill, 1965).

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[95] https://eurl-ecvam.jrc.ec.europa.eu/about-ecvam/.../toxrtool/ToxRTool.xls

For many existing substances, it is acknowledged that some of the available information may have been generated prior to the requirements of Good Laboratory Practice (GLP) and/or prior to the acceptance of the standardised OECD test methods[96]. While such information may still be usable, both the data and the methodology used must be evaluated in order to determine their reliability. Such an evaluation would ideally require an evidence-based evaluation, i.e. a systematic and consistent evaluation following pre-defined, transparent and independently reviewed criteria before making decisions. These should always include justifications for the use of particular data sets on the basis of the criteria-based evaluation.

7.5. Adequacy and relevance of information

The "adequacy" of information defines the usefulness of information for the purpose of hazard and risk assessment, i.e. whether the available information contributes to the decision-making on whether the substance is of low acute toxicity and whether it can be concluded that there is no need to classify the substance for acute oral toxicity. The evaluation of adequacy of test results, and documentation for the intended purpose, are particularly important for substances for which a number of test results are available but some (or all) of the tests have not been carried out according to current standards. Where there is more than one study, the greatest weight is given to the study(ies) that is (are) the most relevant and reliable (e.g. validated and/or with regulatory acceptance).

7.6. Evaluation of consistency of the data

The consistency of the existing data from various sources is crucial and should therefore be thoroughly evaluated in the WoE approach.

In case the elements of evidence are of comparable weight but give inconsistent evidence, usually the WoE analysis will not be conclusive enough. Consequently *in vivo* and/or *in vitro* testing will have to be considered and conducted. In case the weights of the individual pieces of evidence differ considerably (e.g. inconsistent results obtained from *in vitro* and/or *in vivo* testing and human data), a WoE conclusion may be drawn according to the evidence carrying the highest weight. It is important to evaluate what the reasons for inconsistent data e.g. from *in vitro* methods may be, and whether the lack of metabolic capacity affects the prediction. In case the inconsistency cannot be scientifically explained, the WoE analysis becomes inconclusive and, therefore, the WoE-based adaptation should not be proposed by the registrant.

Conversely, consistent data across several studies and/or sources may be considered sufficient for regulatory purposes, pursuant to Annex XI, section 1.2.

7.7. Conclusions from the WoE analysis

The core element of the WoE approach proposed in this Appendix, and which is a prerequisite for applying the approach, is the sub-acute oral toxicity study performed with the registered substance. Where properly justified and documented, a sub-acute oral toxicity study with an analogue substance may be proposed, according to Annex XI, section 1.2. In addition, one or more other WoE elements are needed and the registrant needs to justify (i) why their

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[96] LD50 test according to the OECD guideline 401 has been deleted, and is no longer in use. In case a registrant provides an **old** OECD 401 study record, it is still considered adequate, because it is scientifically relevant.

combination is sufficient to conclude and (ii) how the uncertainty associated with the WoE approach has been minimised.

In the final analysis of the WoE approach, each element of evidence must be characterised for its quality, relevance, coverage and consistency with other information (see the "Matrix for the Weight-of-Evidence analysis", Table R.7.4-3).

When consistency is seen among "qualified" elements of evidence, the WoE analysis may reach a conclusion that the relevant information requirement has been sufficiently covered and that further *in vivo* testing is not necessary. In that case, a conclusion can also be drawn that the substance does not need to be classified for acute toxicity (Figure R.7.4–4).

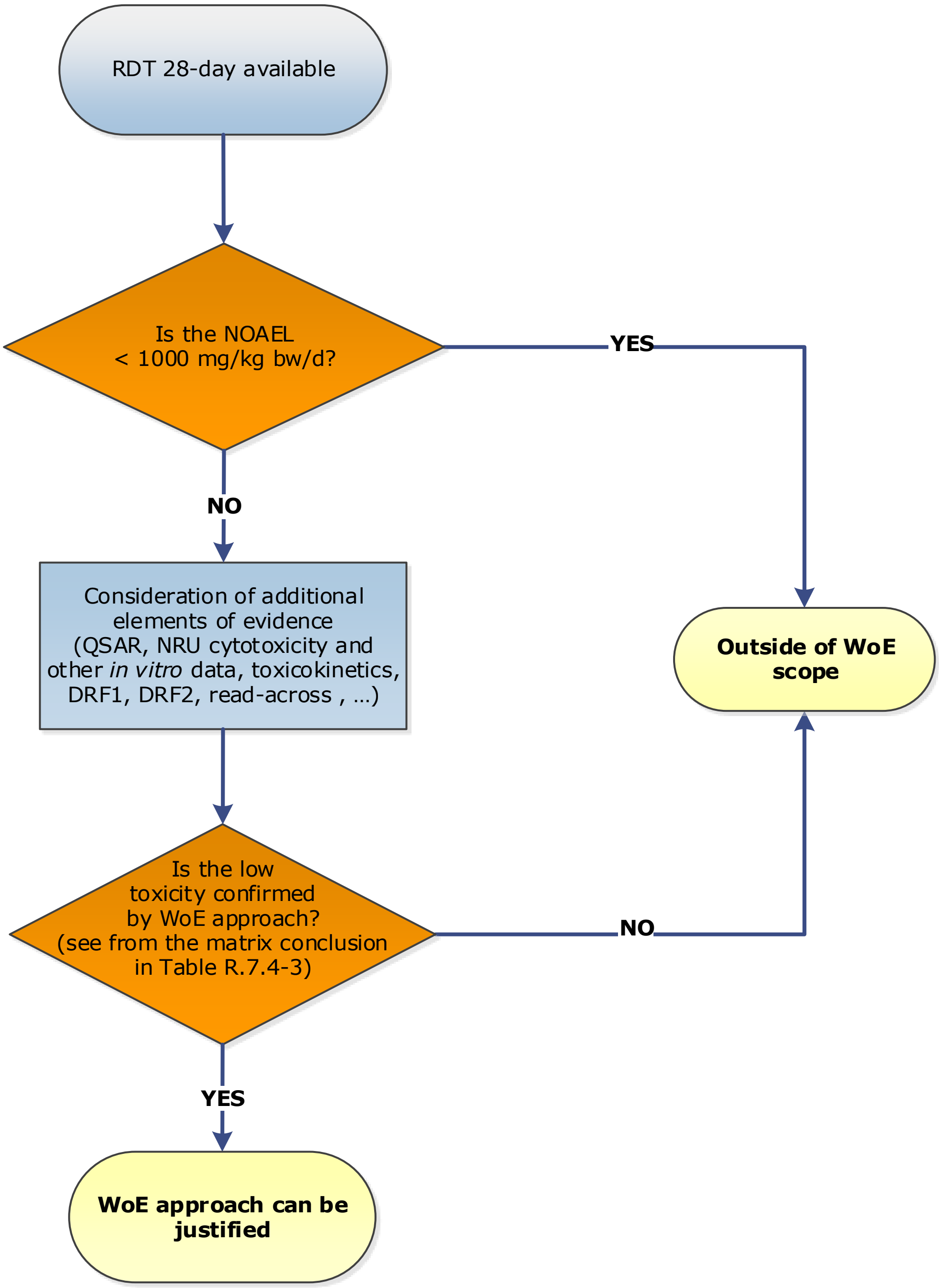

Figure R.7.4–4 Decision tree to assess whether an *in vivo* acute toxicity test is required, when the registrant holds an existing repeated sub-acute oral toxicity study and makes use of the WoE approach.

In case the existing study was performed on an analogue substance, it is the registrant's responsibility to justify the read-across approach proposed. Where the registrant thinks that the justification should be acceptable to ECHA, the study could be used as part of the WoE analysis.

When, on the other hand, insufficient information remains after the "non-qualified" data have been rejected and/or when the remaining information is inconsistent or contradictory, the WoE analysis would reach the conclusion that the relevant endpoint, or information requirement, has not been sufficiently covered and that further *in vivo* testing is necessary, according to the specific legal/regulatory framework.

The WoE justification has to be specific for the registered substance and specific to the set of data information used by the registrant in order to meet the corresponding information requirement.

After collecting and assessing the data, the registrants need to decide how to include the existing information in the registration data set. It is recommended that each element of evidence of the WoE is included in the registration dossier as an individual study record, in Section 7.2 of IUCLID. Furthermore, the WoE analysis and its conclusion may be included in the summary of Section 7.2. The matrix given below can be used for preparing that summary.

8. *In vivo* acute oral toxicity test

Due to the limitations of the methods and types of information described above, there are cases where the acute oral toxicity study will be needed, e.g. when:

- based on the DRF or on the results of the sub-acute toxicity study, the $LD_{50acute,oral}$ is lower or is likely to be lower or equal to the limit of 2000 mg/kg bw (C&L limit) and, therefore, the substance does not fall within the scope of this WoE adaptation, or
- the information obtained and results of the tests performed are inconsistent and this inconsistency cannot be scientifically explained, or
- the registrant has to conclude on classification for acute toxicity category 5, i.e. $LD_{50acute,oral}$ is between 2000 and 5000 mg/kg bw, e.g. because the substance is placed on the market in a country where the authority has implemented that category, or
- the registrant may have some existing information (e.g. structural data) showing that the substance may be acutely toxic and the registrant aims to ensure the proper level of risk management measures.

In all these cases, the registrant is advised to document why the data used in the WoE analysis were not sufficient to fulfill this information requirement and consequently a relevant test according to the OECD/EU guidelines is needed (according to REACH Article 13).

Table R.7.4-3 Matrix for the Weight-of-Evidence analysis.

Fill in the entries for those modules for which data are available or generated. It is recommended that the results of a sub-acute study are always included in the WoE analysis. In addition, one or more other elements of evidence need to be provided. The type of other information (available or which can be generated) will vary depending on the case. For any remaining entries, indicate NA (not available) in the respective column.

Module	**Title of document/full reference or data not available (NA)**	**Study result, evidence obtained**	**Data quality,** according to the Klimisch score, when appropriate	**Adequacy and relevance,** short statement	**Coverage of relevant parameters and observations** [(a)]	**Consistency with other information** [(b)]	**Conclusive remark** [(c)]
1. Sub- acute toxicity study				Highest relevance (prerequisite)			
2. Enhanced DRF1				High relevance (usually)			
3. *In vitro* cytotoxicity assay (NRU)				Only negative results are relevant			
4. (Q)SAR modelling	*i.e. QMRF*	*i.e. Predicted value*		Relevant if applicability domain is considered appropriate			
5. Physico-chemical properties				Relevant when available			
6. Other data (existing human data, read-across)				Case-by-case			
Overall conclusion	1. WoE allows the conclusion that the substance is of low acute toxicity and does not need to be classified as acutely toxic; No additional acute oral toxicity testing is necessary, **or** 2. WoE does not allow the conclusion that the substance is of low toxicity. The registrant needs to consider the most appropriate additional testing, which would usually be an acute oral toxicity test performed according to a relevant OECD test guideline.						

(a) Definition of the relevant parameters for each element of the WoE, when applicable.

(b) For example: *"This element of evidence (any entry except 1 and 2) is consistent with the sub-acute toxicity study".*

(c) For example: *"The existing human data suggest that the substance is not acutely toxic. Due to poor reporting of this data, and low quality in terms of exposure information, the data is inconclusive, and has a low weight in the final evaluation."*

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Appendix R.7.4–2 Background and Analysis supporting the recommended Weight-of-Evidence adaptation

Annex XI specifies several possibilities for adaptation, including e.g. weight of evidence (WoE) (section 1.2), QSAR (section 1.3), and *in vitro* tests (section 1.4), and read-across (section 1.5).

Registrants may use these possibilities as stand-alone adaptations when sufficiently justified. However, the WoE approach for acute oral toxicity as outlined in Appendix R.7.4–1 is recommended as it makes use of combinations of these possibilities. It is based on ECHA's analyses, and it is more likely to result in an adaptation that can be accepted according to Annex XI, section 1.2.

Expectations for the 2018 registration deadline

Of relevance for the analysis is the consideration of the number of *in vivo* acute toxicity studies necessary for the registrants to fulfil their obligations.

It is anticipated that many phase-in substances, which will be registered by the 2018 deadline, will have an *in vivo* acute oral toxicity study already available (the estimate is 65%):

From the second Article 117(3) report published in June 2014 [97]:

- 35% of ca. 5200 substances (to be registered at > 10 tpa, by 2018) are forecast not to have an existing acute oral toxicity study, which represent approximately 1825 studies.
- It is also assumed that approximately 30% of these substances are of low acute toxicity (ie. where the acute oral LD_{50} is higher than 2000 mg/kg bw/day).

Consequently, many registrants will have to conduct a new study to meet the acute toxicity information requirement, or will need to adapt this standard information requirement. Therefore the use of a waiving possibility, instead of performing an *in vivo* oral acute toxicity testing requirement, may have a high impact: if those registrants would follow the alternative approach proposed in this Appendix, the number of acute oral toxicity studies necessary for the 2018 registration deadline could be reduced by approximately 550, i.e. the related *in vivo* acute oral toxicity tests could be avoided.

Supporting background literature

In 2014 EURL ECVAM, part of the Joint Research Centre (JRC) of the European Commission, published a Strategy Document on alternative approaches for acute systemic toxicity testing (Prieto *et al.*, 2014).

EURL ECVAM considered that efforts should be directed towards (i) the reduction and replacement of animal tests for acute systemic toxicity, and (ii) the refinement of *in vivo* studies, according to the Russell and Burch 3Rs principle. By following the approach proposed in this Appendix, registrants would contribute towards these efforts.

Where known, consideration should be given to the mechanistic basis of acute toxicity and the validation of integrated prediction models. EURL ECVAM proposed to evaluate promising components of integrated approaches for testing and assessment (IATA), including the better use of existing alternative methods, such as mechanistically relevant *in vitro* assays.

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[97] Available at http://echa.europa.eu/documents/10162/13639/alternatives_test_animals_2014_en.pdf

Furthermore, according to EURL ECVAM, information on repeated dose toxicity might be useful in supporting classification and labelling for acute systemic toxicity.

In the scientific literature, the value of the acute toxicity test has been discussed and prediction models based on sub-acute toxicity data or *in vitro* cytotoxicity tests that may replace *in vivo* acute toxicity studies, have been developed (Creton *et al.*, 2010; Chapman *et al.*, 2010; Indans *et al.*, 1998; Kinsner-Ovaskainen *et al,*. 2013; Robinson *et al.*, 2008; Siedle *et al.*, 2011; Bulgheroni *et al.*, 2009).

The background and rationale of this guidance for a WoE-based adaptation for the acute oral toxicity study is based on the following:

- There are several initiatives and proposals made by the scientific community suggesting that relevant information on acute oral toxicity can be obtained without performing the standard *in vivo* test.
- In 2015, the JRC launched a survey aimed at exploring waiving opportunities for acute systemic toxicity testing, among experts from different fields (pharmaceutical, chemical industry etc.). From the responses obtained it became evident that some companies have in fact tried to predict the acute effects from repeated dose studies (Graepel *et al.*, 2016).
- Several hundreds of *in vivo* studies can potentially be replaced with the WoE approach.

Analysis of the data provided by 2010 and 2013 registrants

An analysis initiated by JRC (Graepel *et al.*, 2016) and then continued by ECHA has shown that, for **substances of low toxicity**, the prediction of **acute oral toxicity** classification can be based on the data from oral **sub-acute studies** in most cases.

The data used for this analysis were extracted in May 2015 by ECHA, from the whole REACH registration database, from studies reported in sections 7.2.1 (Acute toxicity: oral) and 7.5.1 (Repeated dose toxicity: oral) of the IUCLID dossiers.

<u>Step 1</u>: A preliminary set of filters was used to select the **relevant** experimental data:

- “Test material identity same as registered substance” = “yes”;
- “Study type” = “experimental result” (to select only experimental data and to exclude other study types such read-across or QSAR results);
- Reliability score = “1” or “2”.

<u>Step 2</u>: An additional filter was used to select the **relevant studies** performed according to the following EU/OECD guidelines:

- for acute toxicity: LD_{50} values from EU Method B.1 (bis and tris) or OECD TG 401, 420, 423 or 425;
- for repeated dose toxicity: NOAEL or NOEL from OECD TG 407 or 422, excluding results expressed only in ppm.

<u>Step 3</u>: Another filter was used to select only dossiers containing **relevant studies in both 7.2.1 and 7.5.1 sections**.

As a result, 1256 registration dossiers were selected.

In the remaining registration dossiers, other routes of administration (often inhalation) have been used for the acute and/or sub-acute toxicity tests, or one of these studies was adapted, e.g. by using information on an analogue substance (i.e. read-across adaptation). Hence these study record(s) in the IUCLID dossier could not be used for this analysis.

Step 4: Refinement; ECHA then refined the data set as follows:

- Exclude sub-acute studies reporting a NOAEL < 1000 mg/kg bw;
- If a range was given for a single study, the lowest value was selected;
- If the registrant submitted more than one relevant study per endpoint, the study resulting in the lowest LD_{50} value and/or lowest NOAEL value was selected;
- Furthermore, the information on the identity of the test material was checked in order to exclude cases where another substance than the registered substance could have been tested (i.e. "hidden" read-across)[98].

To summarise, the data included in the final prediction model include dossiers with:

- Relevant acute oral and sub-acute oral toxicity /screening study tests[99]; and
- Sub-acute oral toxicity study which resulted in a NOAEL at or above 1000 mg/kg bw.

Please note that registrant self-classification was not considered.

Substances in 415 dossiers fulfil the above criteria. In addition, all except nine dossiers gave an acute oral toxicity study with an LD50 higher than 2000 mg/kg bw. These cases were manually analysed, and explanations included e.g. the differences in units used or the different modes of administration of the oral dose between the acute and repeated dose studies.

In conclusion, this "prediction model" based on sub-acute toxicity data can be used and constitutes the core element of the WoE approach.

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98 With the term "hidden read-across" ECHA refers to studies marked as "experimental results" where "identity of the test material same as registered substance" is ticked, but the identifiers provided in the test material identifiers table refer to a substance different from the registered one.

99 According to the relevant OECD guidelines, rats and mice are the preferred species. With very few exceptions, the studies used for this "prediction model" were made with these species.

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Appendix R.7.4–3 (Q)SARs for the prediction of acute toxicity

There are several (Q)SAR models for the prediction of acute toxicity. However, so far their use within the regulatory context has been hindered by their limited applicability domain and accuracy of their predictions. While further developments are needed for a wider application of (Q)SARs for acute toxicity, some examples are given below in order to illustrate the prospects for applying (Q)SAR approaches for predictive purposes or to investigate the mechanisms of toxicity.

(Q)SARs for inhalation toxicity

Some simple regression models have been developed for predicting the inhalational toxicity of volatile substances, and these can be used reliably within their domains of applicability. Typically, parameters such as vapour pressure (VP) and boiling point (BP) have been found to be useful predictors of the acute toxic effect (e.g. LC_{50} value). These models are based on the assumption that toxicity occurs by the non-specific mechanism of narcosis, and that the LC_{50} data are based on tests in which a steady-state concentration has been reached in the blood. These models are suitable only for systemically acting volatile compounds.

For example, acute (non-lethal) neurotoxicity data for the neurotropic effects of some common solvents on both rats (whole-body exposures for 4h) and mice (whole-body exposures for 2h), taken from Frantik *et al.* (1994), were subjected to QSAR analysis by Cronin (1996). Stepwise regression analysis of the 4-hr toxicity data causing the 30% depression in response ($\log 1/ECR_{30}$) in rats gave the following equation:

$$\log 1/ECR_{30} = 0.361\ ClogP - 0.117\ {}^{0}\chi - 1.76$$

$$n = 37 \qquad R^2 = 0.817 \qquad s = 0.280 \qquad F = 35.2$$

This relationship demonstrates a partial dependence of neurotoxicity with the octanol-water partition coefficient, logP. The negative correlation with the zero-order molecular connectivity ${}^{0}\chi$ (calculated with the software MOLCONN-X in the original paper) is thought to be an indication that the membrane permeability of blood-brain barrier is reduced for large molecules.

Stepwise regression for mouse neurotoxicity gave the following equation:

$$\log 1/ECM_{30} = 0.212\ ClogP + 0.00767\ BP - 0.176\ {}^{0}\chi - 2.03$$

$$n = 39 \quad R^2 = 0.811 \qquad s = 0.271 \qquad F = 22.4$$

in which BP is the boiling point of the substance (BP is inversely related to vapour pressure).

The application of principal components analysis (PCA), to separate compounds of high neurotoxicity from those of low neurotoxicity, suggested that in addition to partitioning through a membrane (determined by logP and molecular size), aqueous solubility and volatility are also important factors governing neurotoxicity (Cronin, 1996). Metabolism to more toxic compounds is suggested as a possible cause of compounds appearing as outliers in the QSARs.

Regarding baseline inhalation toxicity, Veith *et al.* (2009) developed two models for the prediction of narcosis in rodents using data from inhalation toxicity studies in mice and rats from the US ECOTOX database:

$$\text{Log } LC50_{rat} = 0.69 \text{ logVP} + 1.54$$

$$n = 36 \quad r^2 = 0.94 \qquad \text{Std. Error} = 0.19 \qquad \text{StT test} = 18.35$$

$$\text{Log } LC50_{mouse} = 0.57 \text{ logVP} + 2.08$$

$$n = 28 \quad r^2 = 0.74 \qquad \text{Std. Error} = 0.20 \qquad \text{StT test} = 8.63$$

where VP is the estimated vapour pressure of the substance using EPISUITE v3.2. in mm Hg. For more insight into the results, the reader is recommended to consult the original reference.

The models are not suitable for reactive substances or those exerting receptor mediated toxicity. An approach taken in the development of the models was to exclude those substances identified as reactive by the Russom scheme (Russom *et al.*, 1997).

(Q)SARs for oral toxicity (LD_{50})

There are references in the literature to a few models for predicting LD_{50}, generally for small sets of compounds. For example, Hansch and Kurup (2003) developed the following QSAR to predict the toxicity of barbiturates (LD_{50}) in female white mice, using toxicity data from Cope and Hancock (1939):

$$\log 1/LD_{50} = -1.44 \log P + 0.16 \text{ NVE} - 8.70$$

$$n = 11 \quad R^2 = 0.924 \qquad s = 0.077 \qquad R^2_{cv} = 0.879$$

where NVE is the number of valence electrons (used as a measure of polarisability).

More recently, Koleva *et al.* (2011) developed two nonlinear regression models to quantify the oral LD_{50} for compounds causing only baseline toxicity in rats and mice:

$$\log 1/LD50 \text{ rat} = -1.780 + 0.465 \text{ logP} - 0.111(\text{logP})^2$$

$$n = 55 \qquad \text{rms} = 0.15 \qquad r^2_{adj} = 0.59 \qquad F = 40.3$$

$$\log 1/LD50 \text{ mouse} = -1.841 + 0.503 \text{ logP} - 0.105(\text{logP})^2$$

$$n = 30 \qquad \text{rms} = 0.17 \qquad r^2_{adj} = 0.72 \qquad F = 38.5$$

where logP is the n-octanol/water partition coefficient.

The models were developed with a training set of saturated monohydric alcohols and saturated monoketones. Substances with limited water solubility or potentially undergoing metabolism were considered out of the domain, and excluded from both training and test sets. The authors highlight some classes of reactive substances that are out of the domain since they exert excess toxicity, particularly electrophilic substances that are able to undergo covalent binding to nucleophilic sites.

QSARs for predicting human toxicity

The same descriptors were used to predict the LD_{100} of miscellaneous drugs to humans, using toxicity data from King (1985):

$$\log 1/C = 0.61 \log P + 0.017\ NVE + 1.44$$

$$n = 36 \qquad R^2 = 0.850 \qquad s = 0.438 \qquad R^2cv = 0.817$$

QSARs for predicting *in vitro* effects

A number of QSAR models for predicting *in vitro* effects are cited in the literature (reviewed in Lapenna *et al.*, 2010), but these are not directly relevant to the assessment of acute toxicity for regulatory purposes. In general, these models have been developed to investigate the mechanisms of cytotoxic action and they outline the role of hydrophobicity as well as electronic descriptors, including electrotopological state descriptors (Lessigiarska *et al*., 2006), bond dissociation energies (Selassie *et al*., 1999), and dissociation constants (Moridani *et al*., 2003). While these models are not directly relevant to the assessment of acute toxicity, the fact that reliable QSARs can be developed for the *in vitro* cytotoxicity of defined groups of substances indicates that the approach of modelling *in vitro* data should be further explored with a view to integrating such QSARs into the testing and assessment strategy for acute toxicity. For example, a battery of QSARs could be developed for predicting the *in vitro* data of a validated *in vitro* test and then used to supplement or replace *in vivo* testing.

Computerised models

For heterogeneous groups of compounds, computerised models are available to predict acute toxicity (normally $LD50_{oral}$).

Knowledge-based software (see also Section R.6.1 of Chapter R.6 of the *Guidance on IR&CSA*), such as HazardExpert, are based on rules derived from human expert opinion to estimate toxicity. In statistically based software, such as TOPKAT and MultiCASE, statistical methods are used to derive (Q)SAR models (see also Section R.6.1).

A list of some of the available computerised models with a brief description is provided below:

OECD QSAR Toolbox

The freely available for download OECD QSAR Toolbox software (http://www.qsartoolbox.org/) contains profilers that could be useful in creating mechanistic categories for acute oral toxicity in rats:

- Toxic hazard classification by Cramer, which assigns the substance to a toxicity class (“High”, ”Medium” or “Low”) based on the effects when administered orally.
- Protein binding by OASIS and Protein binding by OECD, which allows identifying electrophilic substances, which are likely to exhibit higher acute toxicity due to their reactivity.
- Repeated dose toxicity (HESS), which was initially developed by the Japanese NITE with a view to help predicting effects in a 28 days study in rats. The profiler would allow to identify some specific modes of action that are also relevant for acute toxicity (e.g. neurotoxicity).

The QSAR Toolbox also contains experimental data on acute toxicity in the following databases:

- ECHA Chem: this database contains non-confidential data from REACH registration dossiers.

- Rodent Inhalation Toxicity Database: it is a compilation of high quality data from rat inhalation studies reported in the literature.

- Toxicity Japan MHLW: it contains experimental results from single dose toxicity test and mutagenicity test results performed under Japan's Existing Chemicals Programme.

The use in combination of profilers and data for analogues could allow the prediction of acute oral toxicity for new substances through a read-across or trend analysis approach.

HazardExpert

HazardExpert is a module of the Pallas software developed by CompuDrug Limited (http://www.compudrug.com). The program works by searching the query structure for known toxicophores, which are stored in the "Toxic Fragments Knowledge Base" and which include substructures exerting both positive and negative modulator effects. Once a toxicophore has been identified, this triggers estimates for a number of toxicity endpoints, including neurotoxicity. The default knowledge base of the system is based on a US-EPA report (Brink and Walker, 1987) and scientific information collected by CompuDrug Limited. This program can be linked to MetabolExpert, another module of the Pallas software, to predict the toxicity of the parent compound and its metabolites. Information on the validity of the model is not available. Investigations on the validity and applicability of HazardExpert are needed before recommendations can be made about its regulatory use.

TOPKAT

The TOPKAT software package employs cross-validated quantitative structure-toxicity relationship (QSTR) models for assessing various measures of toxicity (http://accelrys.com/products/collaborative-science/biovia-discovery-studio/qsar-admet-and-predictive-toxicology.html). The Rat Oral LD_{50} module of TOPKAT includes 19 QSAR regression models for different chemical classes. The models are based on a number of structural, topological and electrophysiological indices, and they make predictions of the oral acute median lethal dose in the rat (LD_{50}).

The TOPKAT rat oral LD_{50} models are based on experimental data from the Registry of Toxic Effects of Chemical Substances (RTECS). Since RTECS lists the most toxic value when multiple values exist, the TOPKAT model tends to overestimate the toxicity of query structures.

The Rat Inhalation LC_{50} module of TOPKAT contains five submodels related to different chemical classes.

TOPKAT models, including the models for acute oral toxicity, were used by the Danish EPA (http://qsar.food.dtu.dk/) in 2005 to evaluate the dangerous properties of around 47,000 organic substances on the EINECS list. An external evaluation of this model using 1840 substances not contained in the TOPKAT database gave poor results (R^2 = 0.31). However, 86% of estimations fall within a factor of 10 from test results (DK EPA study).

The Danish EPA concluded that the TOPKAT model is sufficient to give an indication of the least strict classification for acute toxicity, Xn; R22 (under the former Dangerous Substance Directive (DSD) classification/labelling system used in the EU before the CLP regulation came into force).

CASE Ultra

CASE Ultra software (http://www.multicase.com) contains an acute toxicity module, which consists of a rat LD_{50} model based on 12,262 compounds from compilations by NTP, WHO, RTECS, and other regulatory agencies data. Information on the validity of the model is not available. Investigations on the validity and applicability of CASE Ultra are needed before recommendations can be made about its regulatory use.

T.E.S.T.

The Toxicity Estimation Software Tool (T.E.S.T.), developed by the US EPA allows the prediction of many different endpoints, including oral LD_{50} in rats. Version 4.0 and greater contain a database of 7413 substances with rat acute toxicity data that can be used with different methods to build a model for the prediction of LD_{50}, such as hierarchical clustering, random forest, use of nearest neighbours and a consensus model. The software uses a variety of molecular descriptors to perform the predictions. The accuracy of the predictions for LD_{50} depends on the model used and the type of the substance, but, according to the software documentation, overall it is not as good as for other endpoints.

The software allows visualisation of the closest analogues in the training set and the test set of the models, and accuracy of each model for them, so that the user can use expert judgement to estimate whether a prediction is reliable.

Derek Nexus

Derek Nexus (http://www.lhasalimited.org/products/derek-nexus.htm) contains sets of structural alerts for many human health endpoints. Amongst them there are several alerts for "high acute toxicity", which cannot be used to derive directly an LD_{50}, but can be of use in identifying very toxic compounds. The alerts for other endpoints can be used to identify molecules with specific modes of action which would be expected to be of particular toxicity due to these effects, such as cardiotoxicity or cholinesterase inhibition.

ACD/Percepta

The models contained in the ACD/Percepta suite (http://www.acdlabs.com/products/percepta/) allow the calculation of LD_{50} values for mice under oral, intraperitoneal, intravenous, subcutaneous administration and for rats under oral and intraperitoneal administration methods. All of them are based on fragmental QSARs used to derive baseline toxicity, plus corrections for excess toxicity based on fragments associated with specific modes of action. More than 100,000 compounds were used in the development of the models, although it is unclear on how many data points each model was based. The software provides an automatic assessment of the reliability of the prediction based on the similarity of the compounds in the training set and the accuracy of the predictions for them.

Review papers

The existing QSAR models and software tools for predicting acute (and chronic) systemic toxicity have been investigated and compared in different review papers. In more detail Norlén *et al.* (2012) present an analysis and comparison of the predictive performance of several QSAR tools and the *in vitro* 3T3 NRU method. The review from Lapenna *et al.* (2010) covers literature models, QSAR software and databases available for acute toxicity.

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R.7.5 Repeated dose toxicity

R.7.5.1 Introduction

Repeated dose toxicity studies provide information on possible adverse general toxicological effects likely to arise from repeated exposure to a substance. Furthermore, these studies may provide information on e.g. reproductive toxicity and carcinogenicity, even though they are not specifically designed to investigate these endpoints.

Organs and tissues investigated in repeated dose toxicity studies include vital organs such as heart, brain, liver, kidneys, pancreas, spleen, immune system, lungs etc. Effects examined may include changes in morphology, physiology, growth or life span, behaviour which result in impairment of functional capacity or impairment of capacity to compensate for additional stress or increase in the susceptibility to the harmful effects of other environmental influences. Therefore, it is important that the possible adverse general toxicological effects are assessed for chemical substances that may be present in the environment.

R.7.5.1.1 Definition of repeated dose toxicity

The term *repeated dose toxicity* comprises the general toxicological effects occurring as a result of repeated daily dosing with, or exposure to, a substance for a part of the expected lifespan (sub-acute or sub-chronic exposure) or for the major part of the lifespan, in case of chronic exposure.

The term *general toxicological effects* (in this report often referred to as *general toxicity*) includes effects on, e.g. body weight and/or body weight gain, absolute and/or relative organ and tissue weights, alterations in clinical chemistry, urinalysis and/or haematological parameters, functional disturbances in the nervous system as well as in organs and tissues in general, and pathological alterations in organs and tissues as examined macroscopically and microscopically. Repeated dose toxicity studies may also examine parameters that have the potential to identify specific manifestations of toxicity such as e.g. neurotoxicity, immunotoxicity, endocrine-mediated effects, reproductive toxicity and carcinogenicity.

An *adverse effect* is a change in the morphology, physiology, growth, development, reproduction or life span of an organism, system, or (sub) population that results in an impairment of functional capacity, or an impairment of the capacity to compensate for additional stress, or an increase in susceptibility to other influences (OECD, 2003).

A chemical substance may induce systemic and/or local effects.

- A *local effect* is an effect that is observed at the site of first contact, caused irrespective of whether a substance is systemically available.
- A *systemic effect* is defined as an effect that is normally observed distant from the site of first contact, i.e. after the substance has passed through a physiological barrier (mucous membrane of the gastro-intestinal tract or of the respiratory tract, or the skin) and becomes systemically available.
- It should be noted, however, that systemic effects may occur as a consequence of a local action (i.e. secondary effects where systemic availability of a substance is not necessarily required).

Vice versa, toxic effects on surface epithelia may reflect indirect effects as a consequence of systemic toxicity or secondary to systemic distribution of the substance or its active metabolite(s).

R.7.5.1.2 Objective of the guidance on repeated dose toxicity

The objectives of this Guidance are to address the REACH information requirements related to repeated dose toxicity testing and to inform the registrant about how he can meet these requirements.

The objectives of assessing repeated dose toxicity are to evaluate:

1. adverse effects based on human or non human studies:

    - whether exposure of humans to a substance is associated with adverse toxicological effects occurring as a result of repeated daily exposure for a part of the expected lifetime or for the major part of the lifetime; these human studies potentially may also identify populations that have higher susceptibility;

    - whether administration of a substance to experimental animals causes adverse toxicological effects as a result of repeated daily exposure for a part or a major part of the expected lifespan; effects that are predictive of possible adverse human health effects;

2. the target organs, potential cumulative effects and the reversibility of the adverse toxicological effects;

3. the dose-response relationship and threshold for any of the adverse toxicological effects observed in the repeated dose toxicity studies;

4. the basis for risk characterisation and classification and labelling (C&L) of substances for repeated dose toxicity;

5. the mode of action (MOA) and mechanism data.

R.7.5.2 Information requirements for repeated dose toxicity

Section R.2.1 in Chapter R.2 of the *Guidance on IR&CSA* provides general guidance on the information requirements of the REACH Regulation. For repeated dose toxicity, all available information relevant for the endpoint needs to be evaluated and classification under Regulation (EC) No 1272/2008 on the *Classification, labelling and packaging of substances and mixtures* (CLP Regulation) considered at each tonnage level. The following standard information requirements on repeated dose toxicity are specified in Annexes VII-X to the REACH Regulation:

- In **Annex VII** (≥ 1 t/y), no test requirements on repeated dose toxicity are specified additionally to the available information relevant for repeated dose toxicity.

- In **Annex VIII** (≥ 10 t/y), a short-term repeated dose toxicity study (28 days) is usually required, in one species, male and female, using the most appropriate route of administration, having regard to the likely route of human exposure.

- In **Annex IX** (≥ 100 t/y), a sub-chronic repeated dose toxicity study (90-days) is usually required, in one species (90-day study in rodents), male and female, and a short-term repeated dose toxicity study (28 days) is the minimum requirement, using the most appropriate route of administration, having regard to the likely route of human exposure. It should be noted that a 28-day test is not required at this tonnage level if already provided at Annex VIII level or if a 90-day study is proposed.

- In **Annex X** (≥ 1000 t/y), no specific tests in addition to those required in Annexes VIII-IX for repeated dose toxicity are standard information requirements at this tonnage level. However, following Annex X, column 2, a long-term repeated toxicity study (≥ 12 months) may be proposed by the registrant or required by the Agency in accordance with Articles 40 or 41 if the frequency and duration of human exposure indicates that a longer term study is appropriate and one of the following conditions is met:

  - serious or severe toxicity effects of particular concern were observed in the 28-day or 90-day study for which the available evidence is inadequate for toxicological evaluation or risk characterisation, or

  - effects shown in substances with a clear relationship in molecular structure with the substance being studied were not detected in the 28-day or 90-day study, or

  - the substance may have a dangerous property that cannot be detected in a 90-day study.

Column 1 of Annexes VII-X to the REACH Regulation establishes the standard information required for all chemical substances and Column 2 lists specific rules according to which the required standard information requirements for individual endpoints may be modified (adapted) by waiving the requirement(s) for certain information, or in certain cases, defining the need for additional or different information (for further details see Section R.2.1 in Chapter R.2 of the *Guidance on IR&CSA*).

In addition to the specific rules for adaptation listed in column 2 of Annexes VII to X, the required standard information may also be adapted according to Annex XI, which specifies general rules for adaptation of the standard testing requirements set out in Annexes VII-X in cases where 1) testing does not appear scientifically necessary, 2) testing is technically not possible, and 3) testing may be omitted based on the exposure scenarios developed in the CSR (substance-tailored exposure-driven testing) (see Section R.5.1 "Exposure based waiving" in Chapter R.5 of the *Guidance on IR&CSA*).

It should also be noted that the introductory sections to Annexes VII-X require that *in vivo* testing must be avoided with corrosive substances at concentration/dose levels causing corrosivity.

Factors that can influence the standard information requirements include the results of other toxicity studies, immediate disintegration of the substance, accumulation of the substance or its metabolites in certain tissues and organs, failure to identify a NOAEL in the required test at a given tonnage level, toxicity of particular concern, exposure route, structural relationships with a known toxic substance, physico-chemical properties of the substance, and use and human exposure patterns. These adaptations are detailed in the stepwise Integrated Testing Strategy (ITS) presented in Section R.7.5.6.

R.7.5.3 Information sources on repeated dose toxicity

Toxicological information, including repeated dose toxicity, can be obtained from publicly available study reports (e.g. from NCI) and assessment reports from risk assessment bodies/institutions (e.g. expert panels from EFSA or the European Commission), unpublished studies, databases and publications such as books, scientific journals, criteria documents, monographs and other publications (see Chapter R.3 of the *Guidance on IR&CSA* for further general guidance). Useful databases containing repeated dose toxicity data are available online. Some examples of freely accessible databases are the Fraunhofer ITEM RepDose database (http://fraunhofer-repdose.de/), ToxRefDB by US-EPA (http://www.epa.gov/comptox/toxrefdb/), ECHA CHEM (www.echa.europa.eu). The last three databases are also freely available within the OECD QSAR Toolbox (www.qsartoolbox.org).

Information relevant for repeated dose toxicity can also be obtained from data on other endpoints, structural analogues and physico-chemical properties.

REACH requires that information be generated by means other than vertebrate animal tests whenever possible. Testing on vertebrate animals must be undertaken only as a last resort. Therefore, before new tests are carried out to determine the hazardous properties of a substance, all available information must be collected and assessed, according to step 1 of Annex VI to the REACH Regulation (see Chapter R.4 of the *Guidance on IR&CSA* for general guidance on the evaluation of information).

R.7.5.3.1 Non-human data on repeated dose toxicity

R.7.5.3.1.1 Non-testing data on repeated dose toxicity

Physico-chemical data

The physico-chemical properties of a substance are essential elements to be considered when selecting a suitable vehicle for dilution and dosing of the tested substance, when deciding on the appropriate administration route to be applied in experimental *in vivo* repeated dose toxicity studies as well as when deciding on exemption from testing in cases where testing is technically not possible.

Guidance on the interpretation of physico-chemical data regarding oral, inhalation and dermal absorption can be found in Section R.7.12.2.1 in Chapter R.7c of the *Guidance on IR&CSA*.

(Q)SAR models

Compared with some other endpoints, the possibility to use (Q)SAR models for the prediction of repeated dose toxicity in a regulatory context is limited. This limitation is due to the complexity of the systemic interactions and effects involved in repeated dose toxicity studies. This complexity is difficult to predict with computational tools. Therefore the use of (Q)SAR models should be seen in the context of *Weight-of-Evidence* considerations, where screening and mechanistic information (including the prediction of target organs and metabolites) from (Q)SARs can support available *in vivo* studies. The (mostly commercial) (Q)SAR models for repeated dose toxicity are described in Appendix R.7.5–2 *(Q)SARs for the prediction of repeated dose toxicity*.

More extensive guidance on the availability and application of (Q)SARs is available in Section R.6.1 in Chapter R.6 of the *Guidance on IR&CSA* (see also OECD, 2014) and in ECHA *Practical Guide 5* on "How to use and report (Q)SARs" available on the ECHA website.

Grouping of substances and read-across approach

The concept of grouping, including both read-across and the related chemical category concept has been developed under the OECD HPV programme (OECD 2007a). This is an approach which might be used to fill data gaps without the need for conducting tests when specific conditions, as specified in Section 1.5 of Annex XI to the REACH Regulation, are met.

Extensive guidance on the application of chemical categories/read across is available in Section R.6.2 in Chapter R.6 of the *Guidance on IR&CSA* (see also OECD, 2014).

More detailed advice on the assessment of read-across can be found in ECHA's Read-Across Assessment Framework – RAAF (see http://echa.europa.eu/en/support/grouping-of-substances-and-read-across). Software such as the OECD QSAR Toolbox can be used to find data for analogues and support read-across cases. The OECD eChemPortal (http://www.echemportal.org/echemportal/index?pageID=0&request_locale=en) can be used to collect further data on suitable analogues.

R.7.5.3.1.2 Testing data on repeated dose toxicity

In vitro data

Currently, no available alternatives to animal testing are considered adequate to be used on their own for regulatory purposes for detecting toxicity after repeated exposure. Numerous *in vitro* systems have been developed over the last decades and have been discussed and summarized in EURL ECVAM reports (Worth and Balls, 2002; Prieto *et al.*, 2005; Prieto *et al.*, 2006; Zuang *et al.*, 2015) and publications (Alder *et al.*, 2011). At present, the *in vitro* models listed in these reports are at the research and development level and cannot be used for repeated dose toxicity prediction purposes, although they are very useful to study individual types of organ toxicity or to assess mechanistic aspects of target organ toxicity, at the tissue, cellular and molecular levels. Some of the limitations of these models include for instance the limited capacities of current cell culture systems to account for kinetics and biotransformation, the difficulty to derive values such as NOAELs from *in vitro* systems and the selection of dose/concentration for *in vitro* experiments that would be relevant for extrapolation to human exposure concentrations. Further development and optimisation of current *in vitro* systems as well as the selection of endpoints relevant to general as well as cell-type-specific mechanisms of toxicity or expression of toxic effects *in vivo* is ongoing. New technologies such as genomics, transcriptomics, proteomics and metabolomics could help in the identification of specific markers of toxicity that occur early in the process of long-term toxic responses and that are mechanistically linked to the underlying pathology. An EURL ECVAM workshop report (Prieto *et al.*, 2006) includes a proposed approach to assess repeated dose toxicity *in vitro* by integrating physiologically-based kinetic (PBK) modelling, the use of biomarkers, and omics technologies. However, this integrated approach is still under development and evaluation and it is not ready for regulatory purposes.

The latest information on the status of alternative methods that are under development can be obtained from the EURL ECVAM website (https://eurl-ecvam.jrc.ec.europa.eu/) and those of other international centres for validation of alternative methods. The registrants are also advised to follow any updates to the ECHA webpage concerning Testing methods and alternatives (http://echa.europa.eu/support/testing-methods-and-alternatives) and the OECD website (http://www.oecd.org/chemicalsafety/testing/oecdguidelinesforthetestingofchemicals.htm) for potential new test guidelines and test guideline updates.

In vitro methods may be used to support read-across or a weight-of-evidence approach.

In vitro data using human cell lines, particularly on metabolism, may assist in study interpretation thereby avoiding the need for unnecessary animal experimentation.

At present, available *in vitro* test data from well-characterised target organ and target system models on, e.g. mode(s) of action / mechanism(s) of toxicity may be useful in the interpretation of observed repeated dose toxicity. In this respect, approaches like the Adverse Outcome Pathways (AOPs) as developed under the OECD chemicals programme assist in the integration of different pieces of evidence, including those derived from the use of *in vitro* methods (http://www.oecd.org/chemicalsafety/testing/adverse-outcome-pathways-molecular-screening-and-toxicogenomics.htm).

Animal data

The most appropriate data on repeated dose toxicity for use in hazard characterisation and risk assessment are primarily obtained from studies in experimental animals conforming to internationally agreed test guidelines. In some cases repeated dose toxicity studies not conforming to conventional test guidelines may also provide relevant information for this endpoint.

It should be noted that repeated dose toxicity studies, if carefully evaluated, may provide information on potential reproductive toxicity and on carcinogenicity (e.g. pre-neoplastic lesions).

The information that can be obtained from the available EU/OECD test guideline studies for repeated dose toxicity is briefly summarised below.

Table R.7.5–1 summarises the parameters examined in these OECD test guideline studies in more detail and gives an overview of the similarities and differences between the various studies. It is to be noticed that a full study using 3 dose levels may not be considered necessary if in a limit test using a single dose level of at least 1000 mg/kg bw/day or a single limit concentration no adverse effects are observed.

It should be noted that the test guidelines given in the Annex to the EU Test Methods (TM) Regulation (Council Regulation (EC) No 440/2008) are initially comparable to the OECD test guidelines (http://www.oecd.org/env/ehs/testing/oecdguidelinesforthetestingofchemicals.htm). However, several OECD test guidelines for repeated dose toxicity (e.g. OECD TGs 407, 412, 413) have recently been updated with significant new information but those changes have not yet been implemented in the EU TM Regulation. As alignment of the test guidelines of the EU TM Regulation with updated OECD test guidelines requires some time, the latest update of a test guideline (OECD TG and/or EU method) should be used for conducting new tests. Further details of the study protocols are described in the respective test guidelines.

- Repeated dose 28-day toxicity studies:

Separate guidelines are available for studies using oral administration (OECD TG 407 / EU B.7), dermal application (OECD TG 410 / EU B.9) and inhalation (OECD TG 412 / EU B.8). The principle of these study protocols is identical although the OECD TG 407 protocol includes additional parameters compared to those for dermal and inhalation administration, enabling the identification of a neurotoxic potential, immunological effects or reproductive organ toxicity. In addition, OECD TG 407 allows certain endocrine mediated effects to be put into context with other toxicological effects.

The 28-day studies provide information on the toxicological effects arising from exposure to the substance of young adult animals during a relatively limited period of the animals' life span.

Supplementary information on persistence and reversibility of effects can be gained by the use of additional control and top dose satellite groups.

- Repeated dose 90-day toxicity studies:

Separate guidelines are available for studies using oral administration (OECD TGs 408 and 409 / EU B.26 and B.27 in rodent and non-rodent species, respectively), dermal application (OECD TG 411 / EU B.28), or inhalation (OECD TG 413 / EU B.29). The principle of these study protocols is identical although the revised OECD TG 408 protocol includes additional parameters compared to those for dermal and inhalation administration, enabling the identification of a neurotoxic potential, immunological effects or reproductive organ toxicity.

The 90-day studies provide information on the general toxicological effects arising from sub-chronic exposure (a prolonged period of the animals' life span) covering post-weaning maturation and growth well into adulthood, on target organs and on potential accumulation of the substance.

Supplementary information on persistence and reversibility of effects can be gained by the use of additional control and top dose satellite groups.

- Chronic toxicity studies:

The chronic toxicity studies (OECD TG 452 / EU B.30) provide information on the toxicological effects arising from repeated exposure over a prolonged period of time covering the major part of the animals' life span. The duration of the chronic toxicity studies should be at least 12 months.

The combined chronic toxicity / carcinogenicity studies (OECD TG 453 / EU B.33) include an additional high-dose satellite group for evaluation of pathology other than neoplasia. The satellite group should be exposed for at least 12 months and the animals in the carcinogenicity part of the study should be retained in the study for the majority of the normal life span of the animals.

Ideally, the chronic studies should allow for the detection of general toxicity effects (physiological, biochemical and haematological effects, etc.) but could also inform on neurotoxic, immunotoxic, reproductive and carcinogenic effects of the substance. However, in 12-month studies, non-specific life shortening effects, which require a long latent period or are cumulative, may possibly not be detected. In addition, the combined study will allow for detection of neoplastic effects and a determination of a carcinogenic potential and life-shortening effects.

- The combined repeated dose toxicity study with the reproduction/ developmental toxicity screening test:

The combined repeated dose toxicity / reproductive screening study (OECD TG 422[100]) provides information on the toxicological effects arising from repeated exposure (generally oral exposure) over a period of minimum 4 weeks for males and approximately 63 days for females (a relatively limited period of the animals' life span) as well as on reproductive toxicity. For the repeated dose toxicity part, OECD TG 422 is in concordance with OECD TG 407 / EU B.7 except for the use of pregnant females, for which exposure duration (of female animals) is longer in OECD TG 422 compared to OECD TG 407 / EU B.7.

It has to be noted that the animals used to test for sub-chronic toxicity (OECD TG 407) are usually younger (juveniles, younger than 9 weeks) than the animals used in a combined repeated dose toxicity /reproductive screening study (OECD TG 422; adults, 10-12 weeks old). This age difference may lead to differences in toxicokinetics and susceptibility to the toxicity of the substance to be tested.

- Neurotoxicity studies:

The neurotoxicity study in rodents (OECD TG 424 / EU B.43) has been designed to further characterise potential neurotoxicity observed in repeated dose systemic toxicity studies. The neurotoxicity study in rodents will provide detailed information on major neuro-behavioural and neuro-pathological effects in adult rodents.

- Delayed neurotoxicity studies of organophosphorus substances:

The delayed neurotoxicity study (OECD TG 419 / EU B.38) is specifically designed to be used in the assessment and evaluation of the neurotoxic effects of organophosphorus substances. This study provides information on the delayed neurotoxicity arising from repeated exposure over a relatively limited period of the animals' life span.

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[100] To date there is no corresponding EU test method available.

Table R.7.5–1 Overview of *in vivo* repeated dose toxicity test guidelines

Test	Design	Endpoints
OECD TG 407 (2008) (EU B.7) Repeated dose 28-day oral toxicity study in rodents	Exposure for 28 days At least 3 dose levels (unless limit test) plus control At least 5 males and 5 females per group Rodents, preferred species: rat	Clinical observations Functional observations Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis (optional) Plasma or serum markers of general tissue damage (optional) Oestrus cycle (optional) T3, T4, TSH (optional) Gross necropsy Organ weights Histopathology
OECD TG 410 (1981) (EU B.9) Repeated dose dermal toxicity: 21/28-day study	Exposure for 21/28 days At least 3 dose levels (unless limit test) plus control At least 5 males and females per group Rat, rabbit or guinea pig	Clinical observations Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis (optional) Gross necropsy Organ weights Histopathology
OECD TG 412 (2009) (EU B.8) Repeated dose inhalation toxicity: 28-day or 14-day study	Exposure for 28 or 14 days At least 3 concentrations (unless limit test) plus control At least 5 males and 5 females per group Rodents, preferred species: rat	Clinical observations Body weight and food/water consumption Haematology Clinical biochemistry Bronchoalveolar lavage (BAL) fluid analysis (optional) Urinalysis (optional) Gross necropsy Organ weights Histopathology

OECD TG 408 (1998) (EU B.26) Repeated dose 90-day oral toxicity study in rodents	Exposure for 90 days At least 3 dose levels (unless limit test) plus control At least 10 males and 10 females per group Rodents, preferred species: rat	Clinical observations Ophthalmological examination Functional observations Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis Gross necropsy Organ weights
OECD TG 409 (1998) (EU B.27) Repeated dose 90-day oral toxicity study in non-rodents	Exposure for 90 days At least 3 dose levels (unless limit test) plus control At least 4 males and females per group Non-rodents, commonly used: dog	Clinical observations Ophthalmological examination Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis Gross necropsy Organ weights Histopathology
OECD TG 411 (1981) (EU B.28) Subchronic dermal toxicity: 90-day study	Exposure for 90 days At least 3 dose levels (unless limit test) plus control At least 10 males and females per group Rat, rabbit or guinea pig	Clinical observations Ophthalmological examination Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis (optional) Gross necropsy Organ weights Histopathology
OECD TG 413 (2009) (EU B.29) Subchronic inhalation toxicity: 90-day study	Exposure for 90 days At least 3 concentrations (unless limit test) plus control At least 10 males and females per group Rodents, preferred species: rat	Clinical observations Ophthalmological examination Body weight and food/water consumption Haematology Clinical biochemistry Bronchoalveolar lavage (BAL) fluid analysis (optional) Urinalysis (optional) Gross necropsy Organ weights Histopathology

OECD TG 452 (2009) (EU B.30) Chronic toxicity studies	Exposure for 12 months (longer or shorter duration can be used, but must be adequately justified) At least 3 dose levels (unless limit test) plus control Rodents: At least 20 males and 20 females per group Non-rodents: At least 4 males and 4 females per group Preferred rodent species: rat Preferred non-rodent species: dog	Clinical observations, including neurological changes Ophthalmological examination Body weight and food/water consumption Haematology Clinical biochemistry Urinalysis Gross necropsy Organ weights Histopathology
OECD TG 453 (2009) (EU B.33) Combined chronic toxicity / carcinogenicity studies	Exposure for 12 months (longer or shorter duration can be used, but must be adequately justified), or majority of normal life span (carcinogenicity part) At least 3 dose levels (unless limit test) plus control Chronic toxicity: At least 10 males and 10 females per group Carcinogenicity: At least 50 males and 50 females per group Preferred rodent species: rat Preferred non-rodent species: dog	Essentially as in TG 452 for chronic toxicity

OECD TG 422[101] (2016) Combined repeated dose toxicity study with the reproduction/developmental toxicity screening test	Exposure from 2 weeks prior to mating for a minimum of 4 weeks (males) or until at least post-natal day 13[102] (females – at least 9 weeks of exposure) At least 3 dose levels (unless limit test) plus control At least 10 males and 12-13 females per group Species: rat	Clinical observations as in TG 407 Functional observations as in TG 407 Body weight and food/water consumption Haematology as in TG 407 Hormonal measurements (thyroid hormone) Clinical biochemistry Urinalysis (optional) Gross necropsy Organ weights
OECD TG 424 (1997) (EU B.43) Neurotoxicity study in rodents	Exposure for at least 28 days At least 3 dose levels (unless limit test) plus control; At least 10 males and 10 females per group Rodents, preferred species: rat	Detailed clinical observations Functional observations Ophthalmological examination Body weight and food/water consumption Haematology (if in combination with a repeated dose systemic toxicity study) Clinical biochemistry (if in combination with a repeated dose systemic toxicity study) Histopathology
OECD TG 419 (1995) (EU B.38) Delayed neurotoxicity of organophosphorus substances: 28-day repeated dose study	Exposure for 28 days At least 3 dose levels (unless limit test) plus control At least 12 birds per group Species: domestic laying hen	Detailed clinical observations Body weight Clinical biochemistry Gross necropsy

- Other studies providing information on repeated dose toxicity:

Although not aiming at investigating repeated dose toxicity *per se*, other available OECD/EU test guideline studies involving repeated exposure of experimental animals may provide useful information on repeated dose toxicity. These studies are summarised in Table R.7.5–2.

The one- and two-generation studies (OECD TGs 415 and 416 / EU B.34 and B.35) and the extended one-generation reproductive toxicity study (OECD TG 443 / EU.B.56) may provide

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101 To date there is no corresponding EU test method available.

102 OECD TG 422 was updated in 2016; according to the previous version of OECD TG 422, exposure was at least until post-natal day 4.

information on the general toxicological effects arising from repeated exposure over a prolonged period of time (about 90 days for parental animals) as clinical signs of toxicity, body weight, selected organ weights, and gross and microscopic changes of selected organs are recorded.

The prenatal developmental toxicity study (OECD TG 414 / EU B.31), the reproduction/developmental toxicity screening study (OECD TG 421[103]) and the developmental neurotoxicity study (OECD TG 426[103]) may give some indications of general toxicological effects arising from repeated exposure over a relatively limited period of the animals life span as clinical signs of toxicity and body weight are recorded.

The carcinogenicity study (OECD TG 451 / EU B.32) will, in addition to information on neoplastic lesions, also provide information on the general toxicological effects arising from repeated exposure over a major portion of the animal's life span as clinical signs of toxicity, body weight, and gross and microscopic changes of organs and tissues are recorded.

No OECD or EU test method is currently available to investigate immunotoxicity. However, the "Health Effects Test Guidelines OPPTS 870.7800 Immunotoxicity" can be referred to.

Table R.7.5–2 Overview of other *in vivo* test guideline studies giving information on repeated dose toxicity

Test	Design	Endpoints (general toxicity)
OECD TG 443 (2012) (EU B.56) Extended one-generation reproductive toxicity study	Exposure of 10 weeks (unless specific reasons to shorten) prior to mating (P) until post-natal day 90-120 (F1). If the extension of Cohort 1B is triggered, then until post-natal day 4 or 21 (F2) At least 3 dose levels (unless limit test) plus control Sufficient mating pairs to produce 20 animals per dose group (P generation), 20 mating pairs for extension of Cohort 1B, if triggered 10 males and 10 females per dose group for each of the Cohorts 2A, 2B, and/or 3, if triggered. Preferred species: rat	Clinical observations Body weight and food/water consumption Clinical chemistry Haematology Thyroid hormones (T4 and TSH) Clinical biochemistry Urinalysis Sperm parameters Gross necropsy (adults) Splenic lymphocyte subpopulation analysis Organ weights Histopathology Certain parameters for endocrine mode of action Specific investigation on developmental neurotoxicity, in cases of a particular concern, and/or developmental immunotoxicity based on a particular concern

[103] To date there is no corresponding EU test method available.

OECD TG 416 (2001) (EU B.35) Two-generation reproduction toxicity study	Exposure before mating for at least 10 weeks until the end of the mating period (males) or until weaning of 2nd generation (females) At least 3 dose levels (unless limit test) plus control Sufficient number of animals to yield preferably not less than 20 pregnant females per dose group Preferred species: rat	Clinical observations Body weight and food/water consumption Oestrus cycle Sperm parameters Gross necropsy (all parental animals) Organ weights Histopathology
OECD TG 415 (1983) (EU B.34) One-generation reproduction toxicity Study	Males: Exposure before mating for at least one spermatogenic cycle until end of mating period Females: Exposure before mating for at least two weeks until weaning of 1st generation At least 3 dose levels (unless limit test) plus control Sufficient number of animals to yield about 20 pregnant females per dose group Species: rat or mouse	Clinical observations Body weight and food consumption Gross necropsy Histopathology
OECD TG 414 (2001) (EU B.31) Prenatal developmental toxicity study	Exposure at least from implantation to one or two days before expected birth At least 3 dose levels (unless limit test) plus control Sufficient number of females to result in approximately 20 female animals with implantation sites Preferred rodent species: rat Preferred non-rodent species: rabbit	Clinical observations Body weight and food/water consumption Macroscopical examination of all dams
OECD TG 421[104] (2016) Reproduction/ developmental toxicity screening test	Males: Exposure before mating for at least two weeks until end of mating period Females: Exposure before mating for at least two weeks until at least post-natal day 13[105]	Clinical observations Clinical observations Body weight and food/water consumption Oestrus cycle Clinical chemistry

[104] To date there is no corresponding EU test method available.

[105] OECD TG 421 was updated in 2016; according to the previous version of OECD TG 421, exposure was at least until post-natal day 4.

	At least 3 dose levels (unless limit test) plus control At least 10 males and 12-13 females per group Species: rat	Thyroid hormones (T4) Gross necropsy Organ weights Histopathology
OECD TG 426[104] (2007) Developmental neurotoxicity study	Exposure at least from implantation throughout lactation (PND 21) At least 3 dose levels (unless limit test) plus control At least 20 pregnant females per group Preferred species: rat	Clinical observations Body weight and food/water consumption
OECD TG 451 (2009) (EU B.32) Carcinogenicity studies	Exposure for majority of normal life span, normally 24 months At least 3 dose levels plus control At least 50 males and 50 females per group Rodents, preferred species: rat	Clinical observations (special attention to tumour development) Body weight and food/water consumption Haematology (optional) Clinical chemistry (optional) Urinalysis (optional) Gross necropsy Histopathology

R.7.5.3.2 Human data on repeated dose toxicity

Human data adequate to serve as the sole basis for the hazard and dose-response assessment are rare. When available, reliable and relevant human data are preferable over animal data and can contribute to the overall *Weight of Evidence*. Nonetheless, lack of positive findings in humans does not necessarily overrule positive and good quality animal data.

Human volunteer studies are not recommended due to practical and ethical considerations involved in deliberate exposure of individuals to chemical substances. However, the following types of human data may already be available:

- Analytical epidemiology studies on exposed populations. These data may be useful for identifying a relationship between human exposure and effects such as biological effect markers, early signs of chronic effects, disease occurrence, or long-term specific mortality risks. Study designs include case control studies, cohort studies and cross-sectional studies.

- Descriptive or correlation epidemiology studies. They examine differences in disease rates among human populations in relation to age, gender, race, and differences in temporal or environmental conditions. These studies may be useful for identifying priority areas for further research but not for dose-response information.

- Case reports describe a particular effect in an individual or a group of individuals exposed to a substance. Generally case reports are of limited value for hazard identification, especially if the exposure represents single exposures, abuse or misuse of certain substances.

- Controlled studies in human volunteers. These studies, including low exposure toxicokinetic studies, might also be of use in risk assessment.

- Information from occupational surveillance (major chemical companies often have a routine medical surveillance system in place to monitor and manage employee health).

- Postmarketing surveillance data (e.g. from certain consumer products, cosmetics).

- Meta-analysis. In this type of study data from multiple studies are combined and analysed in one overall assessment of the relative risk or dose-response curve.

R.7.5.3.3 Exposure considerations on repeated dose toxicity

Information on exposure, use and risk management measures should be collected in accordance with Article 10 and Annex VI (Section 3) of the REACH Regulation.

Such information may lead to an adaptation of the extent and nature of information needed on repeated dose toxicity under REACH; two types of *adaptations* are possible due to exposure considerations: exposure-based waiving of a study or exposure-based triggering of further studies.

More detailed guidance on exposure-based adaptations of the repeated dose toxicity information requirements is given in Sections R.7.5.4 (Evaluation of available information) and R.7.5.6 (Integrated Testing Strategy).

Furthermore, the most appropriate route of administration to be used in animal studies needs to be considered (for further details see Section R.7.5.6.3.4). Non-physiological routes of human exposure, such as i.v., i.m., s.c., i.p., are usually considered non-appropriate routes of administration for animal testing requested under the REACH Regulation. The relevance of available studies using such routes of administration needs to be evaluated case by case.

R.7.5.4 Evaluation of available information on repeated dose toxicity

General guidance on how to evaluate the available information is given in Chapter R.4 of the *Guidance on IR&CSA*.

R.7.5.4.1 Non-human data on repeated dose toxicity

R.7.5.4.1.1 Non-testing data on repeated dose toxicity

Physico-chemical properties

The physico-chemical properties of a chemical substance under registration should always be considered before any new experimental *in vivo* repeated dose toxicity studies are undertaken.

The physico-chemical properties of a substance can indicate whether it is likely that the substance can be absorbed following exposure to a particular route (oral, dermal or inhalation route) and whether it (or an active metabolite) is likely to reach the target organ(s) and tissue(s). The physico-chemical properties are thus essential elements in deciding on the most appropriate administration route to be applied in experimental *in vivo* repeated dose toxicity studies (see Section R.7.5.4.3).

The physico-chemical properties are also important in order to judge whether testing is technically possible. Testing for repeated dose toxicity may, as specified in Section 2 of Annex XI to the REACH Regulation, be omitted if it is technically not possible to conduct the study as a consequence of the properties of the substance (e.g. unstable substances cannot be used, or mixing of the substance with water may cause danger of fire or explosion). Annex XI further emphasises that the guidance given in the test methods referred to in REACH Article 13(3), more specifically on the technical limitations of a specific method, must always be respected.

Additional generic guidance on the use of physico-chemical properties is provided for instance in Section R.7.12 on toxicokinetics, in Chapter R.7c of the *Guidance on IR&CSA*.

Grouping of substances and read-across approach

The grouping of substances and read-across offer a possibility for adaptation of the standard information requirements of the REACH Regulation. If the read-across approach is adequate, unnecessary testing can be avoided. A read-across approach can also support a conclusion for a REACH information requirement using a *Weight-of-Evidence* approach.

Guidance on read-across is provided in Chapter R.6 “QSAR and grouping of chemicals” of the *Guidance on IR&CSA* (see also OECD, 2014). It specifies that the terms *category approach* and *analogue approach* are used to describe techniques for grouping chemicals, whilst the term *read-across* is reserved for a technique of filling data gaps in either approach. This guidance also presents recommendations on the methodology for developing grouping and read-across approaches. Furthermore, ECHA has developed and published a RAAF to provide experts with a transparent and structured methodology to assess read-across approaches. The RAAF description is available on ECHA’s website (http://echa.europa.eu/en/support/grouping-of-substances-and-read-across).

The read-across approach has to be considered per information requirement due to the different complexities (e.g. key parameters, biological targets) of the studies needed to meet the information requirement. This means that read across (and the category approach) is specific for the property under consideration and therefore requires a specific read-across hypothesis and justification for predicting individual properties

In the context of a grouping and read-across approach under REACH, adequate and reliable supporting evidence needs to be provided to substantiate scientific claims or hypotheses constituting the basis for predicting properties of a substance from data on another substance. Supporting evidence is not sufficient on its own to determine the property of the substance under consideration, but rather contributes to strengthening and justifying the read-across hypothesis.There may be several lines of evidence used to justify read-across, with the aim of strengthening the case. The potential of different types of supporting information (e.g. toxicokinetics data, metabolomics, high throughput screening data, ...) to strengthen grouping and read-across approaches is captured in the proceedings from a workshop on the use of new approach methodologies in regulatory science held in ECHA on 19-20 April 2016 (https://echa.europa.eu/view-article/-/journal_content/title/topical-scientific-workshop-new-approach-methodologies-in-regulatory-science).

In principle, it is possible to predict the presence or absence of a property/effect by applying a read-across approach. For prediction of an absence of effect(s), typically no mechanistic insight is available to support such a claim. The absence of effect(s) may however be explained by other arguments, e.g. the absence of exposure of biological target(s) or a lack of biological interaction leading to an adverse outcome. These situations need to be addressed in the read-across hypothesis and read-across justification and should be supported by evidence.

The provisions of Section 1.5 of Annex XI to the REACH Regulation require that the results of grouping and read-across approaches “should be adequate for the purpose of classification and labelling and/or risk assessment”. Repeated-dose toxicity studies are typically used to derive

C&L and DNELs on the basis of the strength of the observed effects (e.g. use the identified NOAEL as point of departure). For a prediction, this requires that the source study(ies) allow(s) for the identification of known value(s) of a property for one or more source substances which is then used to estimate the unknown value of the same property for the target substance. In this situation, it is essential to provide a robust scientific basis and quantitative supporting evidence (e.g. toxicokinetic information) to demonstrate that the type of effect and its strength observed in the source study can be used for C&L and/or risk assessment purposes for the target substance without under-estimating the property of the target substance under consideration.

Information on practical aspects of how to report read-across and/or category approaches in IUCLID is provided in ECHA *Practical Guide 6* on "How to use alternatives to animal testing to fulfil your information requirements for REACH registration".

(Q)SAR

A (Q)SAR analysis for a substance may give indications for a specific mechanism to occur and identify possible organ or systemic toxicity upon repeated exposure. The reliability, applicability and overall scope of (Q)SAR science to identify chemical hazard and assist in risk assessment have been evaluated by various groups and organisations. Guidance on this issue is presented in Section R.6.1 in Chapter R.6 of the *Guidance on IR&CSA* (see also OECD, 2014) and in OECD Monograph No. 69 (OECD 2007b). Application of (Q)SARs should be documented according to the appropriate reporting formats: QSAR model reporting format (QMRF, see Section R.6.1.9) and QSAR prediction reporting format (QPRF, see Section R.6.1.10).

Overall, (Q)SAR approaches are currently not well validated for repeated dose toxicity and consequently no firm recommendations can be made concerning their routine use in a testing strategy in this area. There are a large number of potential targets/mechanisms associated with repeated dose toxicity that today cannot be adequately covered by a battery of (Q)SAR models. Therefore, a negative result from current (Q)SAR models without other supporting evidence cannot be interpreted as demonstrating a lack of toxicological hazard or lack of a need for hazard classification. Another limitation of (Q)SAR modelling is that dose-response information, including the N(L)OAEL, is not provided. Similarly, a validated (Q)SAR model might identify a potential toxicological hazard, but because of limited confidence in this approach, such a result may not be adequate to support hazard classification with respect to repeated dose toxicity.

In some cases, (Q)SAR results could be used as part of a *Weight-of-Evidence* approach, when considered alongside other data, provided the applicability domain is appropriate. Also, (Q)SAR data can be used as supporting evidence when assessing the toxicological properties by read-across within a substance grouping approach, providing the applicability domain is appropriate. Positive and negative (Q)SAR modelling results can be of value in a read-across assessment and for classification purposes.

R.7.5.4.1.2 Testing data on repeated dose toxicity

In vitro data

As mentioned earlier in Section R.7.5.3.1, data from currently available *in vitro* tools are not considered adequate to be used on their own for regulatory decision making with respect to risk assessment and C&L for repeated dose toxicity. However, such data may be helpful in the assessment of repeated dose toxicity, for instance to detect local target organ effects and/or to clarify the mechanisms of action. Since, at present, there are no *in vitro* methods validated and accepted for regulatory purposes (Adler *et al*., 2011; Zuang *et al.*, 2015), the quality of each of these *in vitro* studies and the adequacy of the data provided should be carefully evaluated. Furthermore, the concentrations used in *in vitro* tests should be compared to the exposure conditions *in vivo*.

Generic guidance is given in Chapters R.4 and R.5 of the *Guidance on IR&CSA* for judging the applicability and validity of the outcome of various study methods, assessing the quality of the conduct of a study, reproducibility of data and aspects such as vehicle, number of replicates, exposure/incubation time, GLP-compliance or comparable quality description.

In addition, information from AOPs (http://www.oecd.org/chemicalsafety/testing/adverse-outcome-pathways-molecular-screening-and-toxicogenomics.htm) can assist in the organisation of existing knowledge for a specific toxicological endpoint and the identification of knowlegde gaps, where more research is needed to understand the underlying mechanism. It can also aid in chemical hazard characterisation and guide the development of new testing approaches that use fewer or no animals. AOP approaches can be used within the weight-of-evidence concept.

Animal data

The basic concept of repeated dose toxicity studies to generate data on target organ toxicity following sub-acute to chronic exposure is to treat experimental animals repeatedly for 2-4 weeks, 13 weeks or longer. These studies are mentioned in Section R.7.5.3.1 and summarised in Table R.7.5–1.

In addition, other studies performed in experimental animals may provide useful information on repeated dose toxicity. While at present most alternative methods (e.g. (Q)SAR, *in vitro* tests) remain at the research and development stage and are not ready as surrogates for sub-chronic/chronic animal studies, there are opportunities to improve data collection for risk assessment providing greater efficiency and use of fewer animals and better use of resources. Although not required by REACH, other opportunities include obtaining toxicokinetic data at an early stage, in conjunction with repeated dose toxicity testing, thus ensuring that the maximum amount of information is drawn from the animal studies and for use in the risk assessment process.

The number of repeated dose toxicity studies available for a substance under registration is likely to be variable, ranging from none, a dose-range finding study, a 28-day repeated dose toxicity guideline study, to a series of guideline studies for some substances, including sub-chronic and/or chronic studies. There may also be studies employing different species and routes of exposure. In addition, special toxicity studies investigating further the nature, mechanism and/or dose-relationship of a critical effect in a target organ or tissue may also have been performed for some substances.

The following general guidance is provided for the evaluation of repeated dose toxicity data and the development of the *Weight of Evidence*:

- Studies on the most sensitive animal species should be selected as the significant ones, unless toxicokinetic and toxicodynamic data show that this species is less relevant for human risk assessment.

- Studies using an appropriate route, duration and frequency of exposure in relation to the expected route(s), frequency and duration of human exposure have greater weight.

- Studies enabling the identification of a NOAEL or a Benchmark dose Lower Confidence Limit (BMDL), and a robust hazard identification have a greater weight.

- A BMDL can be used in parallel to derive a NOAEL or as an alternative when there is no reliable NOAEL. In addition, the BMD approach is, when possible, preferred over the LOAEL-NAEL (No Adverse Effect Level) extrapolation.

- Studies of a longer duration should be given greater weight than a repeated dose toxicity study of a shorter duration in the determination of the most relevant NOAEL or BMDL.

- If sufficient evidence is available to identify the critical effect(s) (with regard to the dose-response relationship(s) and to the relevance for humans), and the target organ(s) and/or tissue(s), greater weight should be given to specific studies investigating this effect in the identification of a NOAEL or BMDL. The critical effect can be a local as well as a systemic effect.

While data available from repeated dose toxicity studies not performed according to conventional guidelines and/or GLP may still provide information of relevance for risk assessment and C&L, such data require extra careful evaluation. Annex XI to the REACH Regulation specifically identifies circumstances where use of existing studies not carried out according to GLP or test methods referred to in Article 13(3) (guideline studies) can replace *in vivo* testing performed in accordance with REACH Article 13(3). Data from non-guideline studies must be considered to be equivalent to data generated by the corresponding test methods referred to in REACH Article 13(3) if the following conditions are met:

- adequacy for the purpose of C&L and/or risk assessment;

- adequate and reliable coverage of the key parameters foreseen to be investigated in the corresponding test methods referred to in REACH Article 13(3);

- exposure duration comparable to or longer than the corresponding test methods referred to in REACH Article 13(3) if exposure duration is a relevant parameter; and

- adequate and reliable documentation of the study is provided.

In all other situations, non-guideline studies may contribute to the overall weight of the evidence but they cannot stand alone for a hazard and risk assessment of a substance. Thus, such studies cannot serve as the sole basis for an assessment of repeated dose toxicity or for exempting from the standard information requirements for repeated dose toxicity at a given tonnage level, i.e. they cannot be used to identify a substance as being adequately controlled in relation to repeated dose toxicity.

If sufficient information from existing studies is available on the repeated dose toxicity potential of a substance in order to perform a risk assessment as well as to conclude on C&L under CLP for specific target organ toxicity arising from a repeated exposure (STOT RE Category 1 or Category 2), no further *in vivo* testing is needed. The existing information is considered sufficient when, based on a *Weight-of-Evidence* analysis, the critical effect(s) and target organ(s) and tissue(s) can be identified, the dose-response relationship(s) and NOAEL(s) and/or LOAEL(s) for the critical effect(s) can be established, and the relevance for human beings can be assessed.

It should be noted that potential effects in certain target organs following repeated exposure may not be observed within the span of the 28-day study. Attention is also drawn to the fact that the protocols for the oral and inhalation 28-day and 90-day studies include additional parameters compared to those for the 28-day and 90-day dermal protocols.

Where it is considered that the existing data as a whole are inadequate for providing a clear assessment of this endpoint, the need for further testing should be considered in view of all available relevant information on the substance, including use pattern, the potential for human exposure, physico-chemical properties, and structural alerts. The testing strategy is presented in Section R.7.5.6.3.

Information from existing data on neurotoxicity or immunotoxicity or specific mode of action should be evaluated.

Regarding neurotoxicity and immunotoxicity, standard oral 28-day and 90-day toxicity studies include endpoints capable of detecting such effects. Indicators of neurotoxicity include clinical observations, a functional observational battery, motor activity assessment and histopathological examination of spinal cord and sciatic nerve. Indicators of immunotoxicity include changes in haematological parameters, serum globulin levels, alterations in immune system organ weights such as spleen and thymus, and histopathological changes in immune organs such as spleen, thymus, lymph nodes and bone marrow. Where data from standard oral 28-day and 90-day studies identify evidence of neurotoxicity or immunotoxicity, other studies may be necessary to further investigate the effects.

Additional guidance on immunotoxicity is available from the WHO/IPCS Guidance on Immunotoxicity for risk assessment (WHO, 2012).

More focus has also been put on endocrine disruptors. In relation to hazard and risk assessment, there are currently no test methods available that specifically detect all effects which have been linked to endocrine disruption mechanism. Guidance is available to facilitate the interpretation of hazard data derived from screens and tests in the OECD conceptual framework (see http://www.oecd.org/env/ehs/testing/oecdworkrelatedtoendocrinedisrupters.htm#GD_Standardized_TG) has been published in 2012 (OECD, 2012).

Further Guidance on mode of action analysis is available from the WHO/IPCS framework on Mode of action and human relevance. The framework provides a structured and transparent approach to perform a *Weight-of-Evidence* analysis on mode of action (Meek *et al.*, 2014).

If data are not available from a standard oral 28-day repeated dose toxicity guideline study (OECD TG 407 / EU B.7), the minimum repeated dose toxicity data requirement (28-day study) at tonnage levels from 10 t/y may in certain circumstances be met by results obtained from *the combined repeated dose toxicity study with the reproduction/developmental toxicity screening test* (OECD TG 422[106]). One advantage of this approach is to obtain information on repeated dose toxicity and reproductive toxicity in a single study, providing an overall saving in the number of animals used for testing. In addition, the number of animals is higher (10 per sex compared to 5 per sex in the standard oral 28-day study)[107] and the dosing period is longer in the combined study than in the standard oral 28-day study. Therefore, more information on repeated dose toxicity could be expected from the combined study. Potential complications in using the combined study include the selection of adequate dose levels to examine adequately both repeated dose toxicity and reproductive toxicity. In addition, interpretation of the results may be complicated due to differences in sensitivity between pregnant and non-pregnant animals, and an assessment of the general toxicity may be more difficult especially when serum and histopathological parameters are not evaluated at the same time in the study. Consequently, where the combined study is used for the assessment of repeated dose toxicity, the use of data obtained from such a study should be clearly indicated. Despite such complications, the use of the combined study is recommended for the initial hazard assessment of the repeated dose toxicity potential of a substance when this study is also relevant for reproductive toxicity assessment.

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[106] To date there is no corresponding EU test method available.

[107] Histopathological examination of reproductive organs and of all organs showing macroscopic lesions is required for all adult animals. All other organs are investigated in 5 animals per sex and dose.

In general, results from toxicological studies requiring repeated administration of a test substance (see also Section R.7.5.3.1) such as *reproduction and developmental toxicity studies* can contribute to the assessment of repeated dose toxicity. However, such toxicological studies rarely provide the information obtained from a standard repeated dose toxicity study and, therefore, cannot be used as the sole basis for the assessment of repeated dose toxicity or for exempting from the standard information requirements for repeated dose toxicity at a given tonnage level.

Studies such as *acute toxicity, in vivo irritation* as well as *in vivo genotoxicity studies* contribute limited information to the overall assessment of the repeated dose toxicity. However, such studies may be useful in deciding on the dose levels for use in repeated dose toxicity and may also provide some information on the nature of effects (local, systemic).

Guidance on the dose selection for repeated dose toxicity testing (see also Table R.7.5–1) is provided in detail in the EU and OECD test guidelines. Unless limited by the physico-chemical properties or biological effects of the test substance, the highest dose level should be chosen with the aim to induce toxicity but not death or severe suffering.

Although not required by REACH, toxicokinetic studies may be helpful in the evaluation and interpretation of repeated dose toxicity data, for example in relation to accumulation of a substance or its metabolites in certain tissues or organs as well as in relation to mechanistic aspects of repeated dose toxicity and species differences. Toxicokinetic information can also be used in the selection of the dose levels. When conducting repeated dose toxicity studies it is necessary to ensure that the observed treatment-related toxicity is not associated with the administration of excessive high doses causing saturation of absorption and detoxification mechanisms. The results obtained from studies using excessive doses causing saturation of metabolism are often of limited value in defining the risk posed at more relevant and realistic exposure levels where a substance can be readily metabolised and cleared from the body. It is suggested that a key element in designing better repeated dose toxicity studies is to select appropriate dose levels based on results from useful metabolic and toxicokinetic investigations. Further details on the application of toxicokinetic information in the design and evaluation of repeated dose toxicity studies is available in Section R.7.12 on toxicokinetics, in Chapter R.7c of the *Guidance on IR&CSA*.

R.7.5.4.2 Human data on repeated dose toxicity

Human data in the form of epidemiological studies or case reports or information from surveillance programs can contribute to the hazard identification process as well as to the risk assessment process itself. Criteria for assessing the adequacy of epidemiological studies include an adequate research design, formulation of a proper hypothesis, proper selection and characterisation of the exposed and control groups, adequate characterisation of exposure, sufficient duration of follow-up for the disease to develop as an effect of the exposure, valid ascertainment of effect, proper consideration of bias and confounding factors, proper statistical analysis and reasonable statistical power to detect an effect. These types of criteria have been described in more detail by Swaen (2006) and can be derived from Epidemiology Textbooks (Checkoway *et al.*, 1989; Hernberg, 1991; Rothman, 1998).

The results from human experimental studies are often limited by a number of factors, such as a relatively small number of subjects, short duration of exposure, and low dose levels resulting in poor sensitivity in detecting effects.

In relation to hazard identification, the relative lack of sensitivity of human data may cause particular difficulty. Therefore, negative human data cannot be used to override the positive findings in animals, unless it has been demonstrated that the mode of action of a certain toxic response observed in animals is not relevant for humans. In such a case a full justification is required. It is emphasised that testing with human volunteers is strongly discouraged, but when there are good quality data already available they can be used in the overall *Weight of Evidence*.

R.7.5.4.3 Exposure considerations for repeated dose toxicity

R.7.5.4.3.1 Adaptations

Two types of *adaptations* from testing are possible due to exposure considerations: exposure-based waiving of a study and exposure-based triggering of further studies. More information on exposure-based waiving is available in Section R.5.1 in Chapter R.5 of the *Guidance on IR&CSA*. More detailed guidance on exposure-based adaptations of the testing requirements for repeated dose toxicity is given below and in Section R.7.5.6 (Integrated Testing Strategy).

R.7.5.4.3.2 Most appropriate route

Concerning repeated dose toxicity testing, the oral route is the default one because it is assumed to maximise systemic availability (internal dose) of most substances. However, on a case-by-case basis, the appropriateness of other routes of administration should also be assessed. Depending on the physico-chemical properties of a substance and the most relevant route of human exposure, the dermal or the inhalation route can also be appropriate as specified in Annexes VIII and IX to the REACH Regulation.

The dermal route is appropriate if skin contact with the substance in production and/or use is likely, and the physico-chemical (and toxicological) properties suggest a potential for a significant rate of absorption through the skin, and the criteria provided in Section 8.6.1 of Annex VIII and/or column 2 of Section 8.6.2 in Annex IX to the REACH Regulation for the appropriateness of testing by the dermal route are fulfilled. Guidance on the interpretation of physico-chemical data regarding dermal absorption can be found in Table R.7.12-3 in Chapter R.7c of the *Guidance on IR&CSA.*

The inhalation route is appropriate if exposure of humans *via* inhalation is likely taking into account the vapour pressure of the substance and/or the possibility of exposure to aerosols, particles or droplets of an inhalable size. Guidance on the interpretation of physico-chemical data regarding respiratory absorption can be found in Table R.7.12-2 in Chapter R.7c of the *Guidance on IR&CSA.*

If more than one route is appropriate, a decision on the most appropriate route of administration is required (see also Section R.7.5.6.3.4, under "Selection of the most appropriate route of administration").

To support the selection of the route of administration for repeated dose toxicity studies, information on absorption following oral, dermal and/or inhalation exposure could be considered (EFSA, 2012; SCCS, 2016).

Non-physiological routes of human exposure, such as i.v., i.m., s.c., i.p., are usually considered non-appropriate routes of administration for animal testing requested under the REACH Regulation. The relevance of available studies using such routes of administration needs to be evaluated case by case.

R.7.5.4.3.3 Requirement for further studies

According to Annexes VIII-X to the REACH Regulation further studies must be proposed by the registrant or may be required by the Agency for example if there is particular concern regarding exposure, e.g. use in consumer products leading to exposure levels which are:

- close to the dose levels at which toxicity to humans may be expected (Annex VIII);
- higher than the dose levels at which toxicity to humans may be expected (Annex IX);
- close to the dose levels at which toxicity is observed from animal studies (Annex X).

Any of the exposure-triggered studies proposed by the registrant or required by the Agency should be considered on a case-by-case basis.

R.7.5.4.3.4 Waiving of repeated dose toxicity studies

Various types of exposure considerations are a possible basis for the *waiving* of repeated dose toxicity studies. For instance, it is stated in REACH Article 13 and Section 3 of Annex XI that testing in accordance with Sections 8.6 and 8.7 (i.e. repeated dose toxicity and reproductive toxicity) of Annex VIII and with Annexes IX and X may be omitted based on the exposure scenario(s) developed in the Chemical Safety Report. Adequate justification and documentation must in all cases be provided (see Section R.5.1 in Chapter R.5 of the *Guidance on IR&CSA*).

Annex XI, Section 3.2 (a) sets very stringent boundaries/requirements for waiving a repeated dose toxicity study. Three criteria need to be met: (i) concerns "the absence of or no significant exposure", (ii) concerns about relevance and appropriateness of the DNEL and (iii) requires "that exposures are always well below the derived DNEL".

The second criterion requires that the DNEL used must be "relevant and appropriate both to the information requirement to be omitted and for risk assessment purposes". Considering the parameters and observations covered in a sub-chronic study complying with the respective OECD or EU test guideline, it is very unlikely that other types of study would provide information that is as relevant and appropriate. For example, the test duration or histopathology results in studies other than a sub-chronic study would normally not fulfil this standard information requirement. One exception is a chronic toxicity study, which would in most cases cover the information requirement for a sub-chronic study; however, in case a registrant has access to reliable chronic toxicity study data, an exposure-based adaptation would not be needed because a column two adaptation can be applied (see Annex IX, Section 8.6.2). In the legal text, a footnote to the criterion set out in Annex XI, Section 3.2(a)(ii) explicitly rejects the use of a DNEL derived from a 28-day toxicity study for the purpose of waiving the 90-day toxicity study. Therefore, this second criterion will usually not be met and, even when it is, the adaptation possibility of Annex XI, Section 3.2 cannot be applied.

A potentially more likely adaptation possibility is set out in Annex XI, Section 3.2(b), which requires documentation showing that the substance is only handled under strictly controlled conditions. These conditions, i.e. the techniques, controls and procedures that need to be in place in order for the registrant to use this waiving possibility, are specified in Article 18(4) of the REACH Regulation.

Annex XI, Section 3.2(c) deals with substances "permanently embedded in a matrix" and would only apply to these special cases. It is noteworthy that Sections 3.1 and 3.3 of Annex XI are not self-standing or independent waiving possibilities but general requirements, which apply to all adaptations specified under Section 3.2.

Further, the sub-chronic toxicity study (90-day study) does not need to be conducted according to Annex IX to the REACH Regulation if "*the substance is unreactive, insoluble and not inhalable and there is no evidence of absorption and no evidence of toxicity in a 28-day 'limit test', particularly if such a pattern is coupled with limited human exposure*". In order to omit the study the prerequisites interpreted above have to be considered jointly since the word "*and*" is used in between them. In addition, limited human exposure would strengthen the possibility for waiving.

The term "*unreactive*" in the above quotation from the legal text can relate to the inherent chemical reactivity and, as such, can be interpreted as an indicator of the lack of local effects and mutagenicity. The terms "*insoluble and not inhalable*" can be interpreted as indicators of low exposure potential and should be further defined. The terms "*no evidence of absorption*" imply that there has to be evidence of the lack of absorption in order to omit the study. Further, "*no evidence of toxicity in a 28-day limit test*" can be interpreted as meaning that there has to be at least a 28-day limit test available in order to waive the 90-day study, and this 28-day study should not show any sign of toxicity at a dose of 1000 mg/kg bw.

Interpretation of "limited exposure" should encompass the level of exposure, the frequency and/or the duration of exposure. Therefore, "limited exposure" must be considered on a case-by-case basis.

Finally, according to Annex VIII to the REACH Regulation, testing for repeated dose toxicity (28-day study) does not need to be conducted if "*relevant human exposure can be excluded*".

Relevant human exposure depends on the inherent properties of the substance, if the population comes into contact with the substance or not, and how the substance is used. Thus, waiving might be considered on a case-by-case basis.

The concept of Threshold of Toxicological Concern (TTC) might be applied to reduce the use of animals and other evaluation resources (Kroes *et al*., 2004). Use of the TTC concept may also be seen as a driving force for deriving exposure information of adequate quality. However, there are a number of limitations or drawbacks that should be taken into consideration in deciding if the concept is to be applied for industrial chemical substances and further discussions on the cut-off values are needed before integration into the guidance (see Appendix R.7-1 to Chapter R.7, in Chapter R.7c of the *Guidance on IR&CSA*; TemaNord, 2005*).* A review of the Threshold of Toxicological Concern (TTC) approach and development of new TTC decision tree is available from EFSA/WHO (2016).

R.7.5.4.4 Remaining uncertainty on repeated dose toxicity

The key requirement for a CSA is the derivation of DNELs per exposure scenario (box 5 of Figure R.7.5–1). The DNEL for repeated dose toxicity is an assumed no-effect level in humans, derived from a single appropriate repeated dose toxicity study or in a *Weight-of-Evidence* assessment of all available repeated dose toxicity data, and to which is associated an overall assessment factor (AF) that takes into account uncertainty and variability. The following elements contribute to the uncertainty in determination of a threshold for the critical effects and the selection of the AF (further guidance on deriving a DNEL and application of AFs is provided in Chapter R.8 of the *Guidance on IR&CSA*).

R.7.5.4.4.1 Threshold of the critical effect

In the determination of the overall threshold for repeated dose toxicity all relevant information is evaluated to determine the lowest dose that induces an adverse effect (i.e. LOAEL or LOAEC) and the highest level with no biologically and/or statically significant adverse effects (i.e. NOAEL or NOAEC). In this assessment all toxicological responses are taken into account and the critical effect is identified. The uncertainty in the threshold depends on the strength of the data and is largely determined by the design of the underlying experimental data. Parameters such as group size, study type/duration or the methodology need to be taken into account in the assessment of the uncertainty in the threshold of the critical effect(s).

The NOAEL is typically used as the starting point for the derivation of the DNEL. In case a NOAEL has not been achieved, a LOAEL may be used, provided the available information is sufficient for a robust hazard assessment and for C&L. The BMD may also be used as the starting point for the derivation of the DNEL (see Chapter R.8 of the *Guidance on IR&CSA* and EFSA, 2017).

The selection of NOAEL or LOAEL is usually based on the dose levels used in the most relevant toxicity study, without considering the shape of the dose-response curve. Therefore, the NOAEL/LOAEL may not reflect the true threshold for the adverse effect. On the other hand, the BMD is a statistical approach for the determination of the threshold and relies on the dose-response curve. Alternatively, mathematical curve fitting techniques or statistical approaches exist to determine the threshold for an adverse effect. The use of such approaches (e.g. BMD) to estimate the threshold should be considered on a case-by-case basis. For further guidance see Chapter R.8 of the *Guidance on IR&CSA*.

R.7.5.4.4.2 Overall AF

Variability in sensitivity across and within species is another source of uncertainty for repeated dose toxicity. These inter- and intraspecies differences, respectively, are linked with variations in the toxicokinetics and dynamics of a substance. Information derived from non-testing, *in vitro* or *in vivo* methods may lead to an improvement of the understanding of the relevance of animal data for human risk assessment and may lead to a replacement of adopted standard default AFs for these differences.

The quality of the whole database should be assessed for reliability and consistency across different studies and endpoints and take into account the quality of the testing method, size and power of the study design, biological plausibility, dose-response relationships and statistical association.

Missing test data might be substituted by non-testing data obtained from physico-chemical properties, read-across to structurally or mechanistically related substances (SAR/chemical category). (Q)SAR predictions and AOPs could also provide information to be used as part of a *Weight-of-Evidence* approach (for more details on (Q)SAR models for Repeated Dose Toxicity see Appendix R.7.5–2 *(Q)SARs for the prediction of repeated dose toxicity*). *In vitro* data as well as non-standard *in vivo* tests might be used to fill in data gaps. Such data in combination with toxicity tests according to standard OECD/EU guidelines may in some cases lead to an improved understanding of the toxicological effect resulting in a reduction in the overall uncertainty. On the other hand information solely based on *in vitro* and non-testing data is at present insufficient to be used as a surrogate for repeated dose toxicity data and the uncertainty is sufficiently high that such information is unsuitable for use in a CSA and for C&L. In the case of chemical categories, information from non-testing methods or *in vitro* data may be used to fulfil the data requirements for repeated dose toxicity and lead to improvement in the overall reliability and consistency for the read-across within a category of substances.

Since the adequacy and/or completeness of different data may vary, lack of quality and completeness of the overall database should be compensated for by an assessment factor to cover for the remaining uncertainty.

Besides AFs addressing these differences (inter- and intraspecies, quality of the whole database), other uncertainties relating to differences between human and animal exposure conditions (e.g. route and duration), and dose-response characteristics are described in the more extensive guidance on deriving a DNEL (see Section R.8.4.3 in Chapter R.8 of the *Guidance on IR&CSA*).

R.7.5.4.4.3 Other considerations

Another situation may arise when testing is not technically possible, a waiving option indicated in Section 2 of Annex XI to the REACH Regulation (see also Chapter R.5 of the *Guidance on IR&CSA*). In such cases, approaches such as QSAR, category formation and read-across may be helpful in the hazard characterisation (for further information see Chapter R.6 of the

Guidance on IR&CSA and the OECD *Guidance on Grouping of Chemicals*, Second Edition (OECD, 2014)). These approaches should also be considered for generating information that might be suitable as a surrogate for a dose descriptor. Alternatively, generic threshold approaches, e.g. TTC, might be considered for defining the starting point of a risk characterisation (see Appendix R.7-1 to Chapter R.7, in Chapter R.7c of the *Guidance on IR&CSA*).

R.7.5.5 Conclusions on repeated dose toxicity

The evaluation of all available toxicological information for repeated dose toxicity (step 2 in Figure R.7.5–1)

should include an assessment of whether the available information as a whole (i.e. testing and non-testing, and relevant information from studies addressing other endpoints) meets the tonnage-driven data requirements necessary to fulfil the REACH requirements. A *Weight-of-Evidence* approach should be used in assessing the database for a substance. This approach requires a critical evaluation of the entire body of available data for consistency and biological plausibility. Potentially relevant studies should be judged for quality and studies of high quality given more weight than those of lower quality. The evaluation of individual data on toxicity should follow the principles outlined within Chapter R.4 of the *Guidance on IR&CSA* on the Evaluation of available information. When both epidemiological and experimental data are available, similarity of effects between humans and animals is given more weight. If the mechanism or mode of action is well characterised, this information is used in the interpretation of observed effects in either human or animal studies. A *Weight-of-Evidence* approach is not to be interpreted as simply tallying the number of positive and negative studies, nor does it imply an averaging of the doses or exposures identified in individual studies that may be suitable as starting points for risk assessment. The study or studies used for the starting point are identified by an informed and expert evaluation of all the available evidence. The relevance of absence of effects in animal studies for predicting absence of potential effects in humans has to be addressed. This is especially the case when other types of data indicate an effect.

The available repeated dose toxicity data should be evaluated in detail for a characterisation of the health hazards upon repeated exposure. In this process an assessment of all toxicological effects, their dose-response relationships and possible thresholds are taken into account. The evaluation should include an assessment of the severity of the effect, whether the observed effect(s) is (are) adverse or adaptive, reversible or irreversible, or precursor to a more significant effect or secondary to general toxicity. Correlations between changes in several parameters, e.g. between clinical or biochemical measurements, organ weights and (histo)pathological effects, will be helpful in the evaluation of the nature of effects. Further guidance to this issue can be found in publications of the International Programme on Chemical Safety (IPCS, 1994; 1999), ECETOC (2002) and WHO (2016).

The effect data are also analysed for indications of potential serious toxicity of target organs or specific organ systems (e.g. neurotoxicity or immunotoxicity), delayed effects or cumulative toxicity. Furthermore, the evaluation should take into account the study details and determine if the exposure conditions and duration and the parameters studied are appropriate for an adequate characterisation of the toxicological effect(s).

If an evaluation allows the conclusion that the information of the repeated dose toxicity is adequate for a robust characterisation of the toxicological hazards, including an estimate of a dose descriptor (NOAEL/LOAEL/BMDL), and the data are adequate for risk assessment and C&L, no further testing is necessary unless there are indications for further risk, according to column 2 of Annexes VIII-X to the REACH Regulation.

Another consideration to be taken into account is whether the study duration has been appropriate for an adequate expression of the toxicological effects. If the critical effect involves serious specific system or target organ toxicity (e.g. haemolytic anaemia, neurotoxicity or immunotoxicity), delayed effects or cumulative toxicity and a threshold has **not** been established, then dose extrapolation may not be appropriate and further studies are required. In this case a specialised study is likely to be more appropriate for an improved hazard characterisation and should be considered instead of a standard short-term rodent or sub-chronic toxicity test at this stage.

In the identification of a NOAEL, other factors need to be considered such as the severity of the effect, presence or absence of a dose- and time-effect relationship and/or a dose- and time-response relationship, biological relevance, reversibility, and normal biological variation of an effect that may be shown by representative historical control values (IPCS, 1990).

R.7.5.5.1 Concluding on suitability for Classification and Labelling

According to REACH, the data used (existing or generated) should be adequate for the purposes of C&L and risk assessment (box 3 in Figure R.7.5–1).

Therefore, the data should allow a comparison with the CLP criteria for STOT RE classification in Category 1 or 2. These criteria focus on the strength and severity of the effects and the dose levels at which they occur related to the classification categories.

Basically the following conclusions can be obtained from the assessment of adequacy for C&L for repeated dose toxicity:

- Data are considered adequate for the purpose of C&L if they allow a comparison against the criteria for STOT RE classification under CLP (box 3 in Figure R.7.5–1) [108].
- Data are considered as inadequate for the purpose of C&L and cannot be checked against the CLP criteria (inconclusive or lacking data). In this case testing should be considered.

For further details, see Section 3.9 of the *Guidance on the Application of the CLP criteria*.

R.7.5.5.2 Concluding on suitability for Chemical Safety Assessment

In order to be suitable for CSA (box 3 of Figure R.7.5–1), appropriate DNELs have to be established for each exposure scenario. Typically, the derivation of the DNEL takes into account a dose descriptor, modification of the starting point and application of assessment factors (see Chapter R.8 of the *Guidance on IR&CSA*).

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[108] It should be noted that although the exposure assessment and risk characterisation do not need to be performed when a substance is not classified (see Part A, section A.1.2 of the *Guidance on IR&CSA*), for potency-based endpoints like repeated dose toxicity there could still potentially be a risk. Therefore one might consider performing an exposure assessment and risk characterisation on a voluntary basis, to ensure safe handling and use.

For the identification of the so-called dose descriptor an appropriate threshold dose for the critical effect should be established as the starting point for DNEL derivation, i.e. a NOAEL or BMDL. If a NOAEL can not be identified, the LOAEL may be used instead provided the data are adequate for a robust hazard assessment, however, when possible, the BMD approach is preferred over the LOAEL-NAEL extrapolation.

It is to be noted that the dose descriptor should be route-specific. Thus, in case only animal data with oral exposure are available and humans are exposed mainly *via* skin and/or inhalation, a DNEL for dermal route and/or DNEL for inhalation route are needed: i.e. route-to-route extrapolation is needed, if allowed. Guidance for this route-to-route extrapolation is provided in Section R.8.4.2 in Chapter R.8 of the *Guidance on IR&CSA*.

If this route-to-route extrapolation is not allowed, route-specific information is needed, possibly including testing, as a last resort (see Section R.7.5.6.3).

Derivation of a DNEL from this dose descriptor by applying AFs (to address uncertainty in the available data) is described elsewhere (see Section R.8.4.3 in Chapter R.8 of the *Guidance on IR&CSA*; see also Section R.7.5.4.4).

R.7.5.5.3 Information not adequate

A *Weight of Evidence* approach comparing available adequate information with the tonnage-triggered information requirements by REACH may result in the conclusion that the requirements are not fulfilled. In order to proceed in further information gathering the testing strategy described in Section R.7.5.6.3 can be adopted.

R.7.5.6 Integrated Testing Strategy (ITS) for repeated dose toxicity

R.7.5.6.1 Objective / General principles

The objective in this testing strategy is to give guidance on a stepwise approach to hazard identification with regard to repeated dose toxicity (Figure R.7.5–1).

A principle of the strategy is that the results of all available studies are evaluated before another study is initiated. The strategy seeks to ensure that the data requirements are met in the most efficient and humane manner so that animal usage and costs are minimised.

The core objectives of the Integrated Testing Strategy (ITS) for repeated dose toxicity are to generate sufficient information to allow:

- Characterisation of the hazard profile and the dose-response of a substance upon repeated exposure;
- Performance of a chemical safety assessment for repeated dose toxicity.

Information generated in this strategy should be suitable for C&L according to the criteria given in Annex I to the CLP Regulation.

In addition, information from repeated dose toxicity studies can give valuable information for other endpoints based on repeated exposure (e.g. reproductive and developmental toxicity), and are valuable for other *in vivo* studies.

R.7.5.6.2 Preliminary considerations

On the basis of the objectives outlined above, a framework has been developed so that informed decisions can be made on the need for further testing. If generation of further data is deemed necessary, the information needs should be met efficiently in terms of resources and animal use. This means using the most appropriate study type in accordance with the tonnage-driven requirements stipulated by the REACH information requirements and taking into account modifications due to considerations of exposure, grouping and category formation. The data requirements may be increased or decreased taking into account exposure considerations or the level of concern noted during any of the stages in the testing strategy.

Testing for repeated dose toxicity is not required for substances produced at tonnage levels less than 10 tonnes per year (t/y). At higher production volumes, standard data requirements are, in general, increased with each tonnage band (see Section R.7.5.2). Maintaining flexibility to adopt the most appropriate testing regime for any single substance is a key component of the ITS. However, regardless of whether testing for repeated dose toxicity is required or not at a specific tonnage level, all existing test data and all other available and relevant information on the substance should be collected.

In the previous Section R.7.4, the possibility to use a sub-acute oral toxicity study to adapt the information requirement for the acute oral toxicity has been addressed. This adaptation may be proposed when the NOAEL from the sub-acute study is above 1000 mg/kg and when low acute toxicity can be supported by some additional information, which should then be used in a *Weight-of-Evidence* approach. In case a registrant has some indications that a substance is of low toxicity and intends to "waive" the acute oral toxicity study, he should perform the sub-acute oral toxicity study first, i.e. before the acute oral study. Detailed guidance on this *Weight-of-Evidence* based adaptation of the acute oral toxicity study is given in Appendix R.7.4–1 to Section R.7.4.

R.7.5.6.3 Testing and assessment strategy for repeated dose toxicity

The overall testing and assessment strategy for repeated dose toxicity is outlined in Figure R.7.5–1.

In brief, the strategy starts with gathering all available information relevant for repeated dose toxicity (step 1 of Figure R.7.5–1).

This information is then evaluated (step 2 of Figure R.7.5–1) to determine whether it meets the standard information requirements of Annexes VII-X to the REACH Regulation (Column 1) or can be used to justify a Column 2 adaptation argumentation for the specific endpoints (see also Sections R.7.5.6.3.1, R.7.5.6.3.2 and R.7.5.6.3.3 below). Different descriptors used for repeated dose toxicity in these annexes vary from *limited* (Annex IX) to *no relevant exposure* (Annex VIII). In addition, Annex XI to the REACH Regulation contains basic approaches, or rules for adaptation of the standard testing regime, set out in Annexes VII-IX (see Section R.7.5.6.3.5 below and Chapter R.5 of the *Guidance on IR&CSA*).

The adequacy of the available information needs also to be considered (step 3 of Figure R.7.5–1). Exposure considerations at this stage may trigger a need for additional data if the applications include wide dispersive uses to a large population (e.g. consumer products) and if a particular concern exists for a low margin of exposure. The data to be generated at this stage should aim at improving the risk quotient and could therefore be a trigger for an improved exposure characterisation or an improved hazard characterisation. In the latter case the required information might include a special study leading to an improved characterisation of the critical toxic endpoint thereby decreasing the uncertainty in the NOAEL for repeated dose toxicity. An example of such a testing approach applied to neurotoxicity is given in Appendix R.7.5–1.

Furthermore, before new testing is initiated, the available information should be scrutinised for evidence that may indicate severe effects, serious specific system or target organ toxicity (e.g. neurotoxicity or immunotoxicity), delayed effects or cumulative toxicity. These indications may provide a trigger for specialised study protocols instead of the standard protocols for the short-term and/or (sub)chronic toxicity. These specific protocols should be designed on a case-by-case basis, such that they enable an adequate characterisation of these hazards, including the dose-response, threshold for the toxic effect and an understanding of the nature of the toxic effects. An example of such an approach is given in Appendix R.7.5–1.

Based on all the previous steps, a decision should be made (step 4 of Figure R.7.5–1) as to whether the available information is sufficient and adequate to properly conclude on C&L and to perform a CSA (step 4A), or whether it is insufficient and/or inappropriate and further information needs to be generated (step 4B). Registrants should note that a testing proposal must be submitted for a new *in vivo* study mentioned in Annex IX or X. Following examination of such testing proposal, ECHA has to approve the test in its evaluation decision before it can be undertaken.

The new data generated should then be evaluated (steps 2 and 3 of Figure R.7.5–1) to see whether they allow a conclusion on repeated dose toxicity to be reached.

Figure R.7.5–1 Testing and assessment strategy for repeated dose toxicity

STEP 1

Collect all available information

relevant for repeated dose toxicity including animal, *in vitro*, *in silico* and human data

STEP 2

Evaluate all available information

- if the available information addresses the respective standard information requirement of Annex VIII, 8.6.1 or Annex IX, 8.6.2. *or*
- if a specific column 2 rule for adaptation applies *or*
- if a general rule for adaptation of Annex XI applies (e.g. use of existing data, weight of evidence, read-across, exposure-based adaptation)

STEP 3

Consider the adequacy of the available information:
are ALL of the following criteria met ?

- the available information is adequate for C&L *and*
- the available information is adequate for chemical safety assessment (CSA; e.g. DNEL derivation) *and*
- there are no indications of further hazard beyond those covered by the standard information requirement

YES

STEP 4 A

No further information is required:

Conclude on C&L

Perform CSA

NO

STEP 4 B

Further information needs to be generated:

Perform new *in vivo* study (Annex VIII)

Propose new *in vivo* study (Annexes IX, X)

Utilisation of the different tests at each of the different tonnage levels is summarised below. It should be noted that the latest update of a test guideline (OECD TG and/or EU method) should be used for conducting new tests. In addition, Section R.7.5.6.3.4 should be considered before deciding on the test design for repeated dose toxicity assessment.

R.7.5.6.3.1 10 t/y or more (Annex VIII to the REACH Regulation)

At this tonnage level a short-term (28-day) toxicity test (OECD TG 407 / EU B.7) is usually required. The use of a combined repeated dose toxicity study with the reproduction/developmental toxicity screening test (OECD TG 422[109]) is recommended if an initial assessment of repeated dose toxicity and reproductive toxicity is required. The route of exposure in these tests is oral unless the predominant route of human exposure or the physico-chemical properties indicate that the dermal or inhalational route may be a more appropriate route of exposure to assess the repeated dose toxicity test (requiring OECD TG 410 or 412 / EU B.9 or B.8).

If the results of a short-term rodent toxicity study (OECD TGs 407, 410, 412, 422) are adequate for dose-response characterisation, C&L and risk assessment, and if there are no indications for further risks, no further testing is required (see Section R.7.5.5.2 for a detailed discussion of the criteria for a robust hazard characterisation).

At this tonnage level the short-term toxicity study (28 days) does not need to be conducted if:

- a reliable sub-chronic (90 days) or chronic toxicity study is available, provided that an appropriate species, dosage, and route of administration were used; or
- where a substance undergoes immediate disintegration and there are sufficient data on the cleavage products; or
- relevant human exposure can be excluded in accordance with Annex XI Section 3.

It should be noted that any of the rules for adaptation according to Annex XI also applies (see Chapter R.5 of the *Guidance on IR&CSA*). For further details see Section R.7.5.6.3.5 on Annex XI below.

According to REACH (Annex VIII, Section 8.6.1, column 2), the sub-chronic toxicity study (90 days) must be proposed by the registrant if:

- the frequency and duration of human exposure indicates that a longer term study is appropriate;

and one of the following conditions is met:

- other available data indicate that the substance may have a dangerous property that cannot be detected in a short-term toxicity study; or
- appropriately designed toxicokinetic studies reveal accumulation of the substance or its metabolites in certain tissues or organs which would possibly remain undetected in a short-term toxicity study but which are liable to result in adverse effects after prolonged exposure (see "Indications on (bio)accumulation in animals or from human biomonitoring data" under Point 2 of Appendix R.7.6–2).

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[109] To date there is no corresponding EU test method available.

REACH (Annex VIII, Section 8.6.1, column 2) also specifies that further studies must be proposed by the registrant or may be required by the Agency in accordance with Article 40 or 41 in case of:

- failure to identify a NOAEL in the 28 or the 90 days study, unless the reason for the failure to identify a NOAEL is absence of adverse toxic effects; or
- toxicity of particular concern (e.g. serious/severe effects); or
- indications of an effect for which the available evidence is inadequate for toxicological and/or risk characterisation. In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity); or
- the route of exposure used in the initial repeated dose study was inappropriate in relation to the expected route of human exposure and route-to-route extrapolation cannot be made; or
- particular concern regarding exposure (e.g. use in consumer products leading to exposure levels which are close to the dose levels at which toxicity to humans may be expected); or
- effects shown in substances with a clear relationship in molecular structure with the substance being studied, were not detected in the 28 or the 90 days study (see https://echa.europa.eu/support/registration/how-to-avoid-unnecessary-testing-on-animals/grouping-of-substances-and-read-across).

It should be pointed out that a failure to identify a NOAEL does not lead to a data gap in every case and should not trigger additional studies by default. If the data are sufficient for a robust hazard assessment and for C&L, the LOAEL or BMDL may be used as the starting point for the CSA (see also Sections R.7.5.4.4 and R.7.5.5 and Chapter R.8 of the *Guidance on IR&CSA*).

A specialised study is likely to be more appropriate for an improved hazard characterisation and should be considered instead of a standard short-term rodent or sub-chronic toxicity test at this stage.

R.7.5.6.3.2 100 t/y or more (Annex IX to the REACH Regulation)

At this tonnage level, the following information is required (REACH Annex IX, Sections 8.6.1 and 8.6.2):

- a short-term study (28 days) is the minimum requirement. The preferred route of administration in these tests is oral (OECD TG 407 / EU B.7; TG 422[110]) unless the predominant route of human exposure, physico-chemical properties and/or route-specific toxicokinetic behaviour or toxicity indicate(s) that the dermal or inhalation route (OECD TGs 410, 412 / EU B.9, B.8) is the most appropriate route of administration in the repeated dose toxicity tests.
- a sub-chronic toxicity study (90 days) in a single rodent species is usually required. The preferred route of administration in these tests is oral (OECD TG 408 / EU B.26) unless the predominant route of human exposure, physico-chemical properties and/or route-specific toxicokinetic behaviour or toxicity indicate(s) that the dermal or inhalation route

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110 To date there is no corresponding EU test method available.

(OECD TGs 411, 413 / EU B.28, B.29) is the most appropriate route of administration in the repeated dose toxicity tests.

According to REACH, at this tonnage level the sub-chronic toxicity study (90 days) does not need to be conducted if:

- a reliable short-term toxicity study (28 days) is available showing severe toxicity effects according to the criteria for classifying the substance as STOT RE Category 1 or Category 2, for which the observed NOAEL-28 days, with the application of an appropriate assessment factor, allows the extrapolation towards the NOAEL-90 days for the same route of exposure; or

- a reliable chronic toxicity study is available, provided that an appropriate species and route of administration were used; or

- a substance undergoes immediate disintegration and there are sufficient data on the cleavage products (both for systemic effects and effects at the site of uptake); or

- the substance is unreactive, insoluble and not inhalable and there is no evidence of absorption and no evidence of toxicity in a 28-day limit test, particularly if such a pattern is coupled with limited human exposure.

It should be noted that any of the rules for adaptation according to Annex XI also applies. For further details see Section R.7.5.6.3.5 on Annex XI below.

In case human exposure is limited or different in frequency and duration from that used in the test protocol for repeated dose toxicity, the sub-chronic toxicity study may not be necessary if the data for the short-term toxicity study are adequate for a robust hazard characterisation, a risk assessment and classification and labelling (C&L). This adaptation requires full justification by the registrant.

In case the *Weight of Evidence* indicates that the available information is adequate to characterise the short-term toxicity and sufficiently robust for proper dose-selection of the 90-day study, a dedicated 28-day study is not necessary at this stage.

No further testing is required if the available data, which may include a sub-chronic rodent toxicity study (OECD TGs 408, 411, 413 / EU B.26, B.28, B.29) are adequate for a dose response characterisation and C&L and risk assessment.

In case data are inadequate for hazard characterisation and risk assessment further studies must be proposed by the registrant or may be required by the Agency in accordance with REACH Articles 40 or 41: according to REACH Annex IX, Section 8.6.2, column 2, such a situation may arise if there is:

- failure to identify a NOAEL in the 90 days study unless the reason for the failure to identify a NOAEL is absence of adverse toxic effects; or

- toxicity of particular concern (e.g. serious/severe effects); or

- indications of an effect for which the available evidence is inadequate for toxicological and/or risk characterisation; In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity); or

- particular concern regarding exposure (e.g. use in consumer products leading to exposure levels which are high relative to the dose levels at which toxicity to humans occurs).

A specialised study is likely to be more appropriate for an improved hazard characterisation and should be considered instead of a standard short-term rodent or sub-chronic toxicity test. An example of such an approach is given in Appendix R.7.5–1.

It should be pointed out that a failure to identify a NOAEL does not lead to a data gap in every case and should not be a default trigger for additional studies. If the data are sufficient for a robust hazard assessment or for C&L, the LOAEL or BMD may be used as the starting point for the CSA (see also Sections R.7.5.4.4 and R.7.5.5 and Chapter R.8 of the *Guidance on IR&CSA*).

R.7.5.6.3.3 1000 t/y or more (Annex X to the REACH Regulation)

There is no default testing requirement for repeated dose toxicity at this tonnage level beyond those recommended for the level 100 t/y or more (see above). However, in accordance with REACH Articles 40 and 41, if the frequency and duration of human exposure indicate that a long-term study is appropriate and one of the following conditions is met, a long-term repeated toxicity test (≥12 months) may be proposed:

- serious or severe toxicity effects of particular concern were observed in the 28-day or 90-day study for which available evidence is inadequate for toxicological evaluation or risk characterisation; or
- effects shown in substances with clear relationship in molecular structure with the substance being studied were not detected in the 28-day or 90-day study; or
- the substance may have a dangerous property that cannot be detected in a 90-day study.

In addition, further studies must be proposed by the registrant or may be required by the Agency in accordance with REACH Articles 40 or 41, in case of:

- toxicity of particular concern (e.g. serious/severe effects); or
- indications of an effect for which the available evidence is inadequate for toxicological evaluation and/or risk characterisation; In such cases it may also be more appropriate to perform specific toxicological studies that are designed to investigate these effects (e.g. immunotoxicity, neurotoxicity); or
- particular concern regarding exposure (e.g. use in consumer products leading to exposure levels which are close to the dose levels at which toxicity is observed).

In some cases, a specialised study might be the most appropriate study if an improved hazard characterisation is necessary and should be considered instead of a standard sub-chronic or chronic toxicity test. An example of such an approach given in Appendix R.7.5–1.

No further testing is required if the results of a sub-chronic rodent toxicity study (OECD TGs 408, 410, 411, 412, 413 / EU B.26, B.9, B.28, B.8, B.29) are adequate for a robust hazard characterisation and suitable for risk assessment and C&L (see Sections R.7.5.4.4 and R.7.5.5 for a detailed discussion of the criteria for a robust hazard characterisation).

Also, the testing requirements can be adapted if any of the rules according to Annex XI apply. For further details see Section R.7.5.6.3.5 on Annex XI below.

As there is no standard test requirement at this tonnage level, column 2 does not contain any waiving options.

R.7.5.6.3.4 Further considerations for studies that will be performed

In case a new study needs to be generated, the test has to be conducted in accordance with an appropriate test method, according to the principles of good laboratory practice and in line with animal welfare principles. In addition, several considerations are required to ensure that the results will be appropriate for hazard identification. These are important for the selection of the most appropriate route of administration.

Selection of the most appropriate route of administration

A repeated dose toxicity study must be performed by either the oral, inhalation or dermal route. To decide on a specific route, it requires first to identify the appropriate routes. If more than one route is appropriate, a decision on the most appropriate route of administration is required.

Concerning repeated dose toxicity testing, the oral route is the default one because it is assumed to maximise systemic availability (internal dose) of most substances. However, on a case-by-case basis, the appropriateness of other routes of administration should also be assessed. Depending on the physico-chemical properties of a substance and the most relevant route of human exposure, the dermal or the inhalation route can also be appropriate as specified in Annexes VIII and IX to the REACH Regulation.

Non-physiological routes of human exposure, such as i.v., i.m., s.c., i.p., are usually considered not appropriate routes of administration for animal testing to be requested for the REACH Regulation.

It has to be noted that *in vivo* testing with corrosive substances at concentration levels causing corrosivity must be avoided.

- Appropriateness of the dermal route of administration

Testing for repeated dose toxicity by the dermal route is appropriate if skin contact with the substance in production and/or use is likely and the physico-chemical properties suggest a potential for a significant rate of absorption through the skin (for further details, see Table R.7.12-3 in Chapter R.7c of the *Guidance on IR&CSA*). Testing for sub-acute toxicity (28 days) by the dermal route requires furthermore that inhalation of the substance is unlikely. Testing for sub-chronic toxicity (90-days) by the dermal route further requires that one of the following conditions is met:

  - toxicity is observed in the acute dermal toxicity test at lower doses than in the oral toxicity test; or
  - systemic effects or other evidence of absorption is observed in skin and/or eye irritation studies; or
  - *in vitro* tests indicate significant dermal absorption; or
  - significant dermal toxicity or dermal penetration is recognised for structurally-related substances.

If the substance is a severe irritant or corrosive, testing by the dermal route should be avoided unless it can be performed at doses that do not cause severe iriation or corrosion and provided that such doses are still toxicologically relevant for evaluating systemic toxicity and the outcome can be used in risk assessment.

A study by the dermal route might especially be required if route-to-route extrapolation is problematic, e.g. where a study with oral or inhalation administration does not allow reliable route-to-route extrapolation due to significant qualitative differences in metabolism in comparison with dermal exposure. In practice, the differences are most likely due to differences in first pass metabolism or sensitivity to hydrolysis by stomach acid.

- Appropriateness of the inhalation route of administration

Testing for repeated dose toxicity by the inhalation route is appropriate if exposure of humans *via* inhalation is likely, taking into account the vapour pressure of the substance and/or the possibility of exposure to aerosols, particles or droplets of an inhalable size (for further details, see (See Table R.7.12-2 in Chapter R.7c of the *Guidance on IR&CSA*).

Testing by the inhalation route is the default route for gases and the preferred route for liquids of high to very high vapour pressure at ambient temperature (>25 kPa or boiling point below 50°C) for which inhalation is usually the predominant route of human exposure.

For liquids of lower vapour pressure and for dusts (including nanomaterials), testing by the inhalation route is appropriate if human inhalation exposure is likely taking into account the possibility of exposure to aerosols, particles or droplets of an inhalable size (aerodynamic diameter below 100 µm). Further guidance on nanomaterials is available in Appendix R.7-1 *Recommendations for nanomaterials applicable to Chapter R.7a* of the *Guidance on IR&CSA.*

- Selection of the most appropriate route of administration

In case more than one route of administration are appropriate, it is necessary to consider which is the ***most appropriate*** route of administration. This requires evaluating the advantages and disadvantages of all appropriate routes of administration.

Balancing of different routes of administration can include the following aspects:

- Preferred routes of administration, i.e.:
    - inhalation for gases and liquids of very high vapour pressure (>25 kPa or boiling point below 50°C),
    - inhalation, if effects may occur for which oral-to-inhalation extrapolation will not be appropriate; e.g.:
        - if there is some concern for systemic effects following inhalation exposure which might not be detected following oral administration[111]
        - if there is some concern for local effects in the respiratory tract for which a qualitative assessment might not be sufficiently robust to demonstrate safe handling and use of the substance[112]

[111] Systemic effects that could occur following inhalation exposure might not be appropriately detected in a study with oral administration in case there are relevant route-specific toxicokinetic differences. For example, in case the substance is metabolised in the respiratory tract into reactive metabolites, or the substance undergoes a relevant first pass-effect in the gastro-intestinal tract or the liver after oral administration, the oral administration can be expected not to reflect the toxicity of the substance following inhalation exposure.

- oral for all other substances;

- Human exposure, e.g.:
  - route with presumed highest human exposure considering physico-chemical properties of the substance and its uses, with particular attention to exposure of professionals and/or consumers;

- Intrinsic properties/database, e.g.:
  - availability of route-specific information,
  - clarification of a concern for route-specific effect(s),
  - requirement of route-specific information to decide on the design of further test(s) like the extended one-generation reproduction toxicity study;

- Risk assessment, e.g.:
  - requirement of specific DNEL(s),
  - requirement of qualitative assessment,
  - application of risk management measures,
  - uncertainties in the database,
  - proportionality of study types (e.g. economic arguments might be considered in case two routes, e.g. the oral and the inhalation routes, are of equal appropriateness);

- Feasibility (e.g. testing by the inhalation route might be technically difficult for some substances).

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[112] A concern for local effects in the respiratory tract might be assumed *inter alia* for substances that are corrosive or irritating for the skin and/or eyes, substances that are hydrolysed/metabolised in the respiratory tract into reactive metabolites or insoluble inhalable dusts that accumulate in the lungs.

Additional investigations

To adequately identify the hazard of a substance it might be necessary to perform additional investigations, which are either described as optional in the test methods or which are additional to the requirements of the test methods. Additional investigations can be triggered by existing information on the substance or on structurally analogous substances derived from animal studies or non-animal tests that provide an indication for specific effects expected from the administration of the substance (i.e. in relation to neurotoxicity, immunotoxicity, endocrine disruption).

The possibility to explore several parameters within the design of the repeated dose toxicity study could be considered (toxicokinetic data generation, micronucleus formation, neurotoxicity, immunotoxicity) taking into account potential limitations when modifying test protocols in order to investigate specific effects. However, care should be taken when an OECD compliant study design is altered in such a way that validity of that study is compromised.

Specific Investigations that can be important for nanomaterials (e.g. lung burden and Bronchoalveolar lavage (BAL) measurements) are indicated in Appendix R.7-1 *Recommendations for nanomaterials applicable to Chapter R.7a* of the *Guidance on IR&CSA*.

- Toxicokinetics

Toxicokinetic data should be considered in the light of other toxicity data (i.e. repeated dose toxicity) to assist in the estimation of internal exposure to the substance and/or its metabolites and the correlation of the effects observed with internal dose estimates. This is of particular importance for characterising a dose-response relationship and determining whether administered doses caused saturation kinetics resulting in a non-linear dose-response. Such information is valuable for the derivation of assessment factors, route-to-route extrapolation and derivation of DNELs.

In addition, generation of toxicokinetic data (including metabolism characterisation) is considered essential for the application of read-across approaches when common metabolic pathways are part of the similarity justification.

OECD TG 417 provides the protocol for the conduct of toxicokinetic studies either as stand-alone test or in combination with repeated dose toxicity studies.

In recent years, progress has been made in the development of alternatives for the generation of TK data including *in silico* metabolism simulators (OECD Toolbox, commercial solutions) and PBPK modelling. Further details on the use of *in silico* methods for kinetic modelling are available in Section R.7.12 in Chapter R.7c of the *Guidance on IR&CSA*.

- Bronchoalveolar lavage (BAL) optional for inhalation studies

OECD TGs 412 and 413 for sub-acute and sub-chronic inhalation studies provide the option that, when there is evidence that the lower respiratory tract (i.e. the alveoli) is the primary site of deposition and retention, then bronchoalveolar lavage (BAL) may be the technique of choice to quantitatively analyse hypothesis-based dose-effect parameters focusing on alveolitis, pulmonary inflammation, and phospholipidosis. This allows an assessment of dose-response and time-course changes of alveolar injury. BAL measurements generally complement the results from histopathology examinations but cannot replace them. Guidance on how to perform lung lavage can be found in OECD GD 39 (OECD, 2009). OECD TGs 412 and 413 are currently under revision and will include further guidance on BAL measurements.

- Neurotoxicity and immunotoxicity

Information on the mode of action derived from the available data on the substance or data from structurally similar substances should be considered in the design of repeated dose toxicity tests. Such considerations can lead to the inclusion of parameters to be measured for investigating a potential endocrine mode of action, neurotoxicity or immunotoxicity.

It should be noted that endpoints for detailed analysis of neurotoxicity and immunotoxicity are not examined in the standard 28-day and 90-day dermal or inhalation repeated dose toxicity studies. However, it is stated in the OECD TG 413 (90-day inhalation study; 2009) that : "If neurotoxicity is expected or is observed in the course of the study, the study director may choose to include appropriate evaluations such as a functional observational battery (FOB) and measurement of motor activity."

Further Guidance on neurotoxicity is available in Appendix R.7.5–1.

If investigations regarding immunotoxicity need to be performed as part of the repeated dose toxicity test, these should be performed where relevant in a way that allows evaluation of the immunotoxicity potential (e.g. Repeated dose toxicity according to US EPA OPPTS 870.7800 – Health Effects Test Guidelines Immunotoxicity). Reviews of principles for immunotoxicity are available from WHO/IPCS publications and can be considered as additional guidance (WHO, 1996a; 1996b; 1999; 2007; 2012).

- Endocrine mode of action

An endocrine disrupter is an exogenous substance or mixture that alters function(s) of the endocrine system and consequently causes adverse health effects in an intact organism, or its progeny, or (sub)populations (WHO, 2002).

Repeated dose toxicity studies provide information on a broad variety of potential health hazards, including effects on the reproductive, nervous, immune and endocrine system. Depending on the parameters measured, they also add insight that can help elucidate the mechanism(s) of endocrine mediated effects.

OECD GD 150 (OECD, 2012) provides an analysis on the sensitivity and investigations within repeated dose toxicity studies that are considered relevant for endocrine disruption.

Furthermore, the combined repeated dose toxicity / reproductive screening study (OECD TG 422) has been updated with endocrine disruptor relevant endpoints.

- Mode of action / Adverse Outcome Pathway

Further guidance on Mode of action is available from the WHO/IPCS Framework on Mode of Action and Human Relevance (see http://www.who.int/ipcs/methods/harmonization/areas/cancer/en/).

In addition information from the OECD Adverse Outcome Pathway programme (see http://www.oecd.org/chemicalsafety/testing/adverse-outcome-pathways-molecular-screening-and-toxicogenomics.htm) can provide insight into potential pathways relevant for the testing of a substance and the consideration of specific investigations that are likely to be relevant for a particular mode of action.

- Alpha 2u-globulin mediated nephropathy

If a substance leads to kidney effects in male but not in female rats, this may be indicative of an alpha 2u-globulin-mediated nephropathy. It is important to distinguish between a male-specific renal toxicity, which is not mediated by alpha 2u-globulin and which would be presumed relevant for human risk assessment, and alpha 2u-mediated nephropathy. Since humans do not have a functional alpha 2u-globulin gene, this mode of action is considered not relevant to humans (IARC, 1999). The involvement of alpha 2u-globulin in mediating the male rat-specific kidney effects is therefore important for establishing the relevance of the kidney effects for risk assessment. To prove that the effects on the kidney are indeed mediated by alpha 2u-globulin, urinalysis (which is optional in the test methods) is required to investigate kidney function and full histopathological examination is required, including immuno-histochemical investigation to demonstrate the involvement of alpha 2u-globulin in the renal pathology (see for example Hamamura *et al.* (2006) and IARC (1999)).

Due to the extensive database on rats, this species is currently the preferred one to test substances that induce alpha 2u-globulin-mediated nephropathy. However, in case alpha 2u-globulin-mediated nephropathy is limiting the dose that can be applied, use of another species, e.g. the mouse, may be considered and should be justified.

- Additional parameters on reproductive toxicity

Repeated dose toxicity studies may be amended by including reproductive parameters like sperm parameters and/or oestrus cycles measurements. These examinations should be used to ensure the safe use of the substance. Performance of such investigations is at the discretion of a registrant.

Combination of studies

Considering animal welfare, it might be sensible to combine a repeated dose toxicity study with a study that is required to fulfil a different information requirement. Combining studies lies in the responsibility of the registrants and requires careful consideration since a combination of studies also has drawbacks. It needs to be ensured that a combination of studies does not impair the validity and the results of the information of each individual study.

The combined repeated dose toxicity study with the reproduction/developmental toxicity screening test (OECD TG 422) is a combination of a sub-acute toxicity study and the screening study for reproductive/developmental toxicity. The advantages and disadvantages of this test are described above (see Section R.7.5.4.1.2).

For combining a repeated dose toxicity study with an *in vivo* mammalian erythrocyte micronucleus test and/or an *in vivo* mammalian alkaline comet assay, specific considerations and references are provided in OECD TGs 474 and 489.

Combining a sub-chronic toxicity study with the extended one-generation reproductive toxicity study is generally not supported. Information from the sub-chronic toxicity study may be valuable when deciding on the dose levels and the study design for the extended one-generation reproductive toxicity study. However, if the study design of the extended one-generation reproductive toxicity study includes all cohorts and existing information on sub-chronic toxicity has some limitations, then information from the extended one-generation reproductive toxicity study may, together with the existing information, fulfil the information requirement for sub-chronic toxicity.

R.7.5.6.3.5 REACH Annex XI adaptations of the standard testing regime for repeated dose toxicity

General guidance on the application of the Annex XI adaptations of information requirements is given in Chapter R.5 of the *Guidance on IR&CSA*. For repeated dose toxicity the following additional guidance applies.

Testing does not appear scientifically necessary

Some substances may be excluded from testing for repeated dose toxicity if it does not appear scientifically necessary (Annex XI, Section 1). This might be the case for example if:

- existing data on repeated dose toxicity are available from a study that was not carried out according to GLP or the test methods referred to in REACH Article 13(3) but these data adequately and reliably cover the key parameters of the corresponding test method referred to in Article 13(3) and are adequate for classification labelling and/or risk assessment, exposure duration in that study is comparable or longer than that of the standard test method, and adequate and reliable documentation of the study is provided;

- a *Weight-of-Evidence* demonstrates that the available information is sufficient for an adequate hazard characterisation and a CSA where the exposure to the substance is adequately controlled;

- for substances belonging to a group or a category of substances that have a common functionality and/or breakdown products or sufficient information for a qualitative and quantitative understanding of the toxicological properties, testing of all individual category members may not be necessary (Annex XI, Section 1.5). The criteria for application of read-across for a category of substances and detailed guidance can be found in Sections R.4.3.2 and R.6.2 of the *Guidance on IR&CSA* (see also OECD, 2014).

Testing is technically not possible

There may also be cases where it is technically not possible to conduct a repeated dose toxicity test (Annex XI, Section 2). This might be the case if for example:

- The substance ignites in air in ambient conditions;

- The substance undergoes immediate disintegration. In such a case the information requirements for the cleavage products should be assessed following an approach similar to that outlined in this document.

Substance-tailored exposure-driven testing

Annex XI, 3.2 (a) sets very stringent boundaries/requirements for waiving a repeated dose toxicity study. Three criteria need to be met: (i) concerns "the absence of or no significant exposure", (ii) concerns relevance and appropriateness of the DNEL and (iii) requires "that exposures are always well below the derived DNEL". A more detailed explanation of this waiving possibility is given in Section R.7.5.4.3.4 "Waiving of repeated dose toxicity studies".

A potentially more likely adaptation possibility is set out in Annex XI, 3.2(b), which requires documentation showing that the substance is only handled under strictly controlled conditions. These conditions, i.e. the techniques, controls and procedures that need to be in place in order for the registrant to use this waiving possibility, are specified in Article 18(4) of the REACH Regulation.

Annex XI, 3.2(c) deals with substances "permanently embedded in a matrix" and would only apply to these special cases. It is noteworthy that points 3.1 and 3.3 of Annex XI are not self-standing or independent waiving possibilities but general requirements, which apply to all adaptations specified under point 3.2.

R.7.5.7 References on repeated dose toxicity

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